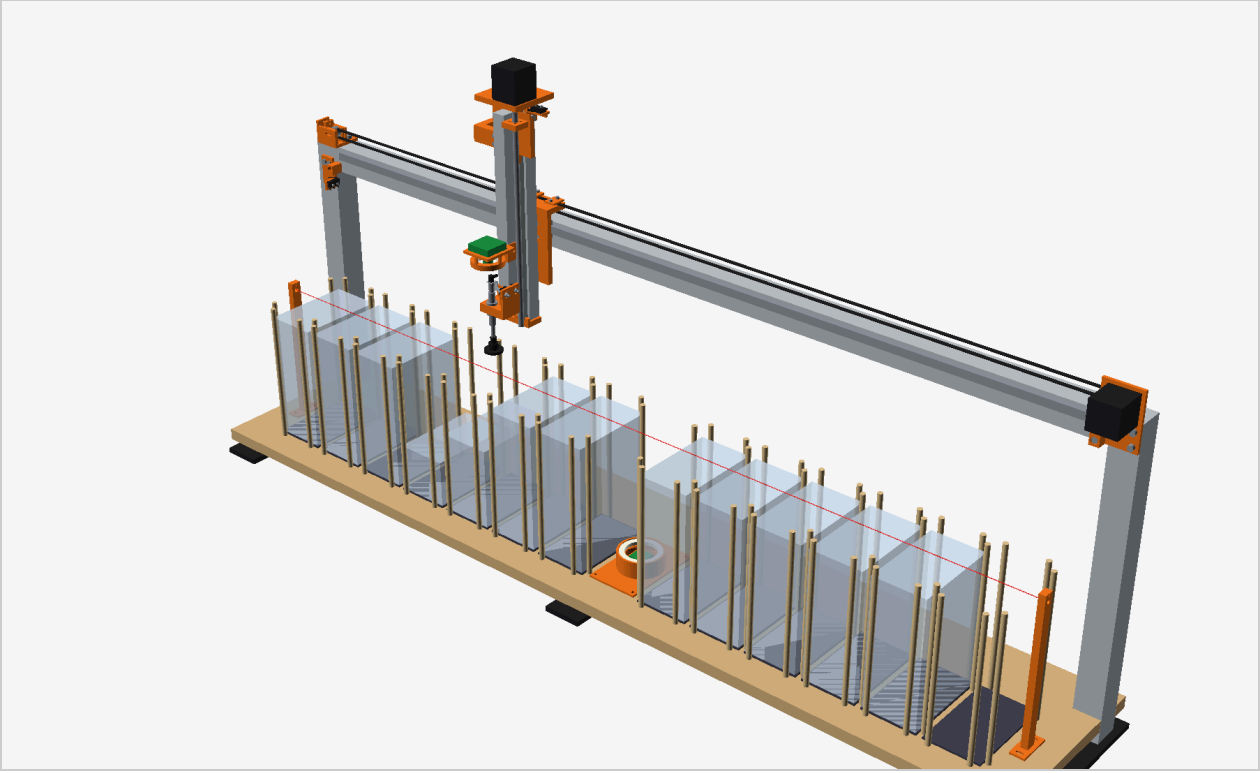
DESIGN REVIEW SET

SlabRail: graded-slab batch scanner

Engineering design package – Rev A, draft for design review



Designed, CAD-checked and simulated. Not yet built.

For reviewers: the reviewer brief, the design rationale and checks, the drawing set, the wiring and the bill of materials.

2026-10-27 · built from git commit `47e3b4b + uncommitted changes`

https://github.com/Icevyy14/codex/tree/claude/psa-slab-batch-scanner-w2e2m9/psa-slab-scanner

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Page numbers are those of this PDF. The drawing sheets are ANSI B landscape and carry their own sheet numbers. Links to files that are not in this PDF open the repository.

SlabRail reviewer brief

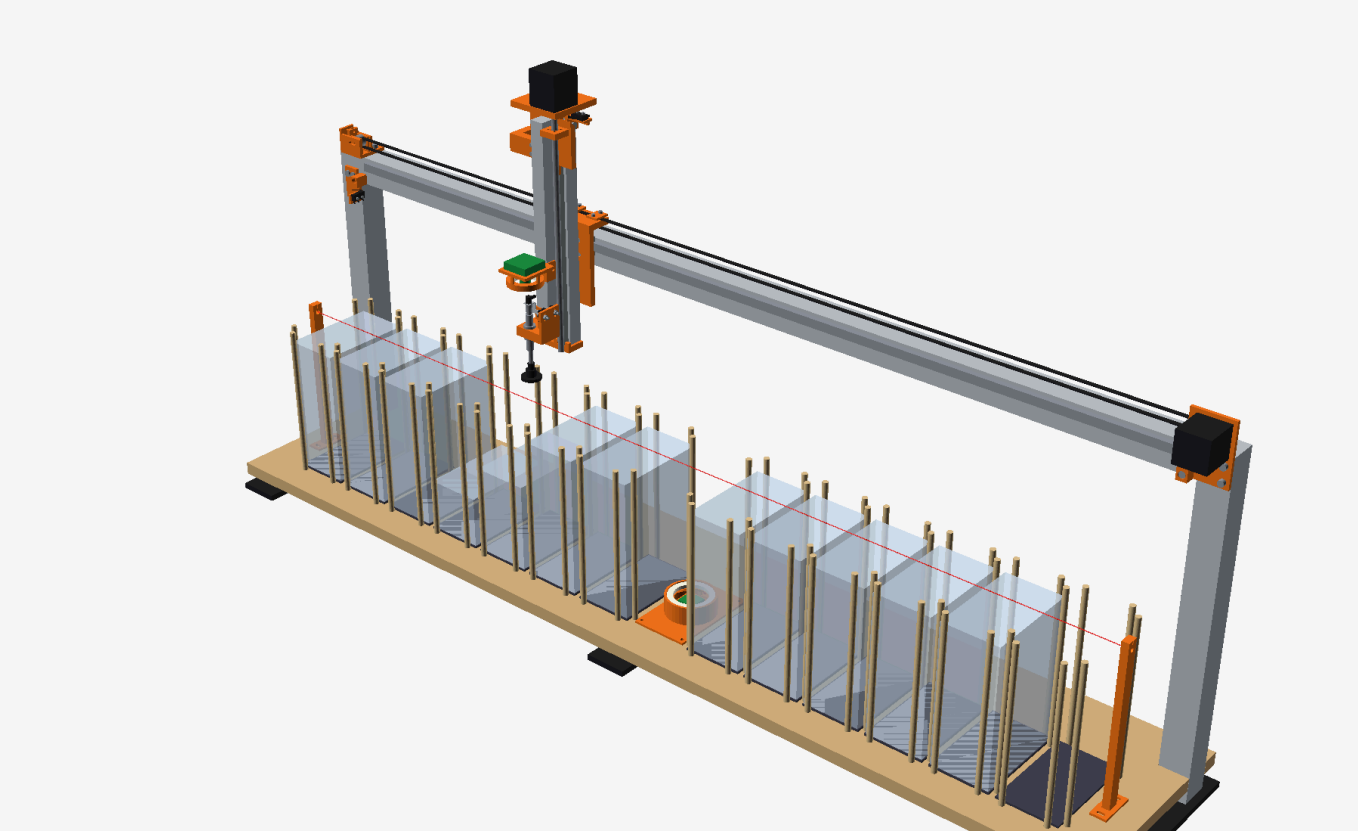
Read this first. It says what SlabRail is, what has been checked and how, where each kind of information lives, what is missing, and what we would like you to review (§9). The files are listed in §10.

Nothing has been built. Every result below comes from CAD checks, software tests, a simulator or public label scans. No real slab, camera, cup, motor or wire has been tested yet.

Confidence tags are copied from the doc each number comes from: **[Certain]** = measured, computed, or checked in the CAD or simulator; **[Likely]** = strong inference; **[Guessing]** = needs a real test. **[Certain] on a calculation or a CAD check means the arithmetic or the geometry is exact for the model and its inputs; it is not a test of the real part**, and many inputs are catalog values or estimates (§8). Units are mm unless stated. "DESIGN §4" is section 4 of `docs/DESIGN.md`; "D-05" is a drawing sheet (§10.2).

1. What it is

SlabRail is designed to read the cert numbers of graded trading-card slabs, PSA and CGC, unattended. One straight row of 14 cells sits under one 4040 gantry beam: 13 pockets formed by wooden dowels, and an upward read camera in cell 7. The owner loads up to 330 slabs face-down in pockets 1–11 (11 stacks of 30). A suction cup lifts the top slab of a stack straight up, a head camera reads the back label, and the cup sets the slab down in the empty pocket to its left, so every stack ends up one pocket left. A slab whose back can't be read is held over the upward camera to read its front label. Anything still unread or doubtful goes to pocket 12, the exception pocket. The output is `certs.csv` (cert, grader, status, pocket and position) and a photo of every label.



The owner is a student building it in a school shop. **[Likely] A $350–650 hands-free 2D scanner gets 80% of the way with no build** (README); the machine buys back about 10–15 minutes per batch and adds a photo of every slab.

2. Requirements as the owner gave them

Requirement	Where written	How the design answers it
Load 200–300+ slabs, walk away, get every cert number back	README; DESIGN §1–2	330 per load, one CSV row per slab, a count check; 79.4 min per load in the simulator. **But** until the learned face-down check is active, every slab without a back-label code stops the run (§5 step 3), and the build guide says not to leave it alone before three clean full runs [Guessing threshold].
PSA first, CGC too	README; CONTRACT	The rules read and stack both. BGS and TAG go to pocket 12 (only if ≤ 82.6 mm wide). SGC (about 86 mm) does not fit the 84 mm pocket.
Fewest moving parts	DESIGN §2	One row, no Y axis: 2 motors, a spring-loaded cup, 2 valves, 1 pump.
Budget about $2,000; $2,500–3,000 acceptable	CONTRACT	$2,181.94 with a 15% allowance [Likely ±10%]; worst case about $2,380 [Guessing] (BOM.md).
Built with school-shop tools	CONTRACT; BUILD_GUIDE	Extrusions bought cut (Misumi, ±0.5 mm); printed parts; rod holes drilled through a printed template that the machine positions (BUILD_GUIDE §5B). 65–90 h of hands-on work [Guessing].

Design rules: the cup touches only a slab's back and no slab slides on another (DESIGN §2–3); a wrong cert is worse than none, so the software never guesses (OPERATING).

3. Status and evidence

The next step is two bench gates that decide whether it gets built:

- **Phase 0A, camera test** (≈ $470, ≈ $281 with a borrowed Pi). Both cameras on a jig that holds the machine's head parts where the machine does, on 30–50 of the owner's slabs. GO: ≥ 95% read, 0 wrong certs, 0 false face-down proofs (BUILD_GUIDE §1.9).
- **Phase 0B, cup test** (≈ $155). The real SMC cup on a hand vacuum pump: seal, leak, and a 24 h marking test. GO: no mark that survives a dry microfiber wipe (BUILD_GUIDE §1.10).

Evidence	What it shows	What it does not show
CAD checks `cad/checks.py`: **177/177 pass** [Certain], 23 Oct 2026; numbers in `cad/out/checks.txt`	Clearance and interference of the model at every pick, place, travel, station, hard-stop and Phase 0 pose (last-slab pick 14.55 from full neighbours, loaded travel 21.4 over the rods, a double pick of two 11.5 holders 5.33); sight lines; switch and sensor trips.	Loads, deflection, belt slack, cables and tube, real shapes of bought parts, switch-lever side load, frame brackets (not in the CAD), print accuracy.
Software tests: **299 passed, 1 skipped** (23 Oct 2026; the skip is the real-image test, which passes alone) [Certain]	Parsing of real PSA and CGC payload strings, grader and face-down rules, carry-sensor logic, the stop paths with a slab in the cup.	Real cameras, GPIO electrics, serial timing. The bench-test tool has run only on the simulator.
Simulator: 330 slabs, seed 7, default mix [Certain, sim]; raises `Crash` on any unsafe move	`ok` 324, `needs_review` 6, no wrong cert or grader in an `ok`. Face-down: 287 by back QR, 41 learned, 2 confirmed. Double pick, seal failure, overfill, Z stall and sensor faults end without a simulated crash.	Real physics (air films, slip, leaks, stalls). The mix is [Guessing]. Every simulated slab is face-down.
Real labels: `tools/real_eval.py` on public scans of 64 real slabs [Certain for these images]; a decode sweep on synthetic labels	**0 wrong certs in an `ok`, 0 false face-down proofs.** Native: 60 `ok`, 2 `needs_review`, 2 `no_read` (BGS and SGC carry no code); at machine-camera scale 60 of 61 `ok`. Back QR reads at ≤ 0.14 mm/px through blur and glare.	The rules were written from these 64 slabs. Flat scans have no slab-plastic reflections.
Hardware	Nothing.	Everything physical.

4. The machine at a glance

Parameter	Value	Source
Overall size	**1436 × 289.5 × 720** (X × Y × Z): X −78 … 1358 over the feet, Y −117.5 … 172 (the bed), Z −49 (feet underside) … 671 (Z motor top); the same in every pose [Certain, CAD bounding box of the assembly STEP]	`cad/out/assembly/*.step`; D-01
Frame	Bed 1416 × 289.5 (18 mm plywood) on two 2040 × 1416 base rails; one portal: two 4040 × 401 uprights on the back base rail and a 4040 × 1416 X beam, centre 400 above the pad; 2040 × 340 Z spine; 2020 × 285 drop arm; joints not in the CAD	`cut_list.csv`, `layout.py`, BUILD_GUIDE §4
Pockets, slab	92 pitch, centres X = 42 + 92·cell; inside 84 × 139: ±1.85 / ±1.75 play on an 80.3 × 135.5 × 6.6 slab (design max 82.6 × 136.5 × 7.0, ≤ 65 g; thick holders 11.5); 82 dowels (5/16 in).	`layout.py`, `dims.py`
Heights	MAX fill line 210, overfill laser 214, rod tops 218; travel c = 248 (empty and loaded); station read c ≈ 171.6	`layout.py`, `machine.toml`
X axis	NEMA17, GT2 20T belt, 80 steps/mm; one MGN12H block on a 1300 rail; soft limits 22… 1253, printed hard stops at cup X 15 and 1262; 15000 mm/min, 2000 mm/s²; 1.0 A	CONTRACT, FIRMWARE
Z axis	NEMA17 (1.5–2.0 A, ≤ 4 mH), integrated T8×8 screw cut to 340; one MGN12H block on a 330 rail; c = −12…250 used, hard stops c = −14 and 262; 6000 mm/min, 1200 mm/s²; run 0.8 A, hold 0.5 A	CONTRACT, FIRMWARE
Drivers	TMC2160, 1/16 microstepping, spreadCycle, no StallGuard, never switched off; `must_home: true`. `r_sense_ohms` ships at 0.022 until the board's resistors are read: the wrong value gives 3.4 × or 0.29 × the set current.	FIRMWARE "Read this first", "What config.yaml sets"
Cup, vacuum	SMC `ZPT25UUK20-B5-A10`: Ø25 urethane cup, M10 spring buffer, 20 mm stroke; probe trips at 2.0 ± 0.5 of travel (about 1.2 N [Likely]). 19.1 N at −50 kPa, SF 30 on a 0.64 N slab; 13.4 N, SF 21 at the −35 kPa seal level (reached within 0.4 s) [Certain, arithmetic; the 78% effective area is an estimate]. Pump, reservoir, valves: BUILD_GUIDE §8.1.	DESIGN §4, `config.py`, BUILD_GUIDE §8.1
Sensors	Fail-safe optical contact probe (Rodent input); carry-height sensor (Pi GPIO 6, blocked from c = 246.5 up); vacuum sensor; overfill laser gate (GPIO 5); X home and Z top NC microswitches; optional lid switch (GPIO 25)	CONTRACT, WIRING §5–8
Cameras	Head: ELP 8 MP USB, 3.6 mm lens, 140 above the held slab, 5 toward the label from the cup axis, 0.055 mm/px. Station: Pi Camera Module 3, 140 below, 0.039 mm/px. The LED rings' mirror image misses the label band, so no polarisers [Certain, mirror geometry; real glare untested until Phase 0A].	CONTRACT, DESIGN §5
Control, power	BTT Rodent (FluidNC v4.1.1+) over USB to a Raspberry Pi 5 4 GB, which switches the pump, valves and rings. LRS-150-24 at 24.5 V behind a fused IEC inlet; E-stop in the +24 V line; fuses 5 / 3 / 1 A; Pi on its own supply.	CONTRACT, WIRING §1, `img/wiring.svg`
Cycle, cost, prints	14.2 s per slab, 79.4 min for 330 [Certain, sim; Guessing until measured]. Parts $1,897.34, **$2,181.94 with 15%** [Likely ±10%]. 25 printed parts (27 prints, 801 cm³, PETG, ≈ 44.5 h).	DESIGN §6, BOM.md, `print_list.json`

5. One slab's trip

Once per run: home (Z up, then X left), refuse to start if the overfill beam is blocked, survey every pocket with the probe (repeated and compared on resume), and in learned mode take one head frame over an empty pocket to check the camera's aim. Then for every slab (`runner.transfer`):

#	Step	Guarded by
1	Move over the source pocket at c = 248.	**Carry-height sensor** must read blocked before any sideways move.
2	Drop to just above the predicted stack top (`G38.3`).	**Contact probe** stops any downward move on contact.
3	One head frame: face-down check, first read of the back.	**Head camera.** Not proven face-down: stop untouched. Front code seen: FACE-UP stop.
4	Probe to the real top, at most 12 past the prediction.	Fail-safe probe (a cut wire reads as touching). > 4.5 taller than predicted stops the run.
5	Grip: seal at −35 kPa within 0.4 s, one retry.	**Carry-height sensor** must read clear; **vacuum sensor**.

#	Step	Guarded by
6	Peel at 2 mm/s through release, 0.5 s low over the stack; lift to c = 248; two head frames.	**Probe hard limit**; a Z stall over about 2.5 stops before any sideways move; vacuum alarm above −25 kPa.
7	If needed, front frames over the station (c ≈ 171.6); decode and OCR over the output stack.	**Station camera**; rules in `slabscan/certs.py`.
8	Probed set-down (1200 mm/min through the rod tops) on the output pocket or pocket 12.	Step ≥ 11.6: DOUBLE PICK; far too big: obstruction; under half a slab: cup empty.
9	Vent 0.25 s; back within 0.5 kPa of atmosphere within 1.5 s; write CSV row, photos, journal.	Failed vent: nothing moves. Count check at the end.

6. Safety and failure handling

No safety standard, safety category or performance level is claimed in any doc, and no risk assessment exists. What the docs have:

Hazard	What the design does	Gap	Source
Mains (120 V)	Fused IEC inlet (T3.15A [Likely]) and rocker in a closed ABS box; 115/230 selector set first; PE to the frame; pre-power checks (PE < 0.5 Ω); instructor check.	Fuse not checked against the LRS inrush. Anodised joints may not carry the earth [Likely].	WIRING §1.1, §12, §14
E-stop	Two latching NC blocks in series in the +24 V line: cuts the Rodent, pump, valves and LED buck, not the Pi, sensors or laser. Motors go limp; Z holds on detent [Likely].	Blocks rated DC-13 1.5 A at 24 V [Certain, BOM.csv]; WIRING §1.3 asks ≥ 5 A at 24 V DC; peak load ≈ 3.4 A [Guessing], capacitive. A weld shows only in the monthly check. **On release the pump and vacuum valve restart by themselves** (the Pi still drives them); motion needs `$H` first.	WIRING §1.3; OPERATING §8.1; BUILD_GUIDE §13.4
Pinch points	Downward moves are probing moves: contact that lifts the cup should stop the head (about 1.2 N at the trip, 1.6 N after a full-speed trip [Likely; never tested against a hand]). A "hands clear?" prompt on resume.	**No guard, enclosure or pinch analysis.** Not sensed: other head parts, the X axis, the open belt and pulleys at the beam ends, the X carriage at its bumpers, the Z carriage at its stops, the drag chain.	BUILD_GUIDE §14; DESIGN §3
Remote start	Start, Stop and Vent on the web page; Ctrl-C twice halts the motors but leaves them powered.	**The web page has no login and can start a run from a phone** [Certain, `web.py`]. The lid switch is a software check between slabs, not a power interlock [Certain, `runner._check_pause`].	OPERATING §3
Laser	≤ 1 mW Class 2 650 nm, 4 above the MAX line; the KY-008 (usually 5 mW, Class 3R [Likely]) excluded by name; labels.	On whenever the Pi is on, E-stop included, at about face height for someone bent over the stacks.	WIRING §8
Held slab, power cut	Both valves normally closed: the cup line stays sealed; the reservoir is on the pump side and does not help.	Hold time is the line's leak; ≥ 60 s is a [Guessing] target.	DESIGN §3; BUILD_GUIDE §13.4
Heat	Motors kept under about 50 °C.	No PETG temperature limit or check.	FIRMWARE step 6

Failure	Detected by	What happens	Designed / tested (bare § = BUILD_GUIDE)
Vacuum loss	Vacuum sensor	Carries the slab back slowly and probes it down. A slab is in the air about 2 s per cycle; a fall from carry height: ≤ about 240 mm, ≈ 0.15 J [Certain, arithmetic]; a corner on a dowel could chip [Guessing].	DESIGN §3 / simulator; BUILD_GUIDE §13.3, §13.4
Double pick	Placement step ≥ 11.6	`needs_review` "DOUBLE PICK" plus an "UNREAD" row; two 11.5 holders clear the rods by 5.33 [Certain, CAD].	DESIGN §2a / simulator; §13.5
Z stall	Carry sensor clear at travel height	Stops before any sideways move; a held slab is lowered onto its own pocket, or held. Does not prevent a stall.	WIRING §6a / simulator; §7.8

Failure	Detected by	What happens	Designed / tested (bare § = BUILD_GUIDE)
Misread, face-up	Consensus rules; hover frame	Doubtful reads go to pocket 12; a slab not proven face-down is not touched.	DESIGN §5 / unit tests, real labels; §13.1, §13.8
Sensor fault, error	Fail-safe probe wiring; the run's guard	FluidNC refuses to probe (`ALARM:4`); the slab is put back, or the pump holds it.	WIRING §6 / simulator; §13.6

7. What a reviewer will look for, and where it is

Item	Where	State
System block diagram	Electrical and control: `docs/img/wiring.svg`, WIRING §1. Vacuum: BUILD_GUIDE §8.1. Software: CONTRACT. Sequence: §5.	No single system diagram.
Moving-mass breakdown	DESIGN §4: CAD roll-up, 0.40 kg head and 1.62 kg X carriage, centre of mass 77 in front of the beam centre; the calculations use 0.75 / 1.9 kg	Heaviest items only, several masses [Guessing]; no full per-part list.
Motor and driver settings	FIRMWARE "What config.yaml sets"; `config.yaml`	Complete on paper; no torque curve for the BOM Z motor.
Power budget	WIRING §1.2	[Guessing] until parts are measured.
Failure modes	§6; DESIGN §3, §7; OPERATING §8.1 (every stop message)	No FMEA with severity or occurrence.
Acceptance criteria	BUILD_GUIDE §1.9, §1.10, §6.7, §7.8, §13.9	Some limits are judgments [Guessing].
Mains, fusing, E-stop	WIRING §1, §12, §14; OPERATING §8.1	See §6.
Guarding, pinch points	Nowhere	**Missing.**
Laser class, eye safety	WIRING §8	Class 2, labelled, no beam stop.
Structural loads	DESIGN §4 (beam sag, twist); BUILD_GUIDE §4.3 (joint plan)	**Missing:** joints, rail moments, drop arm, spine, PETG parts.
Tolerances; cables and tube	D-00 note 6, BUILD_GUIDE §4.1, §4.5; WIRING §11	Several PROPOSED, no stack-up; cables and tube not in the CAD.

8. Key assumptions

Input	Value used	Tag, source	Feeds
Slab mass	≤ 65 g (0.64 N)	`dims.py`	Cup SF, drop
Cup area, lip friction	78% of Ø25 effective; friction coefficient about 0.5	an estimate; a guess (DESIGN §4)	Hold 19.1 / 13.4 / 9.6 N; lateral SF
Buffer spring	SMC: 1.0 N at rest, 3.0 N at full stroke → about 1.2 / 2.4 / 3.0 N at 2 / 14 / 20 mm	[Likely], DESIGN §4	Trip, contact force
Moving masses	0.75 / 1.9 kg assumed (head / X carriage); CAD roll-up 0.40 / 1.62 kg, about 0.53 / 1.94 kg with allowances and a slab	DESIGN §4; allowances [Guessing]	Torque, twist
Screw, rotor	T8×8 at 40% efficiency (50–65% [Likely]); 2 N X friction; 60–80 g·cm² rotor	DESIGN §4; rotor [Guessing]	Torque
Motor torque	Z ≈ 0.15–0.2 N·m at 750 rpm at full current; X ≈ 0.15–0.3 N·m at 375 rpm at the 1.0 A run current	[Likely], DESIGN §4; no curve read	SF ≈ 2–4 (Z), ≈ 4–8 (X)
Detent, back-drive	≈ 0.015 vs ≈ 0.005 N·m	[Likely], DESIGN §4	Z hold unpowered
Beam torsion	J 1.0–2.5 cm⁴, G 26 GPa, both ends fixed	[Guessing on J], DESIGN §4	≈ 0.5 mm twist

Input	Value used	Tag, source	Feeds
Belt tension	15–25 N	[Guessing], BUILD_GUIDE §6.6	PETG loads
Load current	≈ 1.2 A typical, ≈ 3.4 A peak	[Guessing], WIRING §1.2	Fuses, E-stop
Bought parts	Catalog or listing envelopes; head camera 32 mm hole pattern and 20 mm board-to-lens	[Guessing] for the camera, BUILD_GUIDE	Clearances
Printed holes	0.1–0.3 small	[Likely], DESIGN §7	Fits
Label mix, card-window limit	The simulator's default mix; 2% on the holder	[Guessing]; a judgment (DESIGN §6, §7)	Sim results, face-down check

9. What we'd like you to review

A short answer with the question number is enough. Label and face-down risks are left to Phase 0 (DESIGN §7 has the full risk table).

1. **Frame joints.** The CAD has no brackets. BUILD_GUIDE §4.3 proposes [Guessing on the layout] two 4040 L brackets in each beam-to-upright corner, a back plate (none in front: the X rail ends 18 mm from each upright), and uprights held by L brackets bolted M5×30 through the 18 mm plywood into the back base rail. These joints react the head's overhang, and the twist estimate (Q2) assumes they fix the beam ends in torsion; nothing calculates their stiffness. Enough, or end-tap and bolt the uprights? (D-01, D-02)
2. **Beam twist.** ≈ 0.5 mm in Y at the cup at midspan (0.3–0.8 for J 2.5–1.0 cm^4) [Guessing on J]: $\theta = T \cdot L/(4GJ)$, ends fixed, T 1.46 N·m from the CAD centre of mass, G 26 GPa; brackets and rail-block play not included (DESIGN §4). The build check hangs 0.75 kg on the cup over pocket 6 (0.93 N·m [arithmetic: 0.75 kg × g × 127 mm], about 64% of the carriage's own 1.46 N·m), so its reading is not directly the estimate; a rear brace goes on above 1.0 mm (BUILD_GUIDE §6.7). Twist also tilts the head camera in the plane of the aim limits (Q11) and varies along X, while the aim check looks over one pocket (an inference; no doc relates them). Credible? Measure angle? (D-04)
3. **Rail loads.** Each axis runs on one MGN12H block (generic rails; the X listing states no preload). The X block carries the 1.62 kg carriage (CAD; about 1.94 kg with hardware and a slab), centre of mass 77 in front of the beam centre, Z motor top at Z 671. No moment-load, deflection or life check exists. One block enough? (D-04, D-05)
4. **Z torque at 0.8 A.** 0.026 N·m on the T8×8 for 0.75 kg at 1.2 m/s^2 without rotor inertia; with the CAD's 0.53 kg plus inertia, 0.025–0.027 N·m (DESIGN §4). At 0.8 A (40–55% of rating) SF ≈ 2–4 [Likely]; no torque curve. The test passes if a bare-pad probe agrees within 0.1 mm after 100 full-speed lifts; the fixes are 3000 mm/min, then about 60% of rated current (BUILD_GUIDE §7.8). More margin from the start? Is 100 lifts enough for a 330-slab run? (D-05)
5. **Z hold, power off.** Back-drive about 0.005 N·m (0.003–0.005 with the CAD mass) against about 0.015 N·m detent [Likely]. The test hangs 200 g on the cup, presses the E-stop and watches 1 minute; any creep means a brake or counterweight, whose mount is not in the CAD (BUILD_GUIDE §7.8). Design it in now? (D-05)
6. **Switch side load.** Each home switch is worked by a striker sliding across its hinge lever (a 45° lead-in, then a flat holding it 0.6 mm past the click, inside the 1.2 mm overtravel). The lead-in pushes the lever sideways about as hard as down; the datasheet rates only a straight push [Guessing that it holds]. Acceptable, or a roller on a cam? (BUILD_GUIDE §6.8; D-09, D-11)
7. **PETG parts.** The load-bearing parts (5–6 walls, 40–50% infill) carry the Z motor, the head and the belt (15–25 N [Guessing]) beside motors allowed about 50 °C. No doc gives a PETG temperature limit; nothing checks creep or stiffness. Which would you change? (D-08 `x_motor_mount`, D-09 `x_bracket`, D-10 `z_plate`, `z_motor_mount`, D-12 `head_foot`)
8. **Cup margin, marking, power-cut hold.** At the software's thresholds: 13.4 N, SF 21 at the −35 kPa seal; 9.6 N, SF 15 at the −25 kPa transit alarm; lateral SF 49 / 34 on a guessed friction coefficient of 0.5 [Certain,

arithmetic; 78% area an estimate] (DESIGN §4). Marking: 24 h on a hand pump, then 300 cycles + 24 h at mean ≤ −40 kPa (BUILD_GUIDE §1.10, §13.2). After a power cut the slab hangs on the closed cup line only, ≥ 60 s [Guessing target], with a cup-side reservoir as the fallback (BUILD_GUIDE §8.2). Is the marking plan enough for slab plastic, and the hold for unattended runs? (D-17)

9. **E-stop and mains.** The blocks are rated DC-13 1.5 A at 24 V [Certain, BOM.csv]; WIRING §1.3 asks for ≥ 5 A at 24 V DC, and the bus feeds capacitive inputs at up to ≈ 3.4 A [Guessing]. A weld is found only monthly; the pump restarts on release. The T3.15A fuse [Likely] is not checked against inrush; one earth bond crosses anodised joints [Likely]. Acceptable, or the E-stop in the mains feed (the WIRING §1.3 alternative) or a force-guided safety relay (linked contacts that reveal a weld)?
10. **Sensor fail-safety.** The carry sensor's "OK to move" state is dark, so a dead emitter or cut GND is caught only at the next grip [Certain] (WIRING §6a). The probe flag may chatter during the slow peel; the fix is a Schmitt stage [Guessing] (WIRING §6). Acceptable? (D-05, D-06, D-11)
11. **Head camera aim.** The learned check needs the camera within 1.34° toward the cup [Certain, CAD]; on a CGC slab in a pocket corner, about 0.01° toward the label puts over 2% of the card window on the holder [Likely]. The only adjustment is an M2 washer shim (about 0.5° [Likely]). What can the `camera_bracket` / drop-arm stack-up hold, and how would you measure it? (DESIGN §5; D-05, D-12)
12. **Guarding and remote start.** An optional lid switch that pauses between slabs and removes no power; a web page with no login that can start a run. What minimum guard and interlock before unattended runs, and should a start need someone at the machine? (§6)
13. **Tolerances.** D-00 note 6d marks these PROPOSED (no doc states them): printed parts ±0.3, plywood outline ±0.5, sheet-metal outlines ±0.2, dowel cut length ±0.5. The design's own fits: pocket play ±1.85 / ±1.75, rod holes drilled by the machine, Misumi cuts ±0.5 mm, printed holes 0.1–0.3 mm small [Likely]. Right, and holdable in a school shop? (D-00; D-03, D-06, D-07–D-17)
14. **Cables and tube.** Not in the CAD: the 1 m drag chain, the 350 mm Z loop, the vacuum tube (about 1 mm under the head ring at full stroke [Guessing]), about 4 m of head USB [Guessing]. Any routing or flex-life problem? (WIRING §11, §16; D-05)

10. How to read this package

10.1 Start here

Where	What it is
`package/SlabRail_Design_Review.pdf`	The review PDF. Start here.
`package/SlabRail_Complete_Documentation.pdf`	All the docs in one PDF.
`package/SlabRail_CAD.zip`	The CAD files in §10.4: exactly the list in `package/cad_manifest.txt`.
`web/slabrail.html`	One-page overview, as of 23 Oct 2026; where it disagrees with the docs, the docs win.
`web/slabrail_3d.html`	Interactive 3D model generated from the CAD: rotate, click a part to see what it is, hide groups.
`web/slabrail_replay.html`	Animated replay of a simulated 330-slab batch, run by the real machine software, with synthetic labels.
`docs/history/`	Design-round reports (rounds 2–5), the real-label evaluation and its independent check, sourcing data. **Historical: their numbers may be out of date.**

GitHub shows HTML as source: download a `web/` file and open it in a browser (online, for the fonts and the 3D library).

Documents. The README's "What's in this folder" table lists every doc. For review: DESIGN (architecture §2, crash layers §3, calculations §4, verification §6, risks §7), CONTRACT (the binding interface), WIRING,

FIRMWARE, BUILD_GUIDE (Phase 0 §1, acceptance tests §13), OPERATING (every stop message §8.1) and `cad/layout.py` (the master dimensions).

10.2 Drawing set

`cad/out/drawings/SlabRail_drawings.pdf`, 18 sheets, Rev A: a draft for design review, not released for fabrication. Previews are in `cad/out/drawings/png/`; `dims.json` beside them lists every dimension shown and its source, and `cad/drawings.py` re-measures each one on the model.

Sheet	Title	Sheet	Title
D-00	Index and general notes	D-09	Printed parts: X carriage
D-01	General arrangement	D-10	Printed parts: Z carriage plate, Z motor mount
D-02	Frame and extrusion cut list	D-11	Printed parts: Z end cap, Z switch mount, carry-height sensor
D-03	Bed, pockets and guide rods	D-12	Printed parts: head
D-04	X axis	D-13	Printed parts: station mount, drill template
D-05	Z axis and head stack-up	D-14	Printed parts: awl adapter, fill gauge
D-06	Flat parts (probe flag, carry flag)	D-15	Printed parts: Phase 0 jig (base, tray)
D-07	Printed parts: gate posts	D-16	Printed parts: Phase 0 jig (top, occluder mount)
D-08	Printed parts: X drive	D-17	Phase 0 test jig

10.3 Coordinates

`cad/README.md` and D-00 note 3: X left to right along the row (machine X is the cup axis), Y front (−) to back (+) with Y = 0 on the row centre line, Z up with **Z = 0 at the top of the pocket floor pads**; c is the cup-face height above it.

10.4 CAD files

The CAD zip holds the 64 files in `package/cad_manifest.txt`:

File	What it is
`cad/README.md`	Units, coordinates, poses, part names, how to open and regenerate the files
`cad/out/assembly/slabrail_<pose>.step` (4)	Full assembly: `home` (head over pocket 1 at c = 248, stacks as loaded), `pick_bottom` (cup at c = 6.6 on the last slab of pocket 11), `carry` (held slab at c = 248 between pockets 1 and 0), `station` (held slab at c ≈ 171.6). Measure on these.
`cad/out/assembly/slabrail_home.glb`, `slabrail_carry.glb`	The same assemblies as binary glTF meshes for any glTF viewer (metres, Y up, as glTF expects); for looking, not measuring
`cad/out/printed/`: 25 STEP, 25 STL, `print_list.json`	Printed parts in print orientation, with sizes and print notes. The `station_mount`, `phase0_base` and `phase0_tray` STLs are not clean meshes: check them in the slicer preview (`cad/README.md`).
`cad/out/cut_list.csv`, `probe_flag_blank.dxf`, `.svg`	Stock cut lengths (extrusions, lead screw, flags, 82 dowels); the probe flag blank at 1:1
`cad/out/checks.txt`, `cad/out/drawings/SlabRail_drawings.pdf`, `docs/BOM.csv`	All 177 check results; the drawing set; every bought part with price and link

In the repo only: the CadQuery source `cad/*.py`. Bought parts are envelopes, accurate where they meet other parts. Not in the CAD: frame brackets, cables, the vacuum tube, the electronics panel, the optional light roof and rear brace.

11. Glossary

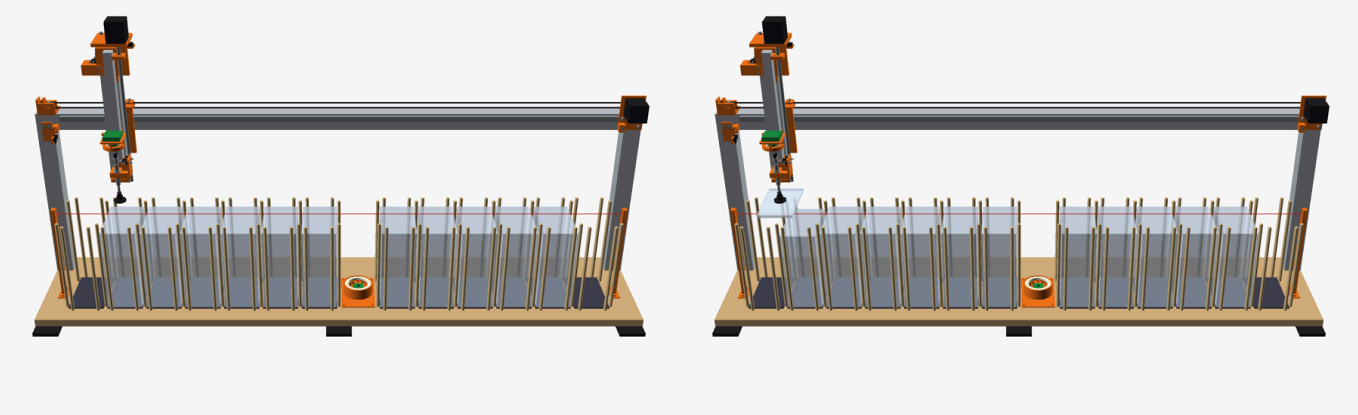
Term	Meaning
c, c_safe	Cup-face height above the pad top, the machine's Z; the free cup hangs 2 below it (`CUP_HANG`). Travel height c_safe = 248: rod tops 218 + two 11.5 holders + 5 + 2.
Cell, pocket, station	14 positions along the row: cell 7 is the station (upward camera), the rest are pockets 0–12; pocket 12 is the exception pocket.
Contact probe, probe window	Fail-safe optical sensor tripped by a flag on the spring buffer at 2.0 ± 0.5 of travel; a probing move goes at most 12 past the predicted contact.
Probe hard limit	FluidNC `probe_hard_limit`: a probe trip during a move that is not a probing move is a critical alarm.
Carry sensor	Carry-height sensor on the Z spine, blocked from c = 246.5 up.
Double pick	Two slabs lifted together, found by a placement step ≥ 11.6.
Back-label code	A code printed only on the back label (PSA back QR, graded CGC QR, CGC 2020–23 back barcode).
Face-down check, learned	Before the cup touches a slab, a head frame must prove it face-down: a back-label code, or (learned mode) its card window, a 55 × 5.5 mm strip of card back, matching references from two proven slabs. Active after a Phase 0 GO (threshold ≥ 0.55), 5 references and a clear camera-aim check over an empty pocket.
Statuses	`ok` (cert and grader confirmed), `needs_review` (a read the rules can't confirm), `conflict` (codes disagree), `no_read`, `other_grader` (a BGS, SGC or TAG cert URL).
QR, Code128, ITF	A 2D code and two 1D barcodes; ITF (interleaved 2 of 5) is on PSA labels.
J	The torsion constant of the beam section, used for the twist estimate.
Rodent, TMC2160, FluidNC	BTT controller board; its stepper drivers (StallGuard is sensorless stall detection, off here); the CNC firmware on it, driven by G-code over USB.
G38.3, `ALARM:4`	A probing move that stops on contact and does not alarm on a miss; FluidNC's alarm when the probe already reads tripped at the start (a cut fail-safe wire).
DC-13	IEC category for DC contacts switching electromagnets (the E-stop blocks' rating).

SlabRail: design rationale and engineering checks

Confidence tags: **[Certain]** = measured, computed, or checked by the CAD/simulator here. **[Likely]** = strong inference. **[Guessing]** = needs a real-slab test (see "Phase 0").

1. What it does

You load up to **330 slabs**: 11 stacks of up to 30, face-down. The machine reads every one, unattended, and writes a CSV of cert numbers with a photo of each label.



1. Load: 11 stacks of up to 30, face-down, label toward you. Pocket 0 (left) and pocket 12 (right) stay empty.
2. The cup lifts the TOP slab of stack 1 straight up, reads it, and sets it down in empty pocket 0.
3. Repeat until stack 1 is gone - then stack 2 moves into the now-empty pocket 1, and so on.
4. Done: every slab read, every stack one pocket to the left (order reversed). Rejects wait in pocket 12.

The run works like this:

1. The cup lifts the top slab of stack 1 straight up. A camera on the head reads the back of the label (QR code / barcode).
2. The cup puts the slab down in the empty pocket next to it.
3. The next slab is now on top, and the cycle repeats.
4. When stack 1 is gone, stack 2 moves into pocket 1, and so on down the row.

Slabs whose back has no readable code (pre-2016 labels, damage, glare) take a detour. They are held over an **upward camera** in the middle of the row, which reads the front label (barcode + printed digits). Anything still unreadable, read only by OCR, or from a grader other than PSA/CGC, goes to the **exception pocket** at the right end.

2. Why this architecture

Option	Verdict	Reason
Buy something	None exists	[Likely] No product feeds 200–300 graded slabs unattended. Raw-card sorters (Roca, PhyzBatch, SortSwift) exclude slabs. Sheet-fed scanners are rated "Slabs: No". See RESEARCH.md.
Hands-free 2D presentation scanner (~$350–650)	Best no-build option	[Likely] 20–30 slabs/min by hand, ~10 min per 250. The machine saves that time and adds a photo of every slab.
Gravity hopper / friction feeder	Rejected	[Certain, arithmetic] The bottom slab gets dragged out under the whole stack. A 300-slab column is about 2 m tall and 15–18 kg, so that's ~60 N of face-on-face drag across a glossy front. PSA slabs have no stacking ridges.
Slotted racks, lift-and-read in place (concept A)	Rejected	[Certain, panel geometry] A straight lift hits the neighbouring slab. It only works if every slab leans at 10°, which leaves 2.2 mm margins. Loading is one slot at a time.
Desktop robot arm (concept D)	Rejected	[Likely] Reach covers about 80–160 slabs. The arm that reaches 300 costs about twice the budget.
Short stacks + top pick by vacuum, one-pocket shuffle (concept B)	**Chosen**	Two research agents and the design panel arrived at it independently. No slab ever slides on another. The cup touches only the back. Loading is fast because you drop in handfuls.
B on a 2-D grid (4 motors) vs **one row (2 motors)**	**One row**	You asked for the fewest moving parts. One row removes the whole Y axis: 2 motors, 2 rails and the gantry squaring. Cost: a footprint of about 1.44 m × 0.29 m (CAD, feet included; the bed is 1416 × 289.5 mm). The grid version is in git (`grid-variant-v1`).

2a. What the independent design panel said (and what changed because of it)

Five agents each designed a different concept in depth. Three judges then scored all five, each through one lens: slab safety, reliability/buildability, and practical value.

Scores (1–10)

Concept	Safety	Buildability	Practical
A, slot racks	6.5	6	5.5
B, stacks + vacuum (the parent of this design)	5	6.5	4.5
C, shelf modules, pull and read	**8**	4	4
D, robot arm	3	3	2.5
E, "flip-through lanes" page-turner	6	**7**	**6.5**

Why this is still B (as SlabRail), not E.

- [Likely] E's slab physics is unproven. Its own author says so. It needs 300 hand-loaded slabs in a row to peel, pass a 2.6–4.3° balance point, and settle flush, every time.
- Pick-and-place of flat, smooth parts from stacks is what every SMT and packaging line does. Its failure modes are known, detectable and simulated here.
- The judges scored the original B, a 4-motor grid. Their main complaints have been fixed in SlabRail:

Judge's objection to B	Fix in SlabRail
Fingers can't reach slabs in tight corrals	Separate dowels, not walls. There is a 52 mm finger gap at the front and back of every pocket for lowering bundles of 5–10 when loading, and for taking slabs off the top in bundles of up to 5 when unloading. Never lift or flip a whole stack (OPERATING.md §6).
Printed posts can lean	Dowels sit in drilled holes, 15 mm deep, drilled through a printed vertical guide.

Judge's objection to B	Fix in SlabRail
Air-film double picks	The slow leg of the peel (2 mm/s) is placed where the cup really leaves the slab: at the probe trip the spring buffer is already squeezed about 2 mm (`buffer_trip_mm`, measured in BUILD_GUIDE §7.6), so the head first rises that far at 20 mm/s while the cup stays on the slab, then 2 mm/s through the release point and 1 mm past it, then a 0.5 s dwell low over the stack, so a slab clinging by an air film falls back a few mm onto its own stack. Squeeze-film force scales with speed, so it falls about 10× to ~0.1 N, below the slab's weight. Doubles are still detected at placement.
Klipper probe macro fragility	FluidNC native `G38.3` with an explicit bounded target.
Too many moving parts	2 motors instead of 4. No gantry squaring.
"Flipping and un-reversing costs more time than it saves"	Don't flip anything. Order doesn't matter, because the CSV records each slab's exact pocket and position, and unloading goes top-down in small bundles (OPERATING.md §6). Face-down loading is a hard rule: before the cup touches a slab, the head camera must see a back-label code (a PSA or CGC back QR, or the CGC 2020–23 back barcode) or, in the default learned mode, a card back it learned from proven slabs (§5); otherwise the run stops untouched for you to check. The head leaves the card window clear (the lens sits 5 mm toward the label) [Certain, CAD]; until the learned check is active, PSA 2016 and older labels, CGC Authentic, BGS and SGC stop, unless the run is started with `--all-face-down`; then a face-up slab would get the cup on its front.

- **What we keep from the other concepts:**
  - fail-safe sensor wiring;
  - a pre-run survey (every pocket probed while you are still there);
  - the Phase 0 tests before buying motion parts;
  - a photo of every slab;
  - CSV written incrementally;
  - count reconciliation.
- **Honest residuals:**
  - Every slab is lifted and carried. A vacuum loss drops it at most about 240 mm onto foam or a stack.
  - Hands-on time is still ~10 min per 300 for loading and unloading.

3. Everything that touches a slab

Contact	Motion	Force	Worst case
Suction cup on the **back**, centre of the card window	Static. The cup only moves vertically while touching.	Trips at ~1.2 N [Likely, SMC catalog spring force]. Vacuum clamp ~19 N over the 25 mm lip.	A faint cup ring or a grit micro-mark on the **back** (allowed). Mitigations: non-marking PU/HT1 cup, IPA wipe each run, 24 h marking test before use.
Slab front resting on the back of the slab below	Static. Pick = straight lift, 2 mm/s through the point where the cup leaves the slab, then 20 mm/s, with a 0.5 s pause just above the stack. Place = probed set-down at 20 mm/s (`G38.3`).	≤ 2 N at contact. Bottom slab of a 30-stack carries 19 N (1.8 kPa, same as stacking by hand).	Lateral slip at placement if a slab lands crooked. This is the most plausible front-scuff path. [Likely] Mitigated by ±1.75 mm pocket play, rods only at the edges, and slow probed set-down.
Bottom slab front on the pocket pad	Static	≤ 19 N	Grit on the pad. Wipe pads weekly. Microfiber face.
Slab edges on 8 mm wooden dowels	Only if the slab is off-centre by > 1.75 mm	< 1 N	Edge rub on wood. The dowels sit 12 mm from every corner, so corner tips never touch.

Contact	Motion	Force	Worst case
Head / camera / Z parts	None by design	—	[Certain, CAD] ≥ 5 mm at every checked pose, including a held slab at the Z top hard stop and a double pick of two thick holders; much more in normal travel (numbers in `cad/out/checks.txt`, §6). The lead screw is cut so its tip stays above anything the head carries.

Drops. [Certain, arithmetic]

- Losing vacuum at carry height means a fall of ≤ about 240 mm, about 0.15 J, onto a stack or pad. That is roughly a 25 cm desk drop. A flat landing is very unlikely to crack a slab [Likely]. A corner landing on a dowel top could chip one [Guessing].
- Exposure: about 2 s in the air per cycle for a slab whose back reads. A slab that needs the station is held longer: the front frames are taken over the lens, then decoding and OCR (up to 30 s on a Pi) happen with the slab over its planned output stack, not over the camera.
- Both solenoid valves are normally closed, so a power cut (E-stop, mains) seals the cup line. **The reservoir does not help then:** it is on the pump side of the closed vacuum valve. The slab hangs only as long as the cup line's own leak allows, seconds to minutes. [Certain, the plumbing in `hw.py` and BUILD_GUIDE §8.1] Acceptance test 13.4 measures it. While the Pi still has power (Ctrl-C twice, a USB cable out, a park that failed) the software keeps the pump holding a slab until you take it.

Crash prevention, layered.

1. **Height bookkeeping.** Every slab's thickness is measured when it is placed (the probe step).
2. **Every downward move is a probing move.** Each one is a `G38.3`, so an unexpected contact stops the head at once instead of driving on; probe windows never go more than 12 mm past the expected surface. A trip at full Z speed squeezes the 20 mm buffer about 6 mm (about 1.6 N [Likely]).
3. **Fail-safe contact sensor.** A cut wire reads as "touching", so the machine refuses to probe.
4. **Probe hard limit** (firmware). A contact-sensor trip during any move that is not a probe (a lift, a traverse) is an immediate critical alarm.
5. **Second contact sensor.** During the start-up survey, a cup that seals without the optical sensor tripping stops the run.
6. **Overfill laser beam.** It runs 4 mm above the fill line. If anything is too tall, the machine won't start. It cannot see anything above the 218 mm rod tops, such as a slab lying across the rods: the survey catches that, because its first probe window starts at travel height.
7. **Face-down check.** No back-label code seen from above = the cup does not touch the slab, unless the learned check (the default mode, §5) matches its card window with two card backs of proven slabs. A front-label code seen from above stops with a FACE-UP message; back and front codes together, two back labels or two certs in one frame mean a neighbour's label, and stop. Modes: `learned` (default), `stop` (back-label codes only), `off` (or `run --all-face-down` for one batch).
8. **Carry-height sensor (Z stall guard).** A slotted optical sensor on the Z spine and a flag on the Z carriage: blocked from c = 246.5 up to the top stop, clear below. Before every sideways move it must read blocked, and before every grip clear (a dead emitter or dirt reads blocked). A loaded head that stalled more than about 2.5 mm low therefore stops the run before it moves sideways; one that still reads blocked (c ≥ 245.5) carries even a double pick of two thick holders 2.5 mm over the rods. [Certain, CAD and simulator] After a stall the head never moves sideways with a slab: over its own pocket it probes straight down and vents, elsewhere it holds. Empty travel runs at 248 too, so the sensor's single edge sits just under both travel heights (about +0.2 s a slab). Not covered: `jog`, `calibrate` and `mark`.
9. **Plausibility checks.** A stack taller than predicted by > 4.5 mm, a vent that does not reach atmosphere, a vacuum sensor reading outside its range, or an exception pocket that's full stops the run. After any stop the

head re-reads its real position, lifts straight up, and puts a held slab back; where that is unsafe, nothing moves and the pump keeps holding it.

4. Calculations

Quantity	Value	Margin
Cup holding force, Ø25 mm at −50 kPa (78% effective area)	19.1 N	Slab weighs 0.64 N → **SF 30** vertical [Certain, arithmetic; the 78% effective area is an estimate]
Cup holding force at the software's seal level, −35 kPa (`vac_seal_kpa`)	13.4 N	**SF 21** on the 0.64 N slab, 19 with the 1.2 m/s² Z acceleration [Certain, arithmetic; same 78% area]. A grip counts as sealed only at this level or deeper.
Cup holding force at the transit alarm, −25 kPa (`vac_transit_alarm_kpa`)	9.6 N	**SF 15**, 13 with the Z acceleration [Certain, arithmetic; same 78% area]. A carry that has not raised the alarm holds at least this. The slab falls off at about −1.7 kPa (BUILD_GUIDE §13.4).
Lateral grip vs X acceleration 2 m/s², at −50 / −35 / −25 kPa	~9.3 / 6.4 / 4.5 N vs 0.13 N	**SF ≈ 70 / 49 / 34** [Certain, arithmetic: friction on the clamp force less the slab's weight; the friction coefficient of about 0.5 between cup lip and slab is a guess]
Z load (0.75 kg head, 1.2 m/s²)	8.3 N → 0.026 N·m on T8×8 at 40% efficiency (conservative: the screw's 20° lead angle gives about 50–65% at a friction coefficient of 0.15–0.25 [Likely])	A 1.5–2.0 A NEMA17 at 750 rpm on 24 V gives ≈ 0.15–0.2 N·m at full current, but Z runs at 0.8 A (40–55% of rating) → ≈ 0.06–0.1 N·m, **SF ≈ 2–4** [Likely]. Spinning up the rotor and screw adds about 0.006–0.009 N·m at 1.2 m/s² [Guessing: a 60–80 g·cm² rotor]; the CAD mass row below shows that the 0.75 kg covers it. The carry-height sensor catches a stall before the next sideways move but does not prevent it, so the 100-lift test (BUILD_GUIDE §7.8) must pass. A high-impedance 0.4–0.6 A "integrated" motor would stall: the BOM excludes it.
Z back-drive when unpowered	~0.005 N·m (0.75 kg at a back-drive efficiency of about 0.5; 0.4–0.7 for a friction coefficient of 0.2–0.1 gives 0.004–0.0065 N·m)	NEMA17 detent ≈ 0.015 N·m → holds [Likely]. Hold current is kept on anyway.
X drive (1.9 kg moving, 2 m/s² + 2 N friction)	5.8 N belt → 0.037 N·m on the 20T pulley (+ about 0.003 N·m for the rotor [Guessing])	The 17HS19-2004S1 (0.59 N·m holding at 2.0 A) runs at 1.0 A: about 0.3 N·m holding, less at 375 rpm, so ≈ 0.15–0.3 N·m → **SF ≈ 4–8** [Likely; no torque curve read]
X beam sag, 4040 resting on the uprights (1376 mm between their centres), 20 N at midspan	0.18 mm (simply supported; E 69 GPa; I = 8.89 cm⁴, Misumi's catalog value for HFS5-4040 [Likely, read from a catalog extract])	Vertical: every set-down is probed, so it uses none of the pocket play; BUILD_GUIDE §4.5 checks ≤ 0.3 mm. The beam's slope under the moving load tilts the head by under 0.02°, under 0.1 mm at the cup in X against ±1.85 mm of play. The beam's own weight (1.28 kg/m) adds a fixed 0.10 mm at midspan. [Certain, beam formula]
X beam twist from the head overhang (cup axis 127 mm in front of the beam centre)	≈ 0.5 mm in Y at the cup near the pad, worst at midspan; 0.3–0.8 mm for a torsion constant J of 2.5–1.0 cm⁴ [Guessing on J: no published value found for a slot-6 4040]	θ = T·L / (4·G·J) with both ends fixed in torsion: T = 1.93 kg × g × 77 mm = 1.46 N·m (the CAD puts the carriage's centre of mass 77 mm in front of the beam centre, not 127), L = 1.376 m, G = 26 GPa, times the 400 mm from the beam axis to the cup face. The earlier 0.8 mm put all 1.9 kg on the cup axis. The corner brackets and the rail block's play are not included. Within pocket play. Measure it during build (BUILD_GUIDE §6.7). Add a rear brace if > 1 mm.
Spring buffer force at trip (2 mm) / end of a probe window (14 mm) / bottomed (20 mm)	about 1.2 / 2.4 / 3.0 N [Likely: SMC lists 1.0 N at rest and 3.0 N at full stroke for ø10–32 buffers]	The software never probes more than 12 mm past contact; a trip at 6000 mm/min squeezes about 6 mm. The `ZPT25UUK20-B5-A10` has 20 mm of stroke: SMC sells no 25 mm stroke on a 25 mm pad. [Certain, SMC ZP catalog Table 2]
Stack height, 30 PSA slabs at 6.25–6.9 mm	188–207 mm	MAX line 210, overfill beam at 214, rod tops 218

Quantity	Value	Margin
Carry height (cup), loaded and empty	248 mm	A **double pick** of two 11.5 mm thick holders still clears the rods by 5 mm, counting the free cup's 2 mm hang below its calibrated height [Certain, CAD]. The carry sensor trips at 246.5: a head it still sees as "up" (≥ 245.5) clears them by 2.5 mm.
Moving mass, Z head (CAD: `build.assembly('carry')`, parts in group `head`)	0.40 kg from CAD volumes and catalog masses; ≈ 0.46 kg with the T8 nut and the head's share of cables and tube [Guessing]; ≈ 0.53 kg with a 65 g slab	Under the 0.75 kg used above. With 0.53 kg: 0.0185 N·m + 0.006–0.009 N·m of rotor and screw inertia = 0.025–0.027 N·m, so **SF ≈ 2–4** at 0.8 A still holds [Likely]. The 0.75 kg plus that inertia would need 0.033–0.035 N·m (SF ≈ 1.7–3). Back-drive with 0.53 kg: 0.003–0.005 N·m.
Moving mass, X carriage (groups `carriage` + `head`)	1.62 kg from CAD volumes and catalog masses; ≈ 1.87 kg with T-nuts, screw shanks, the belt and the drag chain's moving share [Guessing]; ≈ 1.94 kg with a slab	Matches the 1.9 kg used above (5.9 N, 0.037 N·m): the X margin is unchanged. Its centre of mass sits 77 mm in front of the beam centre (twist row).

Mass rollup inputs. Volumes are the solids of `build.assembly('carry')` (`cad/build.py`), summed by `Part.group`. Printed parts: PETG at 1.27 g/cm³ × 0.71, the solid fraction BOM.csv gives for the listed walls and infill (about 720 g of 1020 g solid) [Guessing for each part]; belt clamps at 100%. The aluminium flags, the steel buffer and lead screw (a solid Ø8 in the CAD) and the fastener heads (the only part of a screw that is modelled) use their CAD volume at 2.70 and 7.85 g/cm³. Extrusions, rails and carriage blocks are solid envelopes in the CAD, so catalog values replace them: the 2040 and 2020 at Misumi's 324 and 183 mm² sections × 2.70 g/cm³, the MGNR12 rail at 0.65 kg/m, each MGN12H block about 0.05 kg [Likely]. Z motor body 0.36 kg; head camera 20 g, LED ring 5 g, cup 9 g, each sensor board 4 g, M5 elbow 3 g [Guessing]. Heaviest items: Z motor 0.36 kg, Z spine 0.30, Z rail 0.22, drop arm 0.14, lead screw 0.13, printed parts 0.22 (0.15–0.31 kg for a solid fraction of 0.5–1.0).

5. Reading the label

- **Glare geometry.** The station camera sits on the slab's centre axis, the head camera 5 mm from it toward the label end, each with a small LED ring around the lens. **[Certain, mirror geometry]** The mirror reflection of a ring of radius r lands within r/2 of the lens axis, so within about 10 mm for the head's 40 mm ring (11 mm for a 42.5 mm one) and about 16 mm for the station's 63 mm ring. The label band sits about 40–65 mm off the lens axis, so it is glare-free without polarisers. That avoids the slab plastic's stress-birefringence "rainbow" risk that cross-polarisation has. [Likely] Room lights reflecting off the slab plastic are a minor effect. The LED ring is several times brighter on the label than a ceiling light's reflection, so contrast drops a little but nothing saturates. If Phase 0 shows glare anyway, add the optional black foam-board light roof.
- **Resolution.**
  - Head camera (8 MP IMX179, 3.6 mm lens, 140 mm away): 0.055 mm/px.
  - Station (Camera Module 3): 0.039 mm/px.
- **Decode sweep on synthetic labels in the real formats** (`software/tools/decode_sweep.py`):
  - **[Certain for the synthetic labels]** The back QR + barcode reads at ≤ 0.14 mm/px through blur and strong glare.
  - The front barcode needs ≤ 0.05–0.08 mm/px and low glare.
  - **No misreads occurred in any cell of the sweep.**
- **What real labels carry** [Certain, decoded from 64 real slabs' public studio scans; RESEARCH.md §4]. These formats are the rules in `slabscan/certs.py`:

Label	Back label	Front label
PSA current (9-digit certs, some 8)	QR `https://www.psacard.com/cert/<n>/` + Code128 `<n>`	Code128 `<n>`
PSA 2017+ "lighthouse" (8-digit)	QR `http://psacard.com/cert/<n>` (or the https form) + ITF	ITF
PSA 2016	ITF only	ITF
PSA before 2016	no code	ITF
CGC 2023+ (10-digit certs, a few 140 + 10)	QR `https://www.cgccards.com/certlookup/<n>/<grade>/`	Code128 `<n>`
CGC 2020–23 "OG"	QR `…cgccomics.com/certlookup/<n>/<grade>` + Code128 of 26–28 digits ending in `<n>`	no code
CGC Certified Authentic	no code	QR `…/certlookup/<n>/0_0/` + Code128 `<n>`
BGS, SGC	no code	no code (printed cert only)

- **Interpretation rules** (`slabscan/certs.py`, tested on the real payload strings):
  - **A QR names the cert and is `ok` on its own:** it carries its own error correction. PSA and CGC QRs are parsed with a right boundary, so a longer number is rejected, not cut. A CGC cert keeps all its digits: the old rule cut a real 13-digit cert to 12 and called it `ok` (a silent wrong cert, now a test).
  - **Bare digits are typed by symbology and length:** ITF of exactly 8 digits is a PSA cert (the cert zero-padded; any other ITF length is a misread, and a glare-cut `289412` was seen); Code128 of 26–28 digits is a CGC OG back (the cert is the last 10); Code128 of 10 digits not starting with 0, or 140 + 10, is a CGC front; Code128 of 8–9 digits is PSA (every real one). A Code128 in none of these formats (7 digits, or 10 starting with 0 like the BGS cert `0019865140`) is not typed: it is listed as unparsed and proves no label side, so a slab with nothing else ends in the exception pocket. [Certain, `certs.parse_payload`; 0 of the 119 distinct real payloads parse differently after this change] There is nothing left to learn about the ITF: the old padding-or-check-digit model could never learn on real labels and held every 2016 slab back. [Likely] An 8-digit Code128 with a leading zero still parses as a 7-digit PSA cert; the only real one seen is the phantom `09435764`, which the phantom guard needs to see.
  - **A barcode-only result (ITF or Code128, any grader) needs two printed symbols** (back + front label) **or OCR of the printed digits to agree.** One printed symbol is one source: the head camera's hover and held frames count once, and so do the station's. OCR that disagrees makes it `needs_review`.
  - **OCR also checks every barcode-only `ok`.** Both labels of a PSA 2016 slab print one ITF, and current labels one Code128 on both sides, so one fixed misread of a printed pattern can come back from both and agree on the wrong number. OCR that reads other digits makes it `needs_review`; OCR that reads nothing leaves it `ok`. The cost is one station visit and one OCR per slab without a QR read. [Certain, `runner.needs_ocr`; tested]
  - **An `ok` needs its grader confirmed** [Certain, `certs.consensus`, `certs.psa_itf_confirmed`]:

Evidence for the grader	Grader on an `ok`
A code that names it (a cert URL: PSA or CGC QR)	confirmed
Two different printed symbols in its format (back + front label barcode)	confirmed
An 8-digit ITF whose digits OCR of the printed number confirms	PSA (note "grader PSA from its 8-digit ITF")
One barcode in a grader's format, even with OCR agreeing (one Code128)	not confirmed: `needs_review`, grader kept ("grader … not confirmed")
An 8-digit ITF alone, or with OCR that read nothing	`needs_review` ("one printed barcode only")
An 8-digit ITF with OCR that read other digits	`needs_review` ("barcode N but OCR read M")
A code in no grader's format (bare digits in a QR, EAN-8, Codabar, Code39, a keyed "Cert #"), even beside an ITF	never confirms a grader

Evidence for the grader	Grader on an `ok`
OCR only	`needs_review`, grader `unknown`, exception pocket

The ITF rule makes a pre-2016 PSA slab (front ITF only) `ok`. It holds because in the real set ITF is printed only on PSA labels (CGC uses Code128; the 1 BGS and 1 SGC slab showed no code), and OCR reads the printed number on the slab in the cup, independently of the barcode's digits. [Certain, real set: 5 of 7 pre-2016 slabs `ok` native, 7 of 7 at machine scale; a stress run of 316 ITF-rule `ok` frames gave 0 wrong certs] Only a `needs_review` with grader PSA or CGC may stay in the output stack; `unknown`, `PSA/CGC` or no grader goes to the exception pocket. A grader guessed from the length still counts, so another grader's slab whose barcode has a PSA or CGC format can reach the output stack as `needs_review` [Certain, verifier probe: a 10-digit Code128 is typed CGC], and another grader's 8-digit ITF that OCR confirms would read `ok PSA` [Guessing that one exists]. Two settings switch the ITF rule off in effect: `custom_regex` (every ITF is then typed by it) and a `psa_cert_digits` without 7.

- **OCR finds a digit run, not a grader.** Tokens are 8–10 digits or 140 + 10. An OCR-only number is `needs_review` with grader `unknown` and always goes to the exception pocket (OCR also reads card numbers: `0079865140` for a BGS `0019865140`). OCR runs on a crop of the label band (3.2 barcode lengths along, 0.6 across) with `OMP_THREAD_LIMIT=1`: with Tesseract's default threads it timed out on every frame of a loaded 4-CPU computer [Certain, measured].
- **Phantom guard, twice.** `decode.py` drops a linear read lying inside another read of the same symbology and under a third of its thickness (the real `09435764` inside `27435764`). `consensus` then drops a phantom that got through, from every frame of that symbol, but only the thin read inside the other read's box (under half as thick, centre inside), and only when a different symbol (the QR, the other label side) or OCR confirms the other number ("dropped a phantom read …"). If the confirmed number is the thin read, nothing is dropped and the slab is a `conflict`; two reads whose boxes do not overlap are two labels, never a phantom. [Certain, `certs.drop_phantoms`; tests on the real payloads and boxes]
- **Decode passes:** on head-camera frames the CLAHE pass runs before the early exit when only a barcode was found (a QR the plain pass missed can decide the face-down check); station frames get an extra 0.75× / 0.6× downscale pass (CGC front Code128s with ink-spread bars read there).
- **Head frames are cropped to the slab under the cup** (`head_roi_half_mm` = 38.3 mm either side of its centre line), because the 3.6 mm lens also sees the neighbours' labels and, beside the slab, the lower neighbour stacks. The frame taken from above the stack alone never makes an `ok`, and a cert already recorded in the batch becomes `conflict` (exception pocket).
- **The back-label code that proved a slab face-down must be its own:** if the reads of the slab in the cup do not show that cert, or show another one, the slab is `conflict` ("… may have been FACE-UP …, check it by hand").
- **`psa_cert_digits`** (set from Phase 0): a PSA cert of a length Phase 0 never saw is `needs_review`.
- **Codes that disagree** produce `conflict` and the slab goes to the exception pocket. The machine never guesses.
- **CGC** is read and stacked like PSA. A **BGS/SGC/TAG** URL gives `other_grader` (exception pocket); real BGS and SGC labels carry no code, so they end as `no_read` or OCR-only.

Face-down: what a code seen from above proves

Only rules that held on every real label of their kind may prove a side [Certain, 64 real slabs; `certs.SIDE_PROOFS`]:

Code seen by the head camera	Proves	Evidence
PSA QR (`psacard.com/cert/`)	**face-down**	never on 35 PSA front images; on every 2017+ back
CGC QR with a cert and a grade other than `0_0` (cgccards or cgccomics)	**face-down**	0 of 26 graded CGC fronts have a QR; every 2023+ and OG back has one

Code seen by the head camera	Proves	Evidence
CGC 26–28-digit Code128	**face-down**	OG backs only; OG fronts carry no code
CGC QR with grade `0_0`	**face-UP** (Authentic front)	both Authentic fronts; their backs decode nothing. One official render of a 2023+ back also shows `0_0`: a wrong face-up call only stops the run
CGC 10- or 13-digit Code128	**face-UP** (CGC front)	0 of 26 2023+ backs carry a Code128
CGC QR without a cert	nothing (a hint in the message)	press images and mock-ups only [Likely]
PSA ITF or Code128	nothing	printed on both labels

- **Back and front codes together, two different back labels, or codes of two different certs in one frame** mean another slab's label is in view: the run stops.
- **Measured** [Certain, `tools/real_eval.py`]: 50 of 64 real backs proven face-down, 24 of 64 real fronts proven face-up, 0 false proofs either way (native scans); 48 and 23, 0 false, in the machine-equivalent frames. The 14 backs left unknown are exactly the slabs with no back-label code. These rules were written from the same 64 slabs, so this shows they fit them, not that they hold on new ones.
- **Every back-code proof teaches the learned check** (CGC included), once the slab is placed and its own read agrees (below).

The learned face-down check

Why. Slabs with no back-label code (PSA 2016 and older, CGC Authentic, BGS, SGC) stop the plain check every time: 43 of 330 in the simulator's default mix, 14 of 64 real slabs. Most of them show the same card back through the plastic, and a face-up slab shows a card front there instead.

The descriptor (`slabscan/facedown.py`) [Certain, code]:

- Only the card window, `face_down_window_mm` (x −27.5 … 27.5, y −18.5 … −13.0 mm from the head-camera axis = y −23.5 … −18.0 from the cup axis, −y = the label end; 55 × 5.5 mm = 302.5 mm², at least 300 required). It comes from the CAD (`layout.face_down_window_cam()`), placed from the card positions measured on 22 PSA and 27 CGC backs (`layout.HOLDERS`): 11.0 mm clear of the label band and clear of every head part at hover and held [Certain, `checks.py` frustum checks]. On the card plane (3.5 mm under the slab back, rounded corners) a PSA slab keeps it on the card anywhere in its play; a CGC slab pushed into a pocket corner has 1.52% of it on the holder (4.5% if the 3.5 mm were all air), checked against a 2% limit [Likely, `checks.py`]. The window shrank from 632.5 mm² because the round-3 one let 12 of 206 face-up CGC fronts through in an emulation of the real scans with every slab at the cup end of its play [Likely]: the strip of holder above a CGC card looks the same from both sides.
- Each rectangle is cropped with a 4 mm margin and box-resampled to 1 px/mm. That removes noise, focus and JPEG differences and about 1° of yaw, and costs about as much as a thumbnail.
- Glare (a 1 mm average at grey level 245 or more, grown by 2 mm) is masked everywhere.
- Pattern term: the window minus its own blur (σ 4 mm), so exposure and light fall-off drop out; normalised cross-correlation at every shift within ± 4 mm (the slab's play in its pocket).
- Colour term: a hue-saturation histogram (a grey-level histogram on grey frames).
- **Score = the lower of the two, and a slab needs two references** (from two different slabs of the matched back design) at the threshold, not only the best one. A false accept puts the cup on a front; a false reject only stops the run.
- Cost [Certain, measured on the development container]: about 2 ms to describe a full 3264 × 2448 frame and about 2 ms to compare it with 48 references. [Guessing] 10–20 ms on a Pi 5. It adds one head frame per run (the empty-pocket view), none per slab.

Safety layers [Certain, code; tested in the simulator]:

1. **References only from proven slabs,** and only once the slab is placed: a back-label code seen from above, if the slab's read is `ok` with the same cert and the read station saw no back-label code from below; your `--face-down-ok`, with the same placement checks and never with a front-label code in view; the Phase 0 seed. A learned acceptance never becomes one. A back design known only from your confirmations is used once 3 agree.
2. **Two references per match:** a new back design starts with one slab and matches nothing until a second slab joins it, so one odd reference (or one front taught by mistake) is never enough.
3. **`--face-down-ok` confirms one slab:** only after a face-down stop, only the top slab of that pocket. The web tick appears only then and is cleared after each resume.
4. **A texture floor**, and a match needs at least half the matched back design's own texture (a plain front with a glare spot has some texture, far less than a back).
5. **Glare:** a window with more than 2% glare never becomes a reference; more than 30% is not matched.
6. **The empty-pocket guard:** one head frame over an empty pocket per run. A window whose pattern looks like it (similarity ≥ 0.5) shows the head, not the slab: it neither matches nor teaches.
7. **The occlusion check:** on the newest 5 or more slabs proven on this machine, a pixel that tracks the empty-pocket view by the frame's exposure ratio moves with the camera. More than 10% of the window like that: the mode is not active.
8. **A measured threshold:** Phase 0 compares at least 10 usable backs with each other and with at least 15 face-up slabs, each face-up frame also with 4 reflections added in software, all by the same 2nd-best rule; a back that matches none of the others is left out. GO needs a gap of 0.10; the threshold is the midpoint, never below 0.55. It is stored with a tag of the window and descriptor it was measured for (`face_down_threshold_sig`) and refused without it.
9. **At least 5 references** (`face_down_min_refs`).
10. **The camera-aim check** (`facedown.aim_problem`), on the same view over an empty pocket: in the middle ±12 mm of the window along the row, over its full height and 1 mm beyond its cup-side edge, only the pad far below may show. More than 2 mm² of high-passed edges (4 grey levels or more) there means a head part is in the window, or about to be: the mode accepts and learns nothing that run, and the start line, every stop, the summary (`CAMERA AIM:`), `summary.json` (`camera_aim`), `selftest` and `focus head` say to re-aim. [Certain, sim: a square camera passes; with today's window it fires from 1.30°, the part entering at 1.33°] A verdict counts only for the run that took it (`CardBackRefs.stale`). A resume looks over an empty pocket at its survey (as before); if none is empty, it now accepts and learns nothing until it has emptied its first source pocket and looked over that, so a camera knocked while the run was stopped is seen [Certain, sim: a 1° knock caught with and without an empty pocket; the no-empty-pocket tests fail on the round-4 code]. The cost: with no empty pocket, every slab without a back-label code stops until then, so the operator should empty pocket 12 first. Its blind spots:
    - a head part as dark as the pad shows no edge, and layer 7 does not catch it either while it covers a strip of the window (a pad-dark foot over 9–26% of the window read 0.3–0.9% camera-locked) [Certain, sim];
    - only a tilt toward the cup is checked: a tilt away from it or a roll along the row moves the window toward the card top or off its side, and nothing sees that [Certain, code];
    - the outer window beyond ±12 mm also sees this pocket's rods and the neighbouring pockets, so it cannot be checked there [Certain, CAD];
    - the thresholds were set on the simulator's flat, noiseless pad: pad mottling, dust specks of about 1 mm, a tag or a sharp shadow made it fire on a square camera in a synthetic probe [Certain, synthetic]. That is safe but strict; a real empty-pocket frame from Phase 0 is needed to tune it.

Limits.

- [Certain, CAD aim check] **Camera aim:** a head-camera tilt of more than 1.34° toward the cup puts the cup pad into the window (1.31° with the software's own 147 mm mapping, by an independent ray cast). 1.5° is out of

reach: the cup pad sets the limit, and a window that clears it at 1.5° leaves under the software's 300 mm². Layer 10 catches a tilt toward the cup from about 1.3°; its blind spots are above. A head edge about 1 mm inside the window raised face-up scores in the simulator from 0.37 to 0.52 (hence the 0.55 floor). BUILD_GUIDE §12.3 checks the frame; [Likely] an M2 washer shim corrects about 0.5°.

- [Likely, CAD card-plane check] **CGC holders:** a CGC card top lies 93.8–96.2 mm from the holder's far end (27 real backs; PSA 97.2–100.0 on 22), so a CGC slab at the cup end of its play brings its card top about 3.4 mm nearer the cup than a PSA one. The window keeps about zero margin there: up to about 1.5% of it on the holder (4.5% at most), and a tilt of about 0.01° toward the label, or 0.10° of roll, takes it past 2%. The holder lengths and widths come from cropped scans and probably read short (CGC publishes about 80.0 × 134.6 mm) [Likely]; the card top to far end distance is the design input and is not affected.
- [Likely] **Ring glare:** the head ring's mirror image in the slab back covers 0.0% of the window for a slab square to the lens, 0.97 mm clear of the 2 mm glare halo. A slab tilted about 0.45° toward the label brings the halo in; from about 0.77° its frames stop becoming references (over 2% glare).
- [Certain, real images; Guessing for the machine] **Real separation:** on studio scans of 63 real slabs placed as the machine would see them, with two references 0 of 332 face-up frames and 223 of 252 face-down frames (glare included) were accepted at 0.55. With one reference, a Japanese blue Water-type front reached 0.62 under glare against one odd back; glare raised such scores. The real camera through slab plastic, sleeves, stickers and other games are untested.
- [Certain, sim] The simulator's face-up scores keep rising as the window shrinks: 0.24, then 0.43–0.46 with the round-3 window, now 0.57 in the Phase 0 simulation (threshold 0.69, lowest back 0.81; the simulator now places each card where real holders of its grader put it). The margin is synthetic. [Likely, CAD emulation of the real scans] With today's window the highest face-up score is 0.49–0.53 against the 0.55 floor, and about 101–103 of 126 face-down backs are accepted (the round-3 window: 113–116).

6. Verification done here (no real hardware)

Check	Tool	Result
Geometry: exact min-distance, all-pairs interference (including modelled fastener heads), held-slab sweep, hard-stop, rail-margin and line-of-sight checks at every pick, place, travel, station, hard-stop and Phase 0 pose	`cad/checks.py` (OpenCascade)	**177/177 pass** (re-run on 23 Oct 2026, about 12–15 min; the numbers are in `cad/out/checks.txt`) [Certain]. Picking the last slab next to full neighbour stacks clears them by 14.55 mm (the probe board's M3×16 tips); loaded travel clears the rods by 21.4 mm, a double pick of two thick holders by 5.33 mm. Both LED rings are modelled (40 × 30 × 5 and 63 × 47 × 5 mm, with their solder pads and leads): seated with zero overlap, the barb elbow 3.75 mm from the head ring at the full 20 mm stroke, nothing in either camera's view, a 60 mm ring fits the station seat too. The head ring checks run at 39.5, 40 and 42.5 mm (the ribbed 42.8 mm seat; a 42.5 mm ring has 0.15 mm of radial clearance). **Home switches** (Omron SS-5GL geometry): X clicks at cup X 19.00 and Z at c = 258.00; each striker holds the lever 0.6 mm past its click point (at the lever tip) to the hard stop, under the 1.2 mm overtravel; the levers are free in normal travel; the mounts, screws, finger and leads clear everything that moves (closest 1.49 mm, the Z carriage to `z_switch_mount`); the longest lever each mount takes is 27.24 mm (X) and 24.19 mm (Z), against a 23 mm limit. **Not checked:** belt slack, cables and the vacuum tube, the real shapes of bought parts (the buffer, elbow, rings and switches are modelled from catalog or listing dimensions), loads and deflection, the side load the strikers put on the switch levers, and the corner brackets and joining plates.

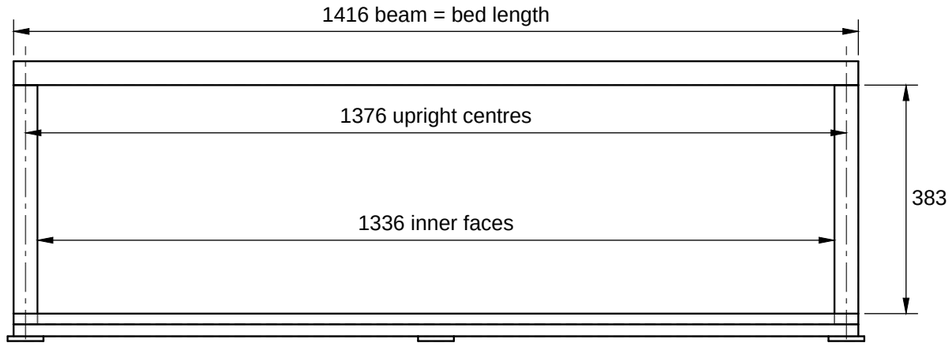
Check	Tool	Result
Face-down card window	`checks.py check_face_down_window`, `check_camera_aim`	[Certain] exact frustums from the lens (5 mm toward the label) at hover over stacks of 1, 15 and 30 and with a slab held: 0.0 mm³ of any part in view, nearest part 1.74 mm (1.94 held); 302.5 mm²; 11.0 mm clear of the label band. [Likely] On the card plane, a slab in a pocket corner: PSA 0.0% on the holder (top +3.39, side +0.22 mm); CGC 1.52% (top −0.03, side −0.23 mm; limit 2%). Aim tolerance before a head part enters: 1.34° toward the cup, over 4° every other way. Before more than 2% of the window is on the holder: 0.49° (PSA corner slab) and 0.01° (CGC corner slab); before it leaves a median card on a centred slab: 1.32°.
Carry-height sensor	`checks.py check_carry_sensor`	[Certain] trips at c = 246.50; beam clear from −14 to 242, fully blocked from 248 to the 262 top stop; flag 1.05 mm from the sensor, flag arm 4.6 mm from every part fixed to the X carriage, its lead 11.0 mm from anything moving; a head stalled just above where it reads clear (245.5) still carries a double thick pick 2.5 mm over the rods.
Probe flag, guide and sensor over the buffer's 20 mm stroke	`cad/checks.py check_probe`	[Certain] trip at 2.00 mm; beam clear for 1.5 mm, fully blocked from 3 to 20 mm; flag 1.05 mm from the sensor and 0.45 mm from its guide; 8 mm guide engagement at full stroke; a rotating (J) buffer stopped at 1.32°, 0.6 mm from the sensor; buffer ± 3 mm on its thread still covered by the ± 4 mm board slots; pad 1.5 mm clear of the buffer body at full stroke; nothing in the label view while reading; a ball-end key reaches the front board screw (1.63 mm clear).
Phase 0 jig vs machine	`cad/checks.py check_phase0`	[Certain] the occluder (foot, buffer, flag, elbow, board) sits 0.0 mm from the machine's relative to the lens and blocks the same share of the slab view; the face-up pass (mount 3 mm up) leaves the free cup 1.0 mm above the slab's front; every jig wall an M5×10 passes through is 4.0 mm.
Unit tests: cert parsing on the real payload strings, grader and side rules (including the 8-digit ITF rule, tried over every combination of reads), phantoms, decode at the design point, OCR fallback, the learned face-down descriptor and the camera-aim check (including a resume with and without an empty pocket), the carry-height sensor, config checks	`software/tests`	**`299 passed, 1 skipped in 1135.84s (0:18:55)`** (full suite on the final code, 23 Oct 2026) and the real-image test on its own: **`1 passed, 61 deselected in 137.29s (0:02:17)`** [Certain, the software fixer's runs]. On the code just before the last fix, the full suite with the real-image test gave `300 passed in 1202.90s (0:20:02)`. The skipped test runs `tools/real_eval.py` on the real image set when `SLABRAIL_REAL_IMAGES` points to it.
End-to-end batches in a simulator that **raises Crash on any unsafe move**	`software/tests/test_runner_sim.py`, `test_safety_fixes.py`, `test_bench_tests.py`, `test_face_down_learned.py`, `test_carry_sensor.py`	pass [Certain]. Scenarios: every slab moved and read; pre-2016 labels via the station; double pick detected; seal failure stops; overfill refused; CGC read, BGS to the exception pocket; face-down check, including FACE-UP and neighbour stops; an unexpected error with a slab in the cup; failed vent; vacuum sensor fault; Ctrl-C hard stop; a Z stall (40 mm and 10 mm losses) caught before the sideways move, a stuck or cut carry sensor, its selftest. Learned check: an old label passes once references exist; a face-up slab still stops; a plain face-up front with glare is refused; the round-2 head and a partly hidden window are refused; one wrong confirmation, or a neighbour's back QR in the frame, does not teach a front; a back code seen from below vetoes a reference; a Japanese back needs 3 confirmations; `--face-down-ok` is refused after another stop; a lower neighbour's back QR beside the slab is cropped out.

Check	Tool	Result
330-slab simulated batch (seed 7, default label mix, `--face-down-check`)	`sim --slabs 330 --face-down-check`	[Certain, sim, 23 Oct 2026, the round-5 code; the last fix touches only reads the simulator never makes, Likely] `needs_review 6, ok 324`: 5 BGS slabs OCR-only (pocket 12) and 1 pre-2016 PSA slab whose OCR read 9 digits (output stack). The other 20 pre-2016 PSA slabs are `ok` from their front ITF and OCR. No `ok` has a wrong cert or a wrong grader (checked against the simulated slabs). Face-down: 287 by back QR, 41 learned (lowest score 0.76), 2 confirmed (the first slabs, before 5 references existed); with back-label codes only, 43 would have stopped. `camera_aim: ok`. 14.2 s per slab, 79.4 min for the load (homing + survey 75 s), 991 head frames. With the head camera tilted 0.8° (`--sim-head-tilt 0.8`, round-3 window, 110 slabs) the aim check fired and every slab without a back code stopped (15 of 110). The mix (PSA current 170, 2017+ 56, 2016 12, older 21; CGC 2023+ 49, OG 12, Authentic 5; BGS 5) is [Guessing], and every simulated slab is face-down: the learned acceptances say nothing about face-up safety (the unit tests and Phase 0 cover that). Phase 0 on the simulator: GO, lowest back-to-back 0.81, highest face-up 0.57, threshold 0.69 (round-3 window: 0.82, 0.43, 0.62). The simulator places each card where real holders of its grader put it (`sim.HOLDERS`, the CAD's `layout.HOLDERS`).
Real labels: the machine's rules on public scans of 64 real slabs	`tools/real_eval.py`	[Certain for these images, 23 Oct 2026] Native (round-5 code): `ok` 60, `needs_review` 2 (pre-2016 PSA: OCR read other digits on `19802594`, nothing on `20908923`), `no_read` 2 (BGS, SGC: no code), **0 wrong certs in `ok`, 0 false face-down proofs**. Machine-equivalent (rescaled to the cameras, degraded, cropped as the runner does; 61 slabs; final code): 60 `ok`, 0 `needs_review`, 1 `no_read` (BGS), 0 wrong, 0 false; all 7 pre-2016 PSA slabs `ok` with "grader PSA from its 8-digit ITF". Before the ITF rule the 7 pre-2016 slabs were `needs_review` on both. With glare on the label (a run before the last fixes): 58 `ok`, 3 `needs_review`, 0 wrong. Before these rules the same images gave 1 silent wrong cert (the 13-digit CGC). **The rules were written from these 64 slabs**, and flat studio scans have no slab-plastic reflections: this shows the rules fit them, not that the machine will read yours.
Real slabs, real cameras, real cup	—	**Not done.** Phase 0 + acceptance tests in BUILD_GUIDE are mandatory before trusting it with your inventory.

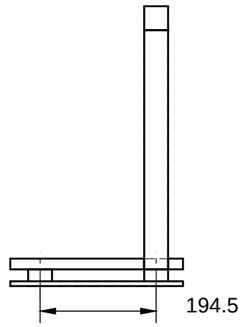
7. Known risks and what retires them

Risk	Retired by
Your labels carry payloads unlike the 64 real slabs seen	Phase 0: scan 30–50 of your slabs with the real cameras, then `tools.real_eval --phase0` (0 wrong certs, 0 false face-down proofs). The report lists ITF lengths and payloads no rule reads. `custom_regex` in config handles a prefix or suffix.
Cup leaves visible marks	Phase 0B 24 h marking test with a hand pump, then the 300-cycle + 24 h test on the machine with the vacuum re-held all the time (BUILD_GUIDE §13.2). Swap cup material if needed (BOM lists two).
Slabs without a back-label code stop the run	The learned check, once Phase 0 says GO and 5 references exist: 2 stops in 330 in the simulator [Certain, sim; Guessing for real backs through plastic]. Until then use `run --all-face-down` only for stacks you checked yourself.
A real face-up front looks like a card back (blue Water-type fronts came closest)	Two references per match; a blue Water-type card in Phase 0's face-up pass; the back photo of every learned row audited in the first batches.
The head camera is aimed off, so a head part or the holder sits in the card window	The camera-aim check (layer 10) refuses a tilt toward the cup from about 1.3°. Not covered: a pad-dark head part, a tilt away from the cup, a roll (a CGC corner slab has about 0.01° toward the label): BUILD_GUIDE §12.3 look at `face_down_window.jpg`, a shim under the board. Software owner: measure aim in both axes from positive evidence (the back-QR box position, the card-top edge).

Risk	Retired by
The camera-aim check fires on a square camera (lint, dust, glue, a shadow on the pad)	Safe: the learned check stays off and slabs without a back code stop. Its thresholds need a real empty-pocket frame (Phase 0, `focus head`'s `focus_head_empty.jpg`). Software owner.
A CGC slab pushed into a pocket corner has up to about 1.5–4.5% of the window on the holder	[Likely] The 2% limit is a judgment, not a margin. Audit learned CGC rows. A +0.5 mm edge margin was tried and is out of reach: one rectangle with that margin leaves 266 mm² (the software needs 300), and the best 3-rectangle window (+0.53 mm at 300 mm²) moves y1 to −17.05, which cuts the aim margin to 0.97° against the 1.25° floor. The window stays as it is [Certain, CAD trade-off search; Likely that no shape does better: the search was a grid].
A neighbour's label in the head camera's crop	The crop to the slab's footprint; the two-cert rule; measuring `cam_head.hfov_deg`; a lens of F2.8 or faster (BOM).
A Z stall drags a slab sideways below the rod tops	The carry-height sensor: a loss over about 2.5 mm is caught before any sideways move (CAD and simulator). The §7.8 100-lift test still has to pass; jog, calibrate and mark do not read the sensor.
A phantom barcode confirms itself or drops the real read	[Certain, tested] Frames of one symbol count once, only the thin read inside the other read's box can be dropped, and a confirmed thin read makes a `conflict`. Two barcodes with the same misread are checked by OCR; if OCR reads nothing the slab stays `ok`.
Another grader's bare barcode is typed by its length (10 digits = CGC)	[Guessing that such labels exist] It is never `ok` (grader not confirmed), but as `needs_review` with a guessed PSA or CGC grader it stays in the output stack. The narrower length rules are applied now: a PSA Code128 is 8–9 digits and a 10-digit CGC one never starts with 0, so a 7-digit Code128 or a BGS-style `0019865140` is not typed at all (exception pocket) [Certain, `certs.py`; tested].
Another grader prints an 8-digit ITF, and OCR reads its printed number	[Guessing that one exists] The ITF rule would give `ok PSA`: the right digits, the wrong grader, in the output stack. In the real set ITF is only on PSA labels, but that set has 1 BGS and 1 SGC slab; BGS and SGC are reported to print a barcode or QR on some holders, type unknown [Likely, second-hand]. A sweep of 216 BGS, SGC and CGC frames found no ITF of any length [Certain, that set]. Put any non-PSA slabs you own through Phase 0 or `tools.real_eval` (OPERATING.md §5.2).
OCR confirms an ITF from a longer zero-padded number	[Guessing that it happens] The ITF rule compares certs with their leading zeros stripped, so OCR reading `0019865140` (a BGS-style number) would confirm an ITF `19865140` and give `ok PSA`. On every real ITF-rule `ok` frame the OCR digits equalled the ITF's exactly (316 of 316) [Certain, stress run]. Software owner: require the raw digits to match.
OCR picks one of two equally long digit runs at random	[Certain, verifier probe] `decode.ocr_digits` keeps the first longest token, and the order of a tie depends on the process's hash seed. Since OCR now decides `ok` versus `needs_review` for pre-2016 PSA slabs, the same slab can come out either way in two runs. Safe both ways (a non-matching token gives `needs_review`). Software owner: break ties by value, or return no read on a tie.
A homing switch lever is longer than the strikers allow, or wears under the sideways push	Measure each lever and cut it back past 23 mm (BUILD_GUIDE §6.8) [Certain, CAD: X takes 27.24 mm, Z 24.19 mm]. The 45° lead-in pushes the lever sideways as hard as down; the datasheet rates only a straight push [Guessing that it holds]: a 100-pass bench test before the build, and FIRMWARE.md step 6's homing repeat test after.
The 42.8 mm head ring bore prints small	[Likely] A 42.5 mm ring has 0.15 mm of radial clearance in the CAD, and printed holes come out 0.1–0.3 mm small: sand the bore for a ring over about 42.2 mm (BUILD_GUIDE §3.3).
The buffer and elbow differ from their catalog dimensions	Measure them on arrival (BUILD_GUIDE §3.3) before cutting the flag; the board slots give ± 4 mm of trip adjustment.
Beam twist larger than estimated	Dial-indicator check during the build. A rear diagonal brace is in the guide.
Double picks (slabs clinging)	Peel dwell. Detected at placement (step ≥ 11.6 mm) and flagged with the exact position.
Mixed weird holders (SGC 86 mm wide doesn't fit; BGS/TAG widths unmeasured; sleeves)	Loading rules in OPERATING.md. SGC is too wide for the pockets. Measure one BGS, TAG and CGC holder before loading them: anything wider than 82.6 mm is "do not load".



FRAME, FRONT (1:10)



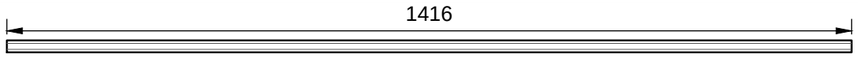
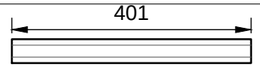
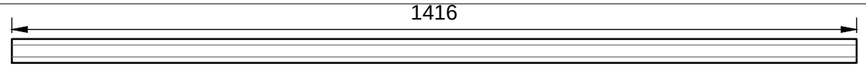
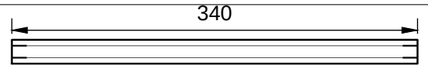
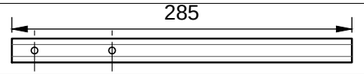
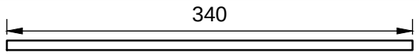
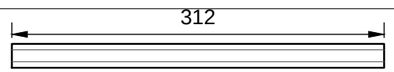
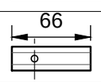
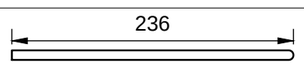
RIGHT (1:10)

Frame set-up (BUILD_GUIDE 4.5; tolerances quoted)

Base rails: centre distance (RIGHT view) +/-0.5 at both ends; diagonals equal within 1. Upright inner faces (FRONT) +/-0.5, equal at the bottom and the top [Certain, CAD]. Bed top to beam bottom +/-0.5, equal at both ends [Certain, CAD]. Uprights stand on the back base rail through notches in the bed (D-03); the beam sits on the uprights, flush with their outer faces. Joints (L brackets, back joining plates) are NOT in the CAD: BUILD_GUIDE 4.3 [Guessing on the exact layout].

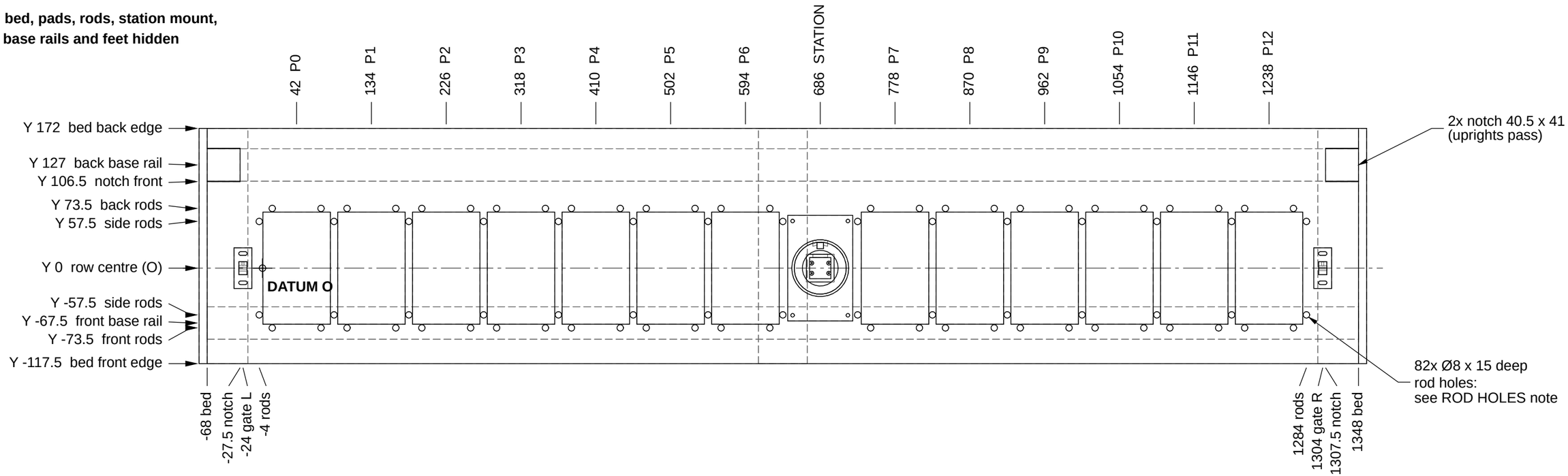
CUT LIST (cad/out/cut_list.csv; extrusion cuts +/-0.5 with square ends, BUILD_GUIDE 4.1)

Extrusions: 20-series, slot 6 (Misumi HFS5, BOM.csv)

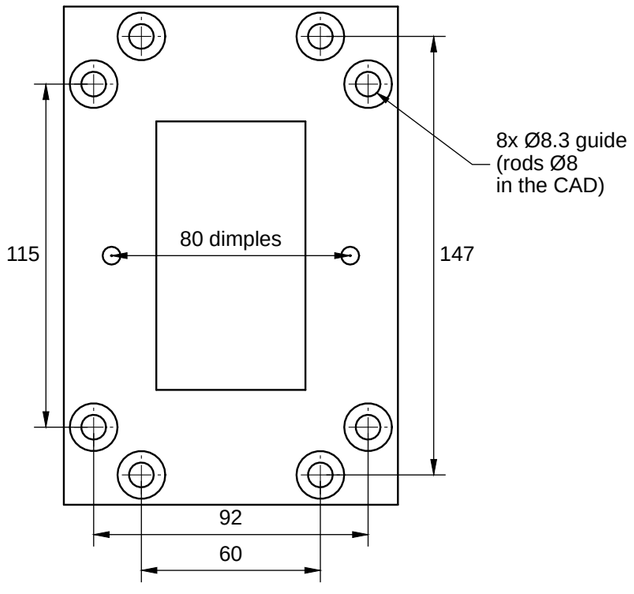
Item	Bar (length from the CAD; scale per row)	Profile / stock	Use, ends, holes	Qty	Tolerance
C-01		2040	base rail, lying flat under the bed; square ends	2	+/-0.5
C-02		4040	upright; square ends	2	+/-0.5
C-03		4040	X beam, on top of the uprights; square ends	1	+/-0.5
C-04		2040	Z spine; BOTH ends: 2x M5 tapped, the two centre bores (20 apart), for 2x M5x12 button heads (bottom) and 2x M5x16 (top), BUILD_GUIDE 7.1, 7.3	1	+/-0.5
C-05		2020	head drop arm; 2x Ø5.5 THRU across the arm (machine X), 19 and 84 from the end in head_foot's socket: match-drill (BUILD_GUIDE 7.5)	1	+/-0.5
C-06		T8x8	T8x8 Z screw, length below the motor face: cut the motor's 400 mm screw; 311-356 works (cut_list.csv)	1	311-356
C-07		1 mm Al sheet	probe flag blank: D-06	1	D-06
C-08		2020	Phase 0 jig post; cut square	2	+/-0.5
C-09		2020	Phase 0 jig arm stub; cross hole 19 from one end, match-drilled through head_foot	1	+/-0.5
C-10		1 mm Al sheet	carry-sensor flag: D-06	1	D-06
C-11		5/16 in dowel	guide rod; dome the top R2-3 (ROD_DOME_R; BUILD_GUIDE 5.4), chamfer the bottom lightly	82	PROPOSED +/-0.5

SlabRail - graded-slab batch scanner	**SHEET D-02 of D-17**
Frame and extrusion cut list	SCALE 1:5, 1:10 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

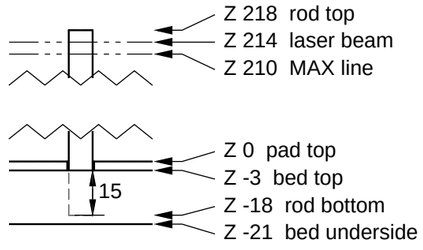
PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails and feet hidden



DETAIL C (1:2): one cell's rod holes = drill_template
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)



DETAIL D (1:2): rod in the bed,
front, broken



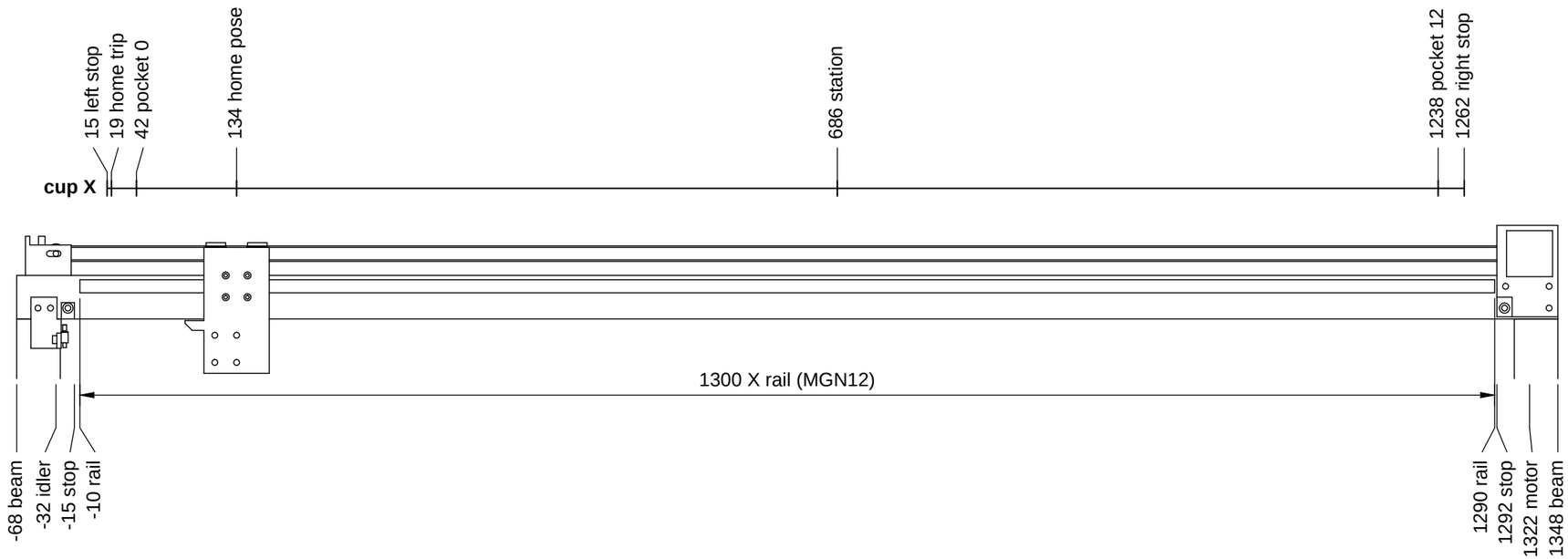
Bed and pads

Item	Size (from the CAD)	Note
Bed, plywood (W)	1416 x 289.5 x 18	cut on site; test-drill for voids first (BUILD_GUIDE 5.1)
Pads (EVA + microfiber), 13	83 x 138 x 3	spray adhesive, after the rods (5.9)
Rods (dowels), 82	D-02 item C-11	glued; plumb within 1 over 200 [Guessing target] (5.7)

Datum O = machine origin, X 0 and Y 0 (the pocket-row centre line), as the software uses them; X ordinates from O. The rod holes are NOT laid out from the bed edges: the machine drills them through drill_template, so beam skew and belt stretch are built in (BUILD_GUIDE 5.6). Station mount: cell 7, four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

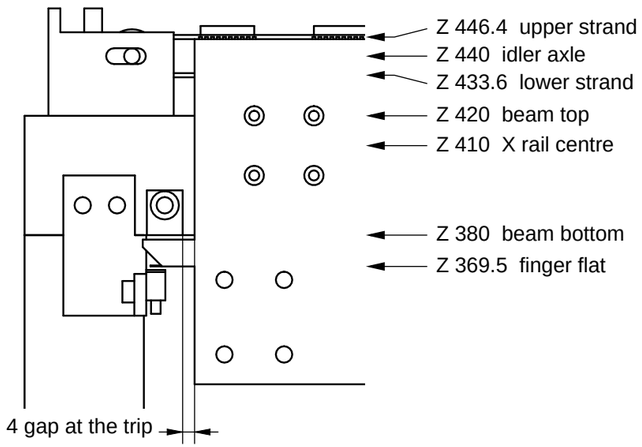
ROD HOLES: 82x Ø8 (CAD ROD_D; drill with the brad-point bit that matches the dowels: 5/16 in for the BOM's 7.94 dowels), 15 deep (ROD_EMBED), square to the bed (guide block, BUILD_GUIDE 5.6). Side rods on the cell wall lines, first X -4, then every 92, at Y ±57.5; front/back rods in pairs 60 apart about each pocket centre, at Y ±73.5; none front/back in the station cell 7.

SlabRail - graded-slab batch scanner	**SHEET D-03 of D-17**
Bed, pockets and guide rods	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**



FRONT (1:5)

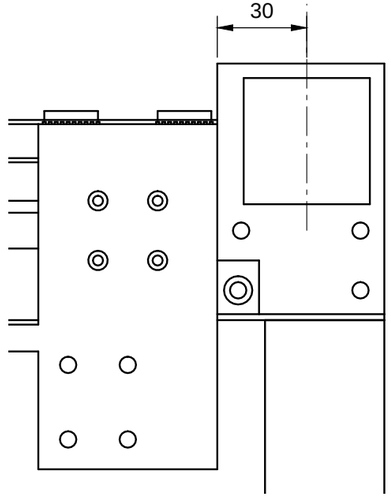
DETAIL A (1:2): left end, carriage at the home-switch trip
(cup X = X_HOME_SWITCH; front view)



Click at cup X 19 (+/-2, BUILD_GUIDE 6.8)

FRONT: X axis with the carriage at home (cup over pocket 1); Z axis and head not shown. cup X = machine X of the suction-cup axis. Belt: GT2 loop over the beam top, both ends clamped on the x_bracket tab (BUILD_GUIDE 6.5).

DETAIL B (1:2): right end, carriage on the right stop
(cup X = X_HARD_STOP[1]; front view)



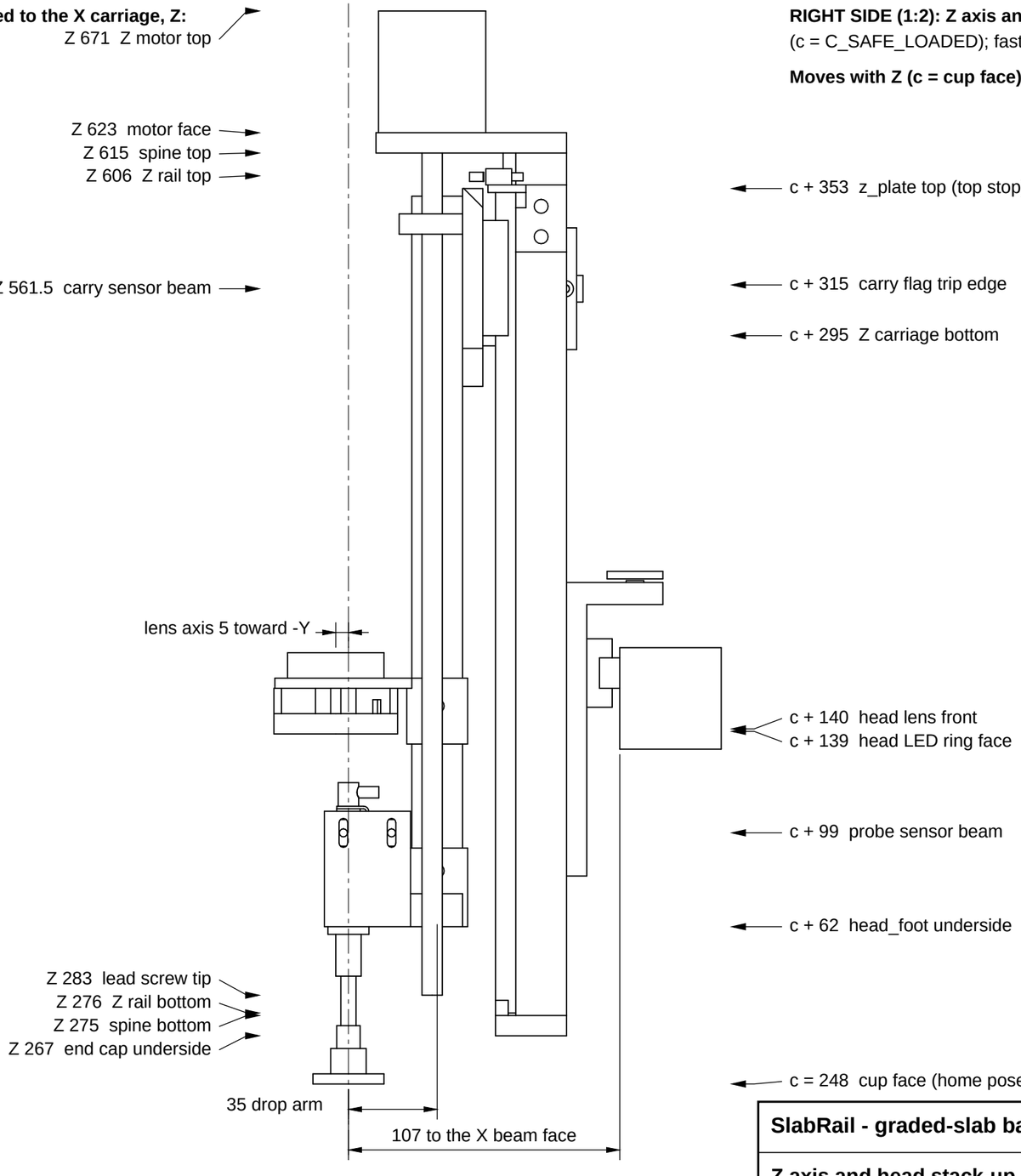
x_bracket lands on the mount's left end

Y stack-up, cup axis to the X beam's front face

Constant (layout.py)	mm	Part
ARM_TO_CUP	35	drop-arm centre to cup axis
EXT20 / 2	10 (of 20)	drop arm (2020): half = 10
Z_PLATE_T	8	z_plate
MGN12_H	13	Z carriage (MGN12H)
EXT20	20	Z spine (2040, 20 deep)
X_BRACKET_T	8	x_bracket
MGN12_H	13	X carriage (MGN12H)
= Y_CUP_TO_BEAM	107	sum: cup axis to beam front face

CAD check: beam front face at Y 107

SlabRail - graded-slab batch scanner	**SHEET D-04 of D-17**
X axis	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

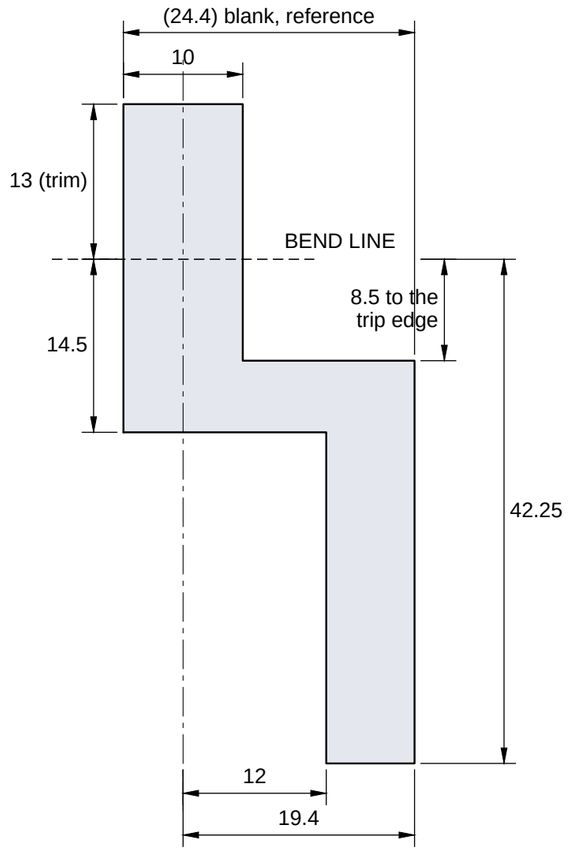


Z-height table (values read from cad/layout.py)

Constant	Value	Meaning
SLAB_T	6.6	c at pick contact = top of the stack: one slab (pose pick_bottom) = SLAB_T; n slabs = n x SLAB_T, nominal
CUP_HANG	2	the free cup hangs this far below c; the probe trips after this much buffer travel
PROBE_TRIP_TOL	0.5	trip tolerance on CUP_HANG (BUILD_GUIDE 7.6)
PROBE_WINDOW	12	max probing travel past the predicted contact
BUFFER_STROKE	20	spring buffer stroke (SMC ZPT25UUK20-B5-A10)
STATION_LENS_Z	23	station lens front above the pad top
STATION_READ_C	171.6	c at the station read = STATION_LENS_Z + CAM_TO_SLAB + SLAB_T + CUP_HANG
CAM_TO_SLAB	140	both cameras: lens front to the slab face
HEAD_CAM_DY	-5	head lens axis, Y from the cup axis (negative = toward the label end)
POST_H	218	guide-rod tops above the pad top
C_SAFE_LOADED	248	carry height, loaded and empty (C_SAFE_EMPTY is the same)
CARRY_SENSOR_C0	246.5	carry sensor: beam blocked from this c up to the top stop
CARRY_BAND	1	carry sensor band: within it either reading is allowed
C_MAX	250	top of the software's Z travel
Z_SWITCH_C	258	Z top (home) switch trip
C_HARD_TOP	262	top hard stop: z_plate meets z_motor_mount
C_MIN	-12	lowest c the design allows
C_HARD_BOTTOM	-14	bottom hard stop: Z carriage skirts on the z_end_cap lips
Z_CARRIAGE_ABOVE_CUP	295	Z carriage bottom edge above the cup face
Z_SPINE_BOTTOM	275	spine bottom end above the pad top (set +/-1, BUILD_GUIDE 7.1)
Z_SCREW_DX	24	lead screw axis, X from the cup axis
Z_SPINE_OFFSET_X	-10	Z spine centre, X from the cup axis

SlabRail - graded-slab batch scanner	**SHEET D-05 of D-17**
Z axis and head stack-up	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

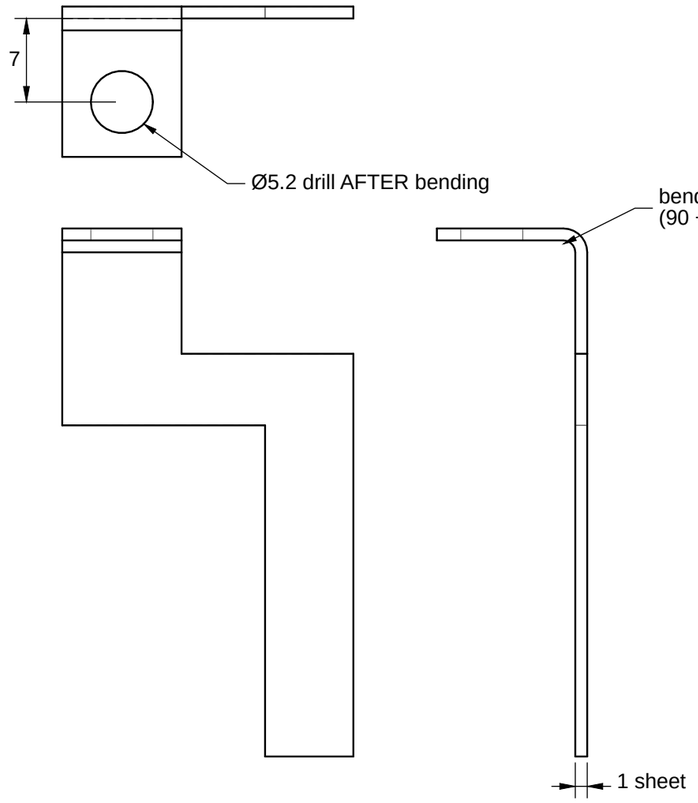
PROBE FLAG BLANK (2:1), flat, before bending
u from the neck centre line (= hole line), v from the bend line



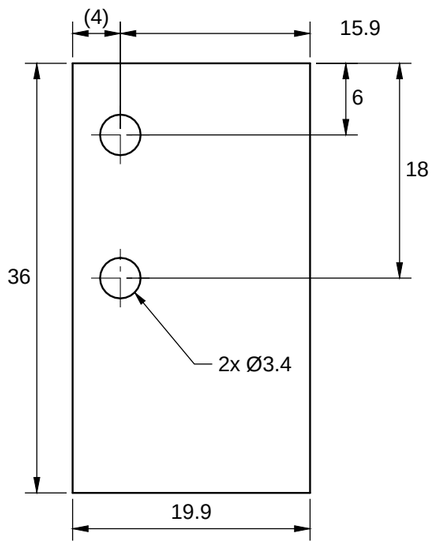
Probe flag (BUILD_GUIDE 7.6; build.py probe_flag_blank.svg notes)

Aluminium sheet, 1 thick (cut_list C-07). 1. Cut and file to the outline; the blade's top edge is the trip edge: straight, square. 2. This view shows the face toward the buffer. Neck in the vise, jaws at the BEND line; bend the tab 90° TOWARD YOU. 3. THEN drill on the centre line, 7 from the plate's inner face. 4. Trim the tab to 2 past the hole. 5. Deburr: it rides in head_foot's guide slot and the sensor slot. Outline: cad/out/probe_flag_blank.dxf (1:1 print: .svg).

PROBE FLAG AS BENT (2:1), for reference
(head frame: X along the row, Z above the cup face)



CARRY FLAG (2:1), flat
(seen from the front; top edge = trip edge)



Carry flag (BUILD_GUIDE 7.4a)

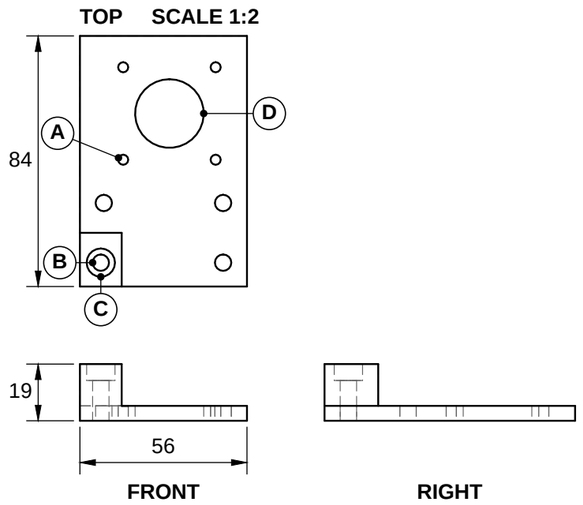
Aluminium sheet, 1 thick, flat (cut_list C-10). File the top edge straight. Fix to carry_flag_arm's leg end with 2 x M3x8, inner edge toward the spine, top edge flush with the arm's top.

Tolerance: outlines PROPOSED +/-0.2 (no doc states one); bend 90 +/-0.5° (BUILD_GUIDE 7.6). The trip points are set by adjustment on assembly (probe 7.6, carry 7.4a), not by these outlines.

SlabRail - graded-slab batch scanner	**SHEET D-06 of D-17**
Flat parts (probe flag, carry flag)	SCALE 2:1
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_motor_mount

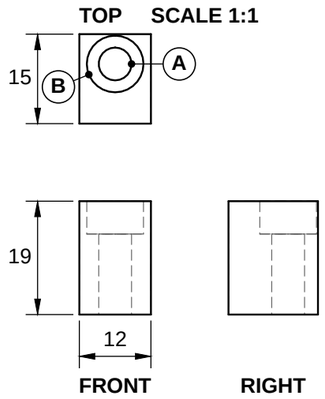
Printed part (P), qty 1, cad/out/printed/x_motor_mount.step. Print: PETG, beam face down (as exported), 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, centres spread over 41 x 20 (X x Y), not a grid: M5 clearance
C	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk
D	Ø23 THRU, axis Z: NEMA 17 pilot clearance
E	Left end = right X hard stop face, 30 from the motor axis; motor axis 20 above the beam top

x_bumper_L

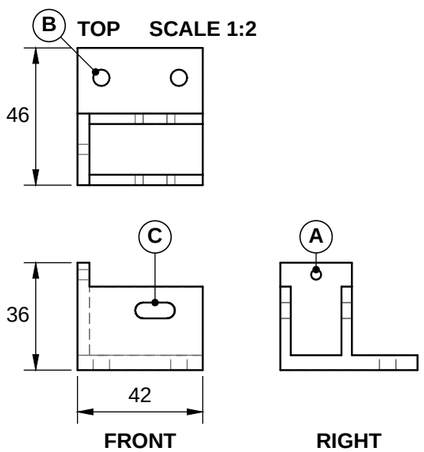
Printed part (P), qty 1, cad/out/printed/x_bumper_L.step. Print: PETG, beam face down (as exported), 4 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Z: M5 clearance
B	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk

x_idler_mount

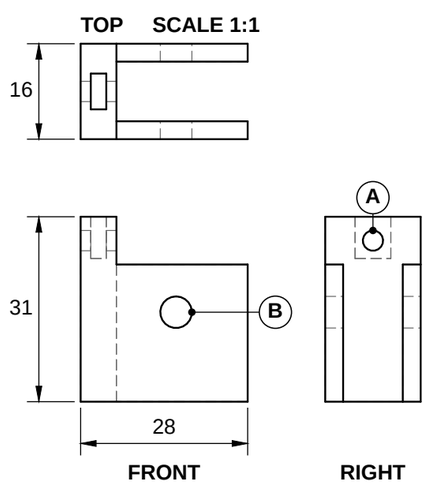
Printed part (P), qty 1, cad/out/printed/x_idler_mount.step. Print: PETG, foot down, 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU, axis X: M3 clearance
B	2x Ø5.5 THRU, axis Z, 26 apart in X: M5 clearance
C	2x slot 5.3 wide x 13.3 long, axis Y: M5 idler axle (tension travel)

x_idler_slider

Printed part (P), qty 1, cad/out/printed/x_idler_slider.step. Print: PETG, as exported, 5 walls, 50% infill



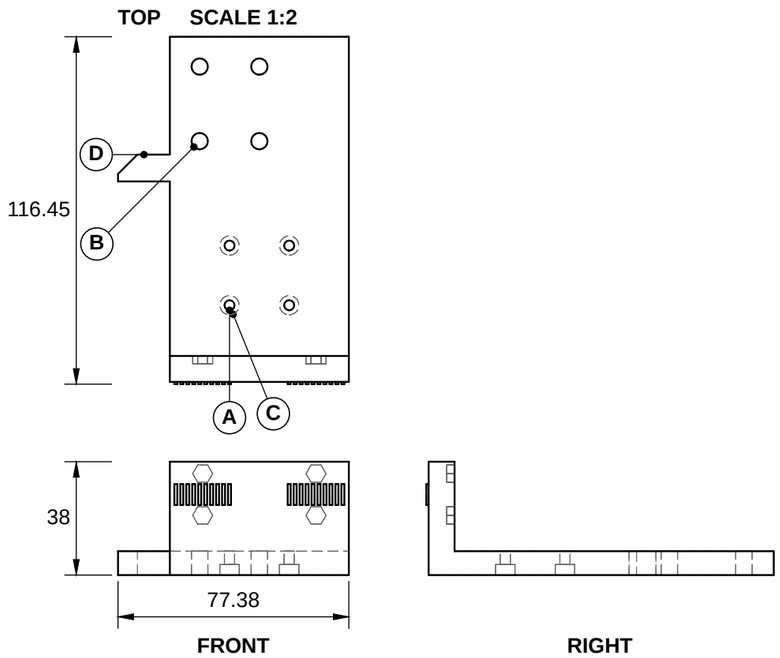
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU (2 walls on one axis), axis X: M3 clearance
B	Ø5.3 THRU (2 walls on one axis), axis Y: M5 axle
C	M3 nut slot 2.6 wide, from the top, in the bridge

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-08 of D-17**
Printed parts: X drive	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

x_bracket

Printed part (P), qty 1, cad/out/printed/x_bracket.step. Print: PETG, print flat on its spine face (as exported), 6 walls, 50% infill; the finger off its left edge works the X home switch



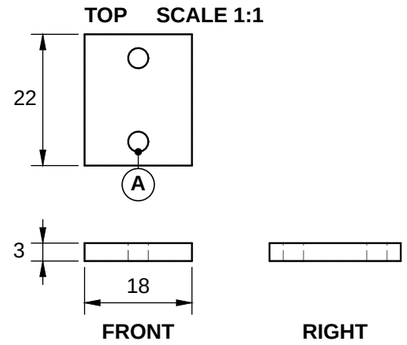
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, on 20 x 25 (X x Y): M5 clearance
C	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
D	X-switch finger 17.38 past the left edge; flat underside at machine Z 369.5; 45° lead-in 6.5 deep
E	GT2 rack on the tab floor, both sides: teeth at 2 pitch (belt teeth down, belt_clamp cap on top)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

belt_clamp

Printed part (P), qty 2, cad/out/printed/belt_clamp.step. Print: PETG, flat, 100% infill

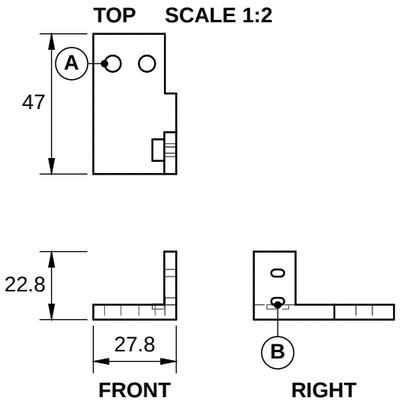
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø3.4 THRU, axis Z, 14 apart in Y: M3 clearance



x_switch_mount

Printed part (P), qty 1, cad/out/printed/x_switch_mount.step. Print: PETG, front plate down (as exported), 4 walls, 40% infill; X home switch on 2 x M2x12, 2 x M5x10 button heads

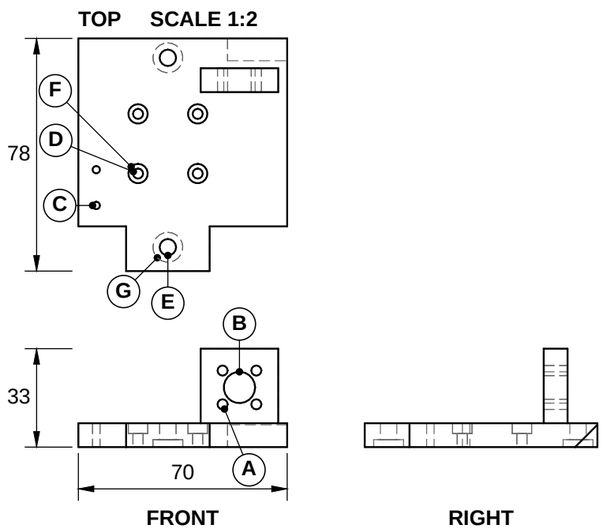
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 11.5 apart in X: M5 clearance
B	2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2



SlabRail - graded-slab batch scanner	**SHEET D-09 of D-17**
Printed parts: X carriage	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

z_plate

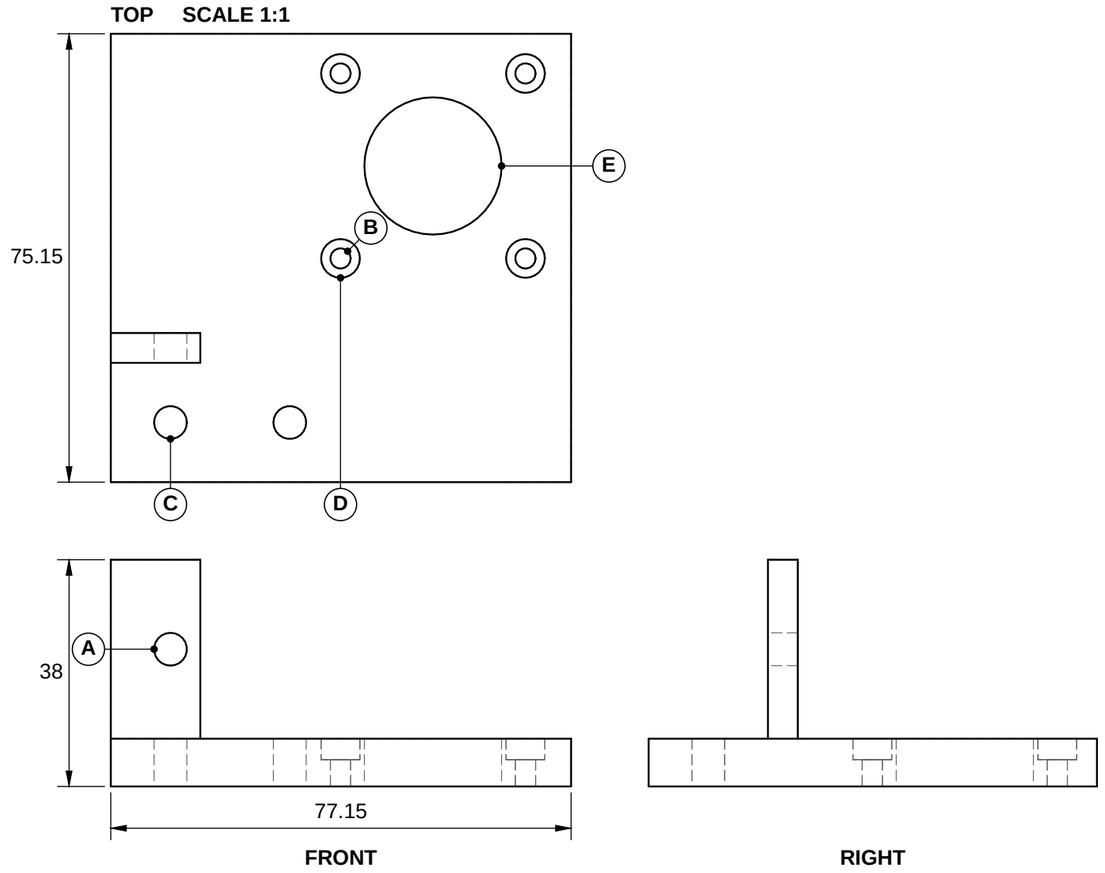
Printed part (P), qty 1, cad/out/printed/z_plate.step. Print: PETG, carriage face down, nut shelf up, 6 walls, 50% infill; the switch chamfer at the top right starts at the bed (45 deg)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Y, on 11.31 x 11.31 (X x Z): M3 clearance
B	Ø10.5 THRU, axis Y: T8 nut body clearance
C	2x Ø2.5 THRU, axis Z, 12 apart in Y: carry_flag_arm, M3x10 self-tapping
D	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
E	2x Ø5.5 THRU, axis Z, 63.5 apart in Y: M5 clearance
F	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
G	2x Ø10 x 2.5 deep, axis Z, 63.5 apart in Y: M5 button-head recess
H	Z-switch chamfer 45°, 7.5 x 7.5, 20 long at the right end of the top back edge
I	Top edge (Z top hard stop) 58 and T8 flange shelf top 48 above the Z carriage's bottom edge

z_motor_mount

Printed part (P), qty 1, cad/out/printed/z_motor_mount.step. Print: PETG, motor face down, tab up, 5 walls, 40% infill



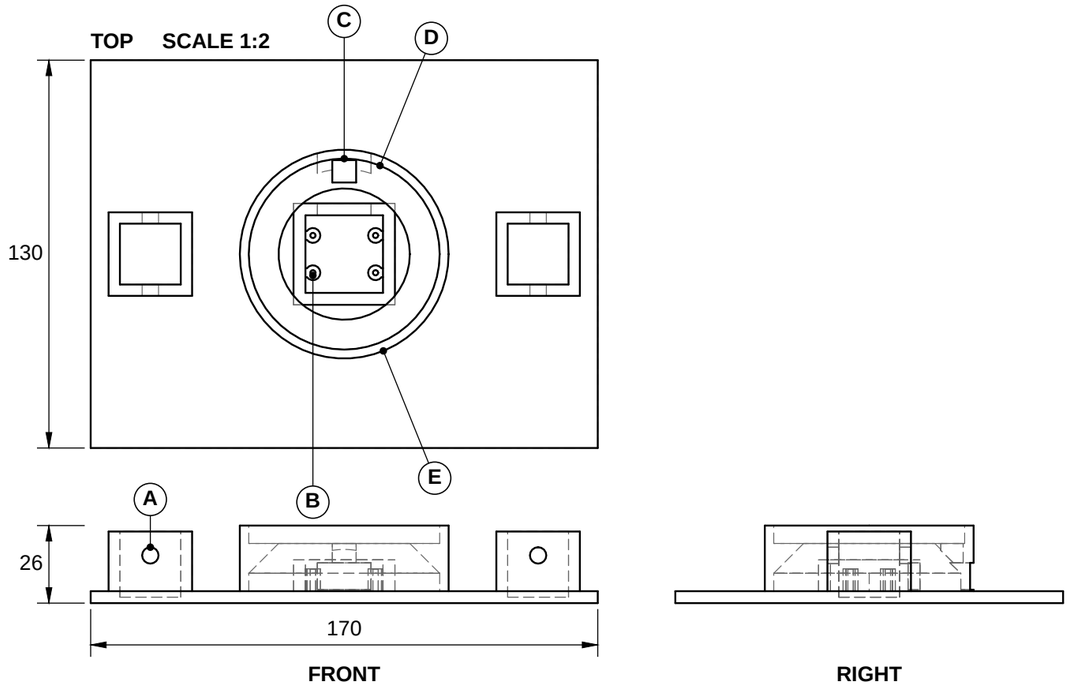
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Y: M5 clearance
B	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
C	2x Ø5.5 THRU, axis Z, 20 apart in X: M5 clearance
D	4x Ø6.5 x 3.5 deep, axis Z, on 31 x 31 (X x Y): M3 head counterbore
E	Ø23 THRU, axis Z: NEMA 17 pilot clearance
F	Plate underside sits on the spine top and is the Z top hard stop; tab hole 15 below it (left slot)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-10 of D-17**
Printed parts: Z carriage plate, Z motor mount	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

phase0_base

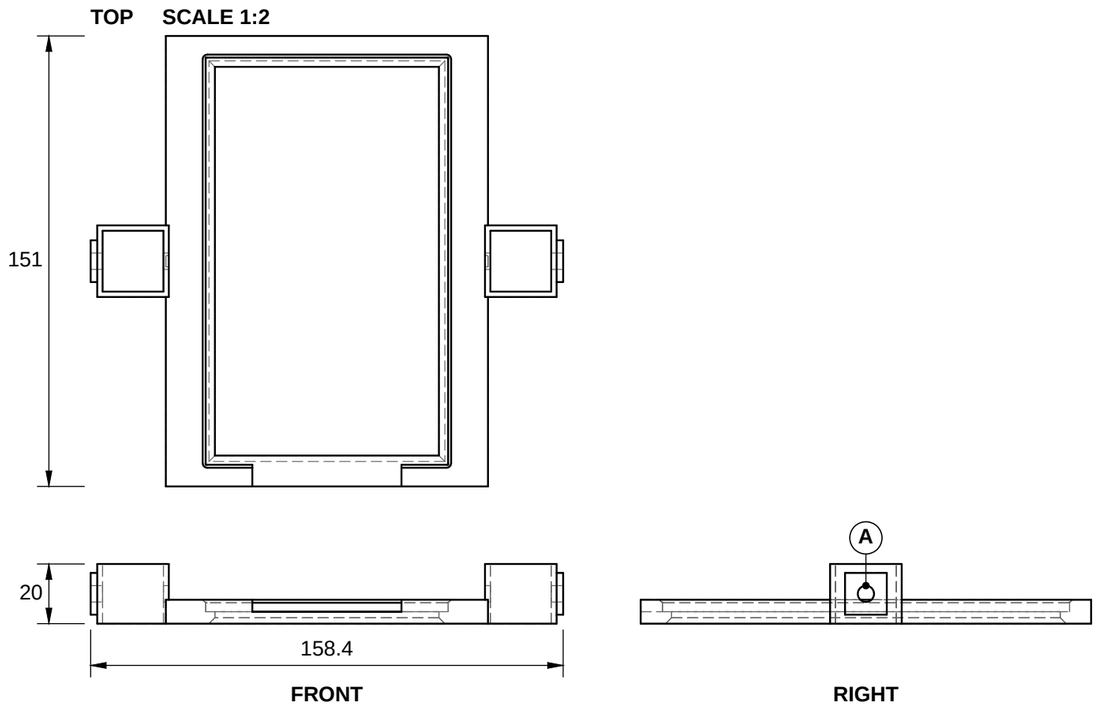
Printed part (P), qty 1, cad/out/printed/phase0_base.step. Print: PLA/PETG black (Phase 0 test jig)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU (2 walls on one axis), axis Y, 130 apart in X: M5 clearance
B	4x Ø1.8 x 7.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping
C	Ø64 x 6 deep, axis Z: hollow under the ring shelf (45° cone roof)
D	Ø64 x 6 deep, axis Z: station LED ring seat bore
E	Ø70 x 22 long (outside), axis Z: ring seat tube
F	Post sockets 20.4 square, floor 2 above the underside, centres 130 apart
G	Ring shelf 20 and socket cross holes 16 above the underside

phase0_tray

Printed part (P), qty 1, cad/out/printed/phase0_tray.step. Print: PLA/PETG (Phase 0 test jig), ledges down



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (4 walls on one axis), axis X: M5 clearance
B	Slab recess 81.3 x 136.5, 4 deep; opening 75.1 x 130.3; the ledges carry the slab's front face
C	Post clamps 20.4 square, centres 130 apart

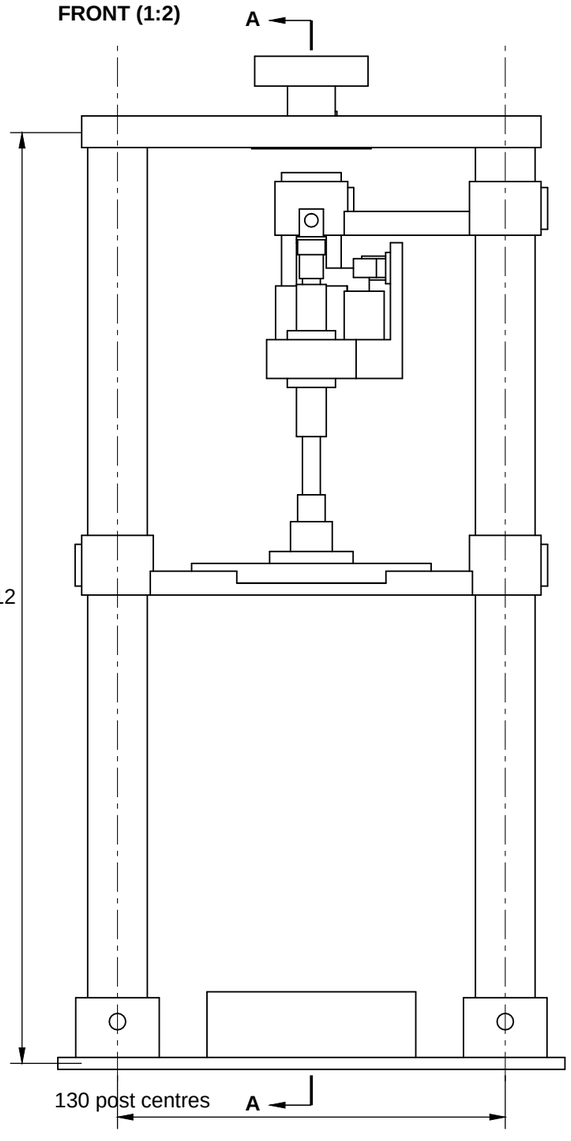
Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-15 of D-17**
Printed parts: Phase 0 jig (base, tray)	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

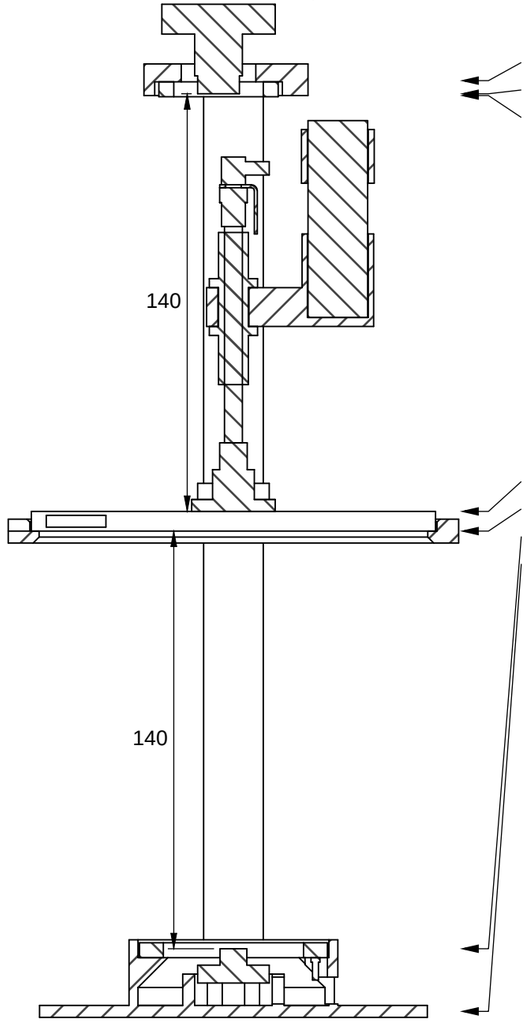
PHASE 0A TEST JIG: one slab between the two cameras, as on the machine (BUILD_GUIDE 1.5); fasteners not shown

Heights above the base underside

FRONT (1:2)



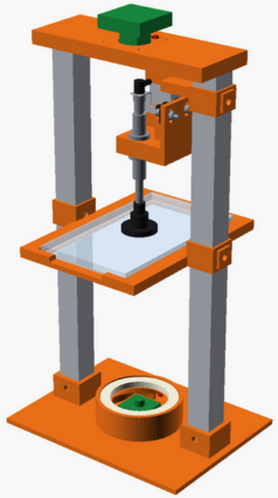
SECTION A-A (1:2)
on both lens axes, seen from +X; cut faces hatched



Z 314 post tops
Z 309.6 head lens front
Z 309 bridge underside

Z 169.6 slab back
Z 163 slab front face (on the ledges)
Z 23 station lens front
Z 2 post bottoms (socket floor)

ISO (render; not to scale)



Part	Type	Qty	Sheet
phase0_base (camera mast, ring seat, post sockets)	P	1	D-15
phase0_tray (the slab on edge ledges)	P	1	D-15
phase0_top (bridge: head camera, head ring)	P	1	D-16
phase0_occluder_mount	P	1	D-16
head_foot, the machine's own (occluder)	P	1	D-12
2020 post, 312, cut square +/-0.5 (C-08)	C	2	D-02
2020 arm stub, 66 (C-09)	C	1	D-02
probe flag (C-07)	S	1	D-06
cameras, LED rings, SMC buffer + cup + elbow, endstop board	B	-	BOM.csv

The posts set the bridge height: cut them square. Both lens-to-slab distances (RIGHT view) equal CAM_TO_SLAB = 140 on the machine (set +/-2, BUILD_GUIDE 1.5). Phase 0 is the go / no-go test before any motion parts are bought.

SlabRail - graded-slab batch scanner	**SHEET D-17 of D-17**
Phase 0 test jig	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ d3c1181 (drawn by cad/drawings.py)	DATE 2026-10-27 **DO NOT SCALE**

Wiring

Every wire in SlabRail, plus how power is laid out and how to check it before power-up. The block diagram is img/wiring.svg. Firmware set-up is in FIRMWARE.md, and pin numbers come from CONTRACT.md.

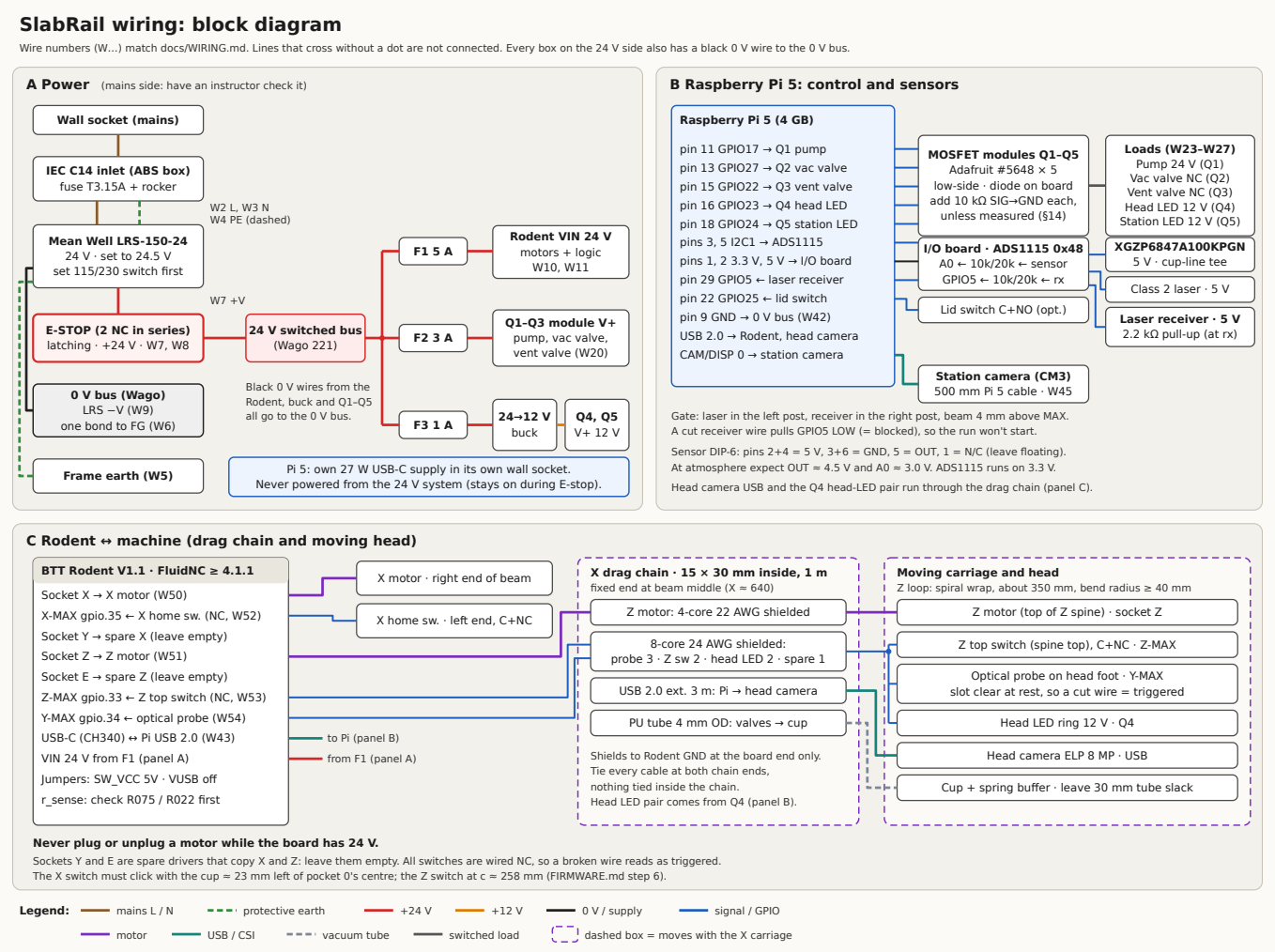


Figure: docs/img/wiring.svg

Confidence tags: **[Certain]** = primary source (datasheet, schematic, FluidNC source), **[Likely]** = strong inference or several secondary sources, **[Guessing]** = a gap I filled. Vendor product pages could not be opened from this session. The module facts come from search-engine extracts of the listings, and every "measure" step below exists because of that.

Safety rules.

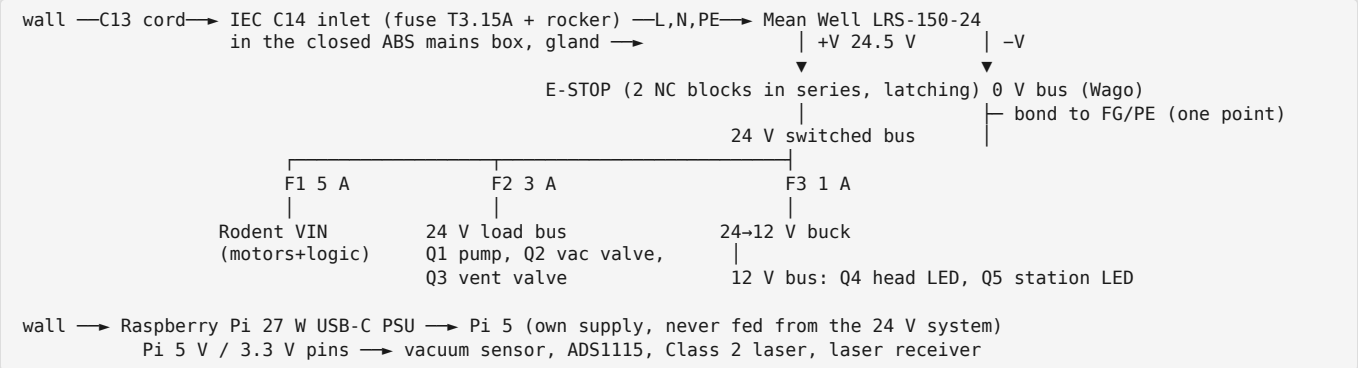
- Have a mechatronics instructor check the mains side (§1.1) before the first power-up.
- Never plug or unplug a stepper motor while the Rodent has 24 V.
- Everything after the 24 V supply is SELV (safe extra-low voltage). Most of the work is there.

Contents

1. Power architecture
2. Board jumpers
3. Wire tables

4. MOSFET modules
5. Vacuum sensor
6. Contact probe
    - 6a. Carry-height sensor
7. Limit switches
8. Overfill laser gate
9. LED rings
10. Cameras
11. Drag chain and Z loop
12. Grounding and shielding
13. Strain relief and connectors
14. Pre-power checklist
15. First power-up
16. Open risks

1. Power architecture



1.1 Mains side (instructor check)

- **Inlet: an IEC C14 inlet with a built-in fuse holder and rocker switch** (BOM "Mains inlet"). One part gives you the inlet, fuse and power switch.
- **It lives in the mains box, not on the plywood panel** (BOM "Mains box": a small ABS box, a cable gland and fully insulated 6.3 mm receptacles or tab boots). [Likely: snap-in IEC modules are made for 1–3 mm panel walls, so on 6 mm plywood they end up glued or hanging, with live tabs open.]
    - Cut the inlet's rectangular hole in the box wall and snap the inlet in.
    - Put fully insulated receptacles (or boots) on every inlet tab, including the fuse-holder side.
    - W2–W4 leave through the cable gland, tightened on the cable jacket, to the LRS terminal strip.
    - Close the lid with its screws. **With the cord plugged in, no metal at mains voltage may be exposed anywhere** (inlet back, fuse holder, LRS terminals). The instructor checks this (§14).
    - An ABS box needs no earth bond. A metal box (Hammond 1590 type) must be bonded to PE.
    - Fuse **T3.15A** slow-blow (5 × 20 mm), as in the BOM. [Likely: suits a 156 W supply on 115 V. Check the LRS-150 datasheet's input current and inrush before final choice]
- **Mean Well LRS-150-24: 24 V, 6.5 A, 156 W, output adjustable 21.6–28.8 V.** [Certain from reseller copies of the datasheet]
    - **It has a 115/230 VAC selector switch.** [Certain, reseller listings] Set it to your mains (115 in the US) **before** the first power-up. On 115 V mains, the 230 position won't run properly. On 230 V mains, the 115 position destroys the supply.

- Wire L, N and PE with 18 AWG (0.75 mm²) mains-rated wire, using the colours in the wire table.
  - **Colour mapping** (the BOM line is the EU set; US 18 AWG MTW is easier to buy, and the instructor decides which): line = brown (EU) or black (US); neutral = blue (EU) or white (US); earth = green/yellow (EU) or green (US). W2–W6 below accept either set. [Certain, IEC 60445 and US NEC conventions]
  - Keep the mains wire inside the mains box and the LRS terminals. On the low-voltage side black means 0 V and white +5 V (the colour table in §3), so never reuse leftover mains wire there.
  - The terminal labels are printed on the supply's terminal strip. Follow them.
  - Fit the supply's terminal cover.
- The Pi's 27 W USB-C supply plugs into its own wall socket, or into the same power strip as the machine cord. **It is never fed from the 24 V system.** That is what keeps the Pi alive when the machine power is cut.

1.2 Why one 24 V supply, and why this one

- **Every motor-side load is a 24 V part.** The Rodent's driver input, the BOM's pump and both Clippard EV-2-24 valves all run on 24 V. Only the LED rings need 12 V, from one small buck.
- **The Rodent's input range starts at 24 V.** The manual says "DC24V to DC56V". [Certain] So 24 V is the bottom of its range. **Set the LRS to 24.5 V** with its V.ADJ trimmer (no load). That leaves margin for wire drop while staying inside the pump and valve ratings (24 V ± 10 % is normal for solenoids [Likely]).
- **The load budget is under 4 A peak against a 6.5 A rating.** [Guessing, before parts are measured]

Load	Typical	Peak	Note
X motor (1.0 A phase, through the TMC2160)	0.3 A	1.0 A	supply current is lower than phase current
Z motor (0.8 A phase)	0.2 A	0.8 A	
Rodent logic	0.1 A	0.1 A	
Vacuum pump (24 V diaphragm)	0.4 A	1.2 A at start	measure yours
2 × valve	0.05 A	0.1 A	about 1 W coils [Guessing]
12 V buck for 2 LED rings	0.1 A	0.2 A	at 24 V
Total	**≈1.2 A**	**≈3.4 A**	LRS-150-24 = 6.5 A

The smaller LRS-100-24 (4.5 A) would also carry it. The 150 W model costs a few dollars more and copes better with the capacitor inrush every time the E-stop is released. [Likely]

1.3 E-stop

What it is: a 22 mm latching mushroom E-stop with NC contacts, in its own box (BOM item). It sits **in the +24 V line straight after the supply**. Pressing it cuts:

- the Rodent (both motors and the controller's logic);
- the pump and both valves;
- the LED buck.

What it doesn't cut: the Pi, which stays alive on its own supply, and the sensors it powers.

What happens on E-stop

- **Motors go limp.** DESIGN.md estimates the Z head holds on detent torque. The test is in FIRMWARE.md's open issues. [Likely]
- **Both valves are normally closed, so both close.** The cup line stays sealed, and a held slab stays on the cup for a while. Acceptance test 13.4 in the build guide checks this.
- **The Rodent switches off**, and its USB-serial chip with it. The Pi sees `/dev/ttyUSB0` vanish.
  - The software must stop and tell you "controller lost power".
  - After you twist the E-stop to release it, run `home` before anything else.

Contact rating and the two contact blocks (required)

- **Use two NC contact blocks wired in series** (BOM: a box with two NC blocks, or add a clip-on 22 mm NC block, about $3). Why: the contact switches 24 V DC straight into the Rodent's and the buck's input capacitors. Every release charges them through nothing but wire and contact resistance, and that inrush can weld a small contact closed. [Likely] A welded NC E-stop is invisible: the button is down and the machine is still live. With two blocks in series, one welded block still leaves the other to open.
- The target is blocks rated for **at least 5 A at 24 V DC**. The BOM's ECX1030 blocks are rated **DC-13 1.5 A at 24 V** [Certain, AutomationDirect], below that target and below the bus's ≈3.4 A peak (§1.2) [Guessing on the peak]; no 22 mm block rated ≥ 5 A at 24 V DC was found. DC-13 is the inductive-load category, so the resistive rating is probably higher [Guessing], but it is not listed. Two blocks in series and the monthly E-stop check are the mitigation for now. **Open issue for review:** the mains-side E-stop below, or a force-guided safety relay, removes it.
- **Do not "fix" this with an ordinary 24 V relay** switched by the E-stop: its contacts see the same inrush and can weld the same way. Only a force-guided safety relay would help.
- The alternative is to put the E-stop in the supply's mains feed (AC-rated contacts, and the LRS's own soft start removes the DC inrush). That means mains wiring in the E-stop box: instructor only.
- **Check it monthly** (OPERATING.md §7): with the machine idle, press the E-stop. The software must report the controller lost, and the 24 V switched bus must read 0 V on a meter.
- Only if F3 actually blows when you release the E-stop, change it to 2 A (the ring wiring takes it).

Mounting: at the front of the machine, where you can hit it without reaching over the bed.

1.4 Branch fuses (ATC blade, inline holders)

Fuse	Feeds	Rating	Why
F1	Rodent VIN	5 A	Above 3.4 A peak, below the 18 AWG wire limit
F2	Pump + both valves (24 V load bus)	3 A	Pump start current. Protects the 22 AWG load wiring.
F3	24→12 V buck (LED rings)	1 A	Protects the 22/24 AWG ring wiring

The BOM's "Branch fuses" line has three inline holders with 1 A, 3 A and 5 A fuses. [Certain, BOM.csv]

1.5 12 V buck for the LED rings

- Use the BOM's 24→12 V 3 A buck module. The Pololu D36V28F12 (12 V, 2.4 A, 12.9–50 V input) is a documented alternative. [Certain, Pololu specs]
- Set or check 12.0 V before connecting the rings.
- Its ground goes to the 0 V bus.

2. Rodent jumpers

Jumper	Setting	Why
SW_VCC (limit-input voltage)	**5V**	The optical probe is powered from V-Lim, and its output stage only works if V-Lim matches its 5 V supply (§6).
V-Probe	**5V**	The Probe connector is unused. 5 V is the safe choice.
NPN EN	**off**	Not needed
VUSB	**off**	BTT: "Do not insert the jumper when the device is in use." [Certain]
DIAG X/Y/Z/E	**none**	They connect driver stall outputs to the limit pins.

3. Wire tables

Wire gauge

- 18 AWG for mains and the 24 V trunk.
- 22 AWG for motors and 24 V/12 V loads.
- 24–26 AWG for signals.

Colours

Colour	Use
red	+24 V
orange or yellow	+12 V
white	+5 V (Pi side)
black	0 V
blue	switched minus (load return through a MOSFET)
green/yellow	protective earth only
other	signals

Labelling: write the W-number on both ends of every cable.

Connector types on the Rodent are 2.54 mm pitch 3-pin (limits, probe) and 4-pin (motors) headers. They look like JST-XH in BTT's renders. [Likely] **Check against your board before you crimp.** Power is on screw terminals. [Certain]

3.1 Mains and 24 V / 12 V power

W	From	To	Wire	Ends	Notes
W1	Wall socket	IEC C14 inlet	C13 cord	moulded	BOM power cord
W2	Inlet L (after fuse and switch)	LRS L	18 AWG brown/black, 300 V	fork crimp or ferrule	**mains**
W3	Inlet N	LRS N	18 AWG blue/white, 300 V	fork or ferrule	**mains**
W4	Inlet PE	LRS FG (⏚)	18 AWG green/yellow (US: green)	fork or ferrule	earth
W5	LRS FG	Frame bond point on the base 2040	18 AWG green/yellow (US: green)	M5 ring + serrated washer into a T-nut	§12
W6	LRS −V	LRS FG	18 AWG green/yellow (US: green), short	ferrules	the **only** 0 V-to-earth bond
W7	LRS +V	E-stop box contact IN	18 AWG red (one core of a 2-core cable, about 1.5 m)	ferrule / contact screw	
W8	E-stop contact OUT	24 V switched bus (Wago 221-415)	18 AWG red (other core of W7's cable)	ferrule / lever	
W9	LRS −V	0 V bus (Wago 221-415, two joined if needed)	18 AWG black	ferrule / lever	
W10	24 V switched bus	F1 5 A holder → Rodent VIN "VCC"	18 AWG red	lever / ferrule into the screw terminal	
W11	0 V bus	Rodent VIN "GND"	18 AWG black	ferrule	
W12	24 V switched bus	F2 3 A holder → 24 V load bus (Wago)	20–22 AWG red	lever	
W13	24 V switched bus	F3 1 A holder → buck VIN+	22 AWG red	lever / solder or screw	
W14	0 V bus	buck GND (input side)	22 AWG black		
W15	Buck VOUT 12 V	12 V bus (Wago 221-413)	22 AWG orange		

3.2 MOSFET modules and loads (Q1–Q5, Adafruit #5648, see §4)

Each module has an input header (load supply **V+**, **GND**, **SIG**) and two output terminals (**+**, which is tied to V+, and **−**, which the MOSFET switches to GND). Use the silkscreen names on your board.

W	From	To	Wire	Ends	Notes
W20	24 V load bus	Q1, Q2, Q3 V+	22 AWG red	JST PH or 0.1" pad	The module V+ **must** be the load's own supply, or the flyback diode can't clamp.
W21	12 V bus	Q4, Q5 V+	22 AWG orange	JST PH or pad	
W22	0 V bus	Q1–Q5 GND	22 AWG black	JST PH or pad	
W23	Q1 out + / −	Pump + / −	22 AWG red / blue, ≤ 0.5 m	screw terminal, ferrules	Check pump polarity: DC diaphragm pumps are polarised.
W24	Q2 out + / −	Vacuum valve coil (EV-2-24 leads)	22–24 AWG	screw terminal; solder + heat-shrink at the valve	Coil polarity doesn't matter for an EV-2 [Likely]
W25	Q3 out + / −	Vent valve coil	22–24 AWG	as W24	
W26	Q4 out + / −	Head LED ring (through the 8-core, cores 6 and 7, §11)	24 AWG	screw terminal / solder at the ring	
W27	Q5 out + / −	Station LED ring	22 AWG 2-core, about 0.8 m	screw terminal / solder	Runs beside the CSI ribbon through the bed slot

3.3 Raspberry Pi 5 header and the I/O board

What the I/O board is: a small perfboard, about 70 × 50 mm, next to the Pi. It carries the ADS1115, the XGZP6847A, the three dividers (vacuum sensor, laser gate, carry-height sensor) and screw terminals. It connects to the Pi header with female Dupont leads.

Physical pin numbers are the standard 40-pin layout. [Certain]

W	Pi pin	Signal	To	Wire	Notes
W30	1	3.3 V	ADS1115 VDD	26 AWG Dupont	Gives the ADS 3.3 V I2C pull-ups
W31	2	5 V	I/O board 5 V rail: sensor VDD, laser, receiver	24 AWG	about 50 mA total
W32	6	GND	I/O board GND rail	24 AWG	
W33	3	GPIO2 SDA	ADS1115 SDA	26 AWG, < 20 cm	
W34	5	GPIO3 SCL	ADS1115 SCL	26 AWG, < 20 cm	ADS ADDR → GND gives 0x48 [Certain, TI datasheet]
W35	11	GPIO17	Q1 SIG (pump)	24 AWG	
W36	13	GPIO27	Q2 SIG (vacuum valve)	24 AWG	
W37	15	GPIO22	Q3 SIG (vent valve)	24 AWG	
W38	16	GPIO23	Q4 SIG (head LED)	24 AWG	
W39	18	GPIO24	Q5 SIG (station LED)	24 AWG	
W40	22	GPIO25	Lid switch C (switch NO → GND, pin 20)	24 AWG 2-core	Optional. Closed = lid shut.
W41	29	GPIO5	Gate divider midpoint (§8)	24 AWG	
W48	31	GPIO6	Carry-height sensor divider midpoint (§6a)	24 AWG	GPIO 6 is used by nothing else [Certain, docs, software and firmware searched]
W42	9	GND	**0 V bus** (one wire)	22 AWG black	Common reference for the MOSFET inputs. Keep it even though USB also joins the grounds.
W43	USB 2.0 (black) port		Rodent USB-C	USB-A → USB-C **data** cable, 1 m	Serial to FluidNC
W44	USB 2.0 (black) port		Head-camera extension (§10)		

W	Pi pin	Signal	To	Wire	Notes
W45	CAM/DISP 0		Station camera	500 mm Pi 5 camera cable	§10

3.4 Rodent inputs and motors

W	From	To (Rodent)	Wire	Ends	Notes
W50	X motor (right end of beam)	Socket **X**	22 AWG 4-core, about 1.3 m, static	4-pin 2.54 mm housing	Along the beam's back slot to the middle, then down with the bundle. The BOM's extension is needed if the motor lead is 1 m.
W51	Z motor (top of Z spine)	Socket **Z**	4-core 22 AWG shielded high-flex, about 2.5 m, through the chain	4-pin housing; shield → Rodent GND at board end only	
W52	X home switch (left end of beam): **C** and **NC** tabs	**X-MAX**: signal and GND	24 AWG 2-core, about 1.3 m, static	3-pin housing (V-Lim pin empty)	§7. Not "X-MIN": the board's names don't matter, the config does.
W53	Z top switch on the spine: **C** and **NC**	**Z-MAX**: signal and GND	8-core cores 4 and 5, through the chain	3-pin housing	§7
W54	Optical probe board: VCC, GND, OUT	**Y-MAX**: V-Lim, GND, signal	8-core cores 1, 2 and 3, through the chain	3-pin housing	§6. Yes, Y-MAX: there is no Y axis.
W55	Carry-height sensor board (Z spine): VCC, GND, OUT	VCC and GND: spliced into cores 1 and 2 at the spine top (the probe's 5 V and GND). OUT: core 8 → **I/O board divider** → **Pi GPIO 6** (not the Rodent)	the board's 3-pin cable, about 0.2 m, then the 8-core through the chain	solder + heat-shrink at the splice; screw terminal at the I/O board	§6a
—	nothing	Sockets **Y** and **E**	—	—	Spare drivers (FIRMWARE.md). Leave them empty.
—	nothing	Probe, E0-MAX, E1-MAX, V-MOS, RS485, spindle	—	—	unused

3.5 Overfill gate (Pi side)

W	From	To	Wire	Ends	Notes
W46	I/O board 5 V and GND	Class 2 (≤ 1 mW) laser module + and − (left gate post)	24 AWG 2-core, about 1.2 m	Dupont / solder + heat-shrink	§8. Along the bed's front edge in clips.
W47	I/O board 5 V, GND, divider input	Receiver VCC, GND, OUT (right gate post)	24 AWG 3-core, about 1.2 m	Dupont / solder	2.2 kΩ OUT-to-VCC pull-up soldered **at the receiver** (§8)

4. MOSFET modules

Pick: Adafruit MOSFET Driver, product #5648, ×5. It switches the pump, the vacuum valve, the vent valve and both LED rings, all low-side. [Certain for the specs below, from Adafruit's listing as quoted by several resellers]

Why this module

- **It is specified for 3 V logic.** The listing says "signal (3V to 20V logic level)". It uses an AO3406 N-MOSFET (30 V, 70 mΩ). A 3.3 V Pi pin turns it fully on, with no guessing about gate threshold.
- **The flyback diode is on the board** (1N4007). The pump and valve coils need one, and an on-board diode can't be soldered in backwards or forgotten.
- **Its current rating covers every load here:** 1.5 A continuous, 3 A peak. The JST PH input is rated 2 A.
- **The 30 V rating is enough on a 24.5 V bus**, because the diode clamps coil spikes to about 25.5 V. [Likely]

If you substitute "D4184" (AOD4184) modules (the BOM keeps 1N5819 diodes as a qty-0 conditional line for this case): those modules switch more current, but they have no flyback diode. Their makers also give no 3.3 V drive figure, so whether they switch fully at 3.3 V is unconfirmed. [Likely] If you use them:

- solder a diode across each pump and valve coil, cathode (stripe) to +;
- with the load on, check that the voltage across the module's output is under 0.2 V.

Don't use IRF520 modules: they aren't logic-level.

Rules for all five modules

- **Module V+ must be the load's supply:** 24 V for Q1–Q3, 12 V for Q4–Q5. The diode returns the coil current to V+.
- **Add a 10 kΩ resistor from SIG to GND on each module** unless you measured one already there. Power off and measure SIG to GND **before the first power-up** (§14 step 13). [Guessing whether #5648 has one; Adafruit's listing does not mention one] The Pi's GPIOs float while it boots or is off, and the pull-down keeps the pump and valves off. Without it, a floating vent gate can open the vent and drop a held slab.
- **If your pump draws more than 1.2 A running or 3 A at start, use a bigger module for Q1 only.** Measure it on the bench with a meter in series.

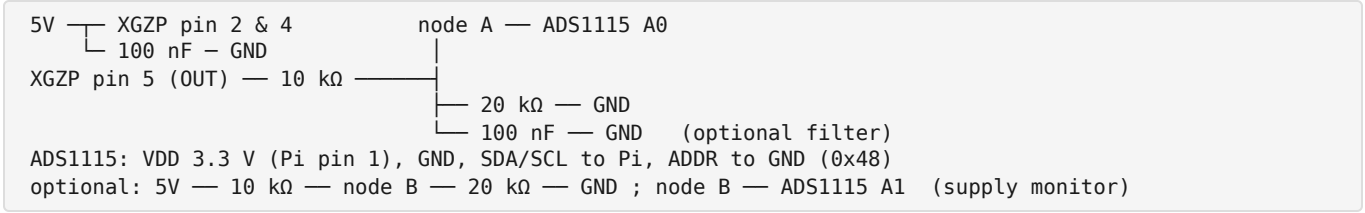
5. Vacuum sensor (XGZP6847A + ADS1115)

Part: CFSensor XGZP6847A100KPGN. Analog, 5 V supply, −100…0 kPa ("GN" = negative gauge), 0.5–4.5 V out. [Certain, CFSensor ordering code via distributor extracts] **A part number ending in "33" is the 3.3 V version. Don't buy that one.**

Pins (DIP-6) [Certain, CFSensor datasheet as quoted]:

Pin	Name	Connect to
1	N/C	nothing. The datasheet says leave it floating.
2	VDD	I/O board 5 V rail
3	GND	I/O board GND
4	VDD	5 V (duplicate of pin 2; connect it too)
5	OUT	10 kΩ → node A
6	GND	GND (duplicate of pin 3; connect it too)

Circuit



Why the divider

- The ADS1115 runs on 3.3 V, so that its I2C pull-ups match the Pi. Its inputs must stay within GND − 0.3 V to VDD + 0.3 V = 3.6 V. [Certain, TI datasheet]
- The sensor's 4.5 V output × 20/(10+20) = **3.0 V**. Even a sensor stuck at its 5.1 V rail gives 3.4 V.
- `hw.py` already uses `DIVIDER = 2/3` and the ±4.096 V range. [Certain]

Expected readings

- **At atmosphere: about 4.5 V at pin 5 and about 3.0 V at A0, so hw.py shows about 0 kPa.**

    - This assumes the standard direction: 0.5 V at −100 kPa, 4.5 V at 0 kPa, as `hw.py` expects. [Likely; CFSensor can calibrate either direction]
    - If you read about 0.5 V at atmosphere, your sensor is reversed. Tell the software owner to flip the formula. Don't rewire.
- **Optional supply monitor on A1:** the output is ratiometric to the 5 V supply, and the Pi's 5 V can sit anywhere from 4.9 to 5.2 V. That error is ±3 kPa, which is fine for the −35 kPa seal threshold. Wiring A1 lets the software correct it later. [Likely]

Tubing: the sensor has a 3 mm barb. Use 2 mm ID silicone tube (BOM) from the barb to a tee on the **cup side** of the vacuum valve, so it reads the cup line, not the reservoir. Mount the I/O board near the valves so this tube stays short.

6. Contact probe (fail-safe optical endstop)

Part: a slotted optical endstop board with a TCST2103 sensor (BOM), on the head foot's sensor post. A 1 mm aluminium flag, clamped under the vacuum elbow on top of the spring buffer, rises through the slot as the cup is pushed up (BUILD_GUIDE §7.6; it is in the CAD as `probe_flag`).

Wiring (W54):

- board **VCC** → Rodent Y-MAX **V-Lim**, which is 5 V with the SW_VCC jumper on 5 V;
- board **GND** → Y-MAX **GND**;
- board **OUT/SIG** → Y-MAX **signal**.

How the Rodent's limit input works [Certain, V1.0.2 schematic]

- The signal pin is the cathode of an opto LED, fed from V-Lim through 1 kΩ.
- **When the sensor output pulls low (sinks about 4 mA), the input reads LOW = "not touching".**
- When the output is high or disconnected, the opto is off and the input reads HIGH = **triggered**. The config uses `pin: gpio.34`, active high.

Mechanical rule (this is what makes it fail-safe)

- **At rest, the slot must be clear**, so the light reaches the detector and the output is LOW.
- **Contact pushes the flag into the slot.** The output goes HIGH, which reads as triggered.
- Set the board in its slots so the flag blocks the beam after 2.0 ± 0.5 mm of buffer travel (BUILD_GUIDE §7.6). The sim assumes 2 mm.

Why this way round: every failure lands on "triggered", and FluidNC then refuses to probe (ALARM:4).

Failure	Input reads	Result
Signal wire cut	open = HIGH	triggered → safe
Sensor GND or VCC cut	output can't sink = HIGH	triggered → safe
Emitter LED dead, or dirt in the slot	dark = output HIGH	triggered → safe
Rodent loses 24 V	everything off	machine stopped anyway

Check your board's polarity first. Most of these boards output HIGH when the slot is blocked, which is what you want. [Likely]

1. Power it from 5 V on the bench.
2. Measure OUT to GND: about 0–0.4 V with the slot clear, about 5 V with a card in the slot.
3. If yours is the other way round, don't use it. Inverting the logic would turn a cut wire into "not touching".

Hysteresis (only if needed): the firmware's `probe_hard_limit` alarms on any probe edge outside a `G38` move. If the flag chatters at the slot edge during the slow peel, BUILD_GUIDE §13.2's 300 cycles will show "Probe Hard

Limit" alarms. Then add hysteresis to the sensor output, keeping the fail-safe rule (a cut wire must still read triggered): for example a 74HC14 Schmitt inverter pair powered from V-Lim, with a 10 kΩ pull-up on its input [Guessing; have the instructor check it], and repeat FIRMWARE.md step 5's fail-safe rows. Or set `probe_hard_limit: false` and live without it.

Indicator LED caution: some boards have an indicator LED arranged so that it holds OUT at mid voltage when "high". The Rodent's opto then stays half on, and the input never reads triggered. FIRMWARE.md Step 5 catches this.

- **Fallback:** wire a bare TCST2103 directly. Detector collector to signal, detector emitter to GND. Emitter LED anode to V-Lim through 220 Ω, cathode to GND. [Likely, standard TCST2103 use]

Why not the board's Probe connector: its opto input is the other way round. The sensor would need to source current at rest, and an emitter failure would read "not touching". (Explained in config.yaml.)

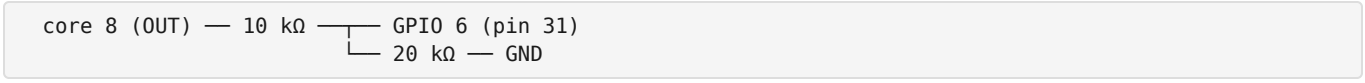
6a. Carry-height sensor (Z stall guard)

Part: the second optical endstop board from the BOM's packs (the same TCST2103 board as the probe), on `carry_sensor_mount` on the Z spine's left face. A flat 1 mm aluminium flag on `carry_flag_arm` (on the `z_plate`) enters its slot from below once the cup is at least 246.5 mm above the pad, and stays in it up to the top stop (BUILD_GUIDE §7.4a). [Certain, CAD]

What reads it: the Pi, not the Rodent: FluidNC knows nothing about it. The runner reads it before every sideways move (it must read blocked), before every grip (it must read clear), and on every Z move with a slab in the cup. [Certain, `runner._carry_check`, `hw.PiCarry`: majority of three reads 1 ms apart]

Wiring (W55, W48):

- At the top of the Z spine, where the 8-core's jacket is already opened for the Z switch (§11): board **VCC → core 1** (+5 V, shared with the probe), **GND → core 2**, **OUT → core 8**. Solder and heat-shrink each splice. The board is fixed to the X carriage, so its lead runs only in the X drag chain and never enters the Z service loop. At the head end, core 8 is cut back and insulated.
- At the panel end: core 8 to the I/O board:



- 5 V × 20/(10 + 20) = 3.3 V at GPIO 6 with a push-pull output, and the 20 kΩ pulls GPIO 6 LOW when the lead is cut. [Certain, arithmetic]
- The board's 5 V comes from the Rodent's V-Lim (core 1). The Pi and the Rodent share a ground through W42 and the USB cable. [Certain, §3.3]

Polarity: HIGH = the flag is in the beam = the head is at carry height. Check it on the bench first, as for the probe (§6): OUT about 0–0.4 V with the slot clear, about 5 V with a card in it. Don't use a board that is the other way round.

Check the level at GPIO 6 once it is wired [Likely: many of these boards pull OUT up through about 10 kΩ, and the 30 kΩ divider then gives only about 2.5 V]: with a card in the slot, GPIO 6 must read at least 2.5 V. If it is lower, solder a 2.2 kΩ pull-up from OUT to VCC **at the board**, as on the laser receiver (§8). At the board end a cut wire still reads LOW.

Failure	GPIO 6 reads	Result
OUT wire (core 8) cut	LOW (the 20 kΩ)	"clear" at travel height: sideways moves are refused ("Z stall suspected … right after homing this is most likely the sensor itself") → safe
Board VCC cut, or the Rodent unpowered	LOW	the same → safe

Failure	GPIO 6 reads	Result
Board GND cut	HIGH (OUT floats up to VCC) [Likely]	"blocked" everywhere: **not safe by itself**; caught at the next grip, where it must read clear ("Carry-height sensor fault …")
Emitter LED dead, or dirt or a card in the slot	HIGH	the same: **not safe by itself**, caught at the next grip
Flag bent away, or the mount slid down	wrong trip height	`selftest --motion` measures the trip; outside 246.5 ± 1 mm it fails

Why a dead emitter is not simply fail-safe here: the flag blocks the beam at carry height, so "dark" is the "OK to move" reading. The grip rule (the cup is far below the trip at every pick, so the sensor must read clear there) is what turns a dark sensor into a stop. [Certain, the CAD report's own warning, implemented in `runner._carry_check`]

7. Limit switches (NC)

Use bare SPDT hinge-lever microswitches with C / NO / NC solder tabs (the Omron SS-5GL size: 19.8 × 10.2 × 6.4 mm body, holes 9.5 mm apart). The printed mounts take a flat lever that ends at most 23 mm from the hinge end; cut a longer one back to 20 mm (BUILD_GUIDE §6.8).

- Solder the cable to **C** and **NC**: closed at rest, open when pressed.
- C goes to the connector's GND pin and NC to its signal pin. Leave V-Lim empty.

Don't use the 3D-printer "mechanical endstop" boards as they come. (The BOM lists bare lever microswitches for this reason.) Their LED-and-resistor circuit differs between makers. On some, it feeds the Rodent's opto LED and hides a pressed switch. [Likely: forum reports]

- If you only have those boards, desolder the switch, or cut the traces and wire straight to its C and NC tabs. Check C, NO and NC with a meter.

How it reads [Certain, schematic + config]

- At rest: switch closed → opto on → LOW → OK.
- Pressed, or a wire broken: open → HIGH → triggered. Outside homing that is ALARM:1, so a broken switch wire stops the machine instead of letting it crash.

Where the switches go [Certain, CAD `x_switch_mount`, `z_switch_mount`, `checks.txt`]

- **X home switch:** on the printed `x_switch_mount`, under the beam's left end, on the front faces of the beam and the left upright (2 × M5×10 button heads into T-nuts in the beam's lower front slot, left of `x_bumper_L`). Lever up, hinge end toward the back. A finger on the `x_bracket`'s left edge slides across the lever (a 45° lead-in, then a flat). It must click when the cup axis is at layout X ≈ 19 mm, about 23 mm left of pocket 0's centre, **± 2 mm**.
  - The printed `x_bumper_L` is the hard stop at cup X 15, so the switch must trip at least 2 mm before the `x_bracket` reaches it. The carriage still has 2.3 mm of rail there. The finger's flat holds the lever 0.6 mm past its click point (at the lever tip) all the way to the bumper.
  - **Lead (W52):** from the terminals straight down to about 50 mm below the beam (below the corner brackets), back under the beam, then up behind it into the beam's **lower rear** slot, and along that slot to the middle (§11). At the left joint, keep the flat joining plate's beam leg on the upper rear slot so it does not cover the spot where the lead enters (BUILD_GUIDE §4.3). [Certain, CAD lead path; Guessing on the brackets and plate, which are not in the CAD]
- **Z top switch:** on the printed `z_switch_mount`, on the Z spine's right (+X) face near the top (2 × M5×10 button heads into T-nuts in that face's slot), fixed to the X carriage. The switch lies on the mount's shelf, lever toward the `z_plate`. A 45° chamfer along the `z_plate`'s top back edge slides across the lever, and the plate's back

face then holds it. It must click at c ≈ 258 mm (the cup face 258 mm above the pad top), ± 2 mm. That is 4 mm before the `z_plate` meets the `z_motor_mount` (the top hard stop at c = 262).
    - **Lead:** cores 4 and 5 of the 8-core (§11), back past the spine, then up behind the `z_motor_mount` to the spine top, where the 8-core's jacket is opened.
- Setting each click point (the switch in its M2 slots, the mount in its M5 slot) is in BUILD_GUIDE §6.8 and §7.4. Exact positions are checked in FIRMWARE.md Step 6.

Lid switch (optional, GPIO25): use **C and NO**, closed when the lid presses it, so a shut lid reads LOW (`Button.is_pressed`). A cut wire then reads "lid open", which is safe.

8. Overfill laser gate

Parts (BOM): a ≤ 1 mW (Class 2) 650 nm laser module with a 6 mm barrel, and a non-modulated laser receiver module. Both are 5 V parts.

- **Do not use the KY-008 laser that comes in the receiver kit.** KY-008 modules are usually sold as 5 mW (Class 3R). [Likely] The laser is on whenever the Pi is on (W46 is the Pi's 5 V), and the beam runs along the open row at about 214 mm, the height of a face bent over the stacks while loading. A Class 2 beam is protected by the blink reflex; a 3R beam is not. The listing must say "≤ 1 mW" or "Class 2 / Class II". Keep the kit's receiver.
- The laser sits in the left gate post and the receiver in the right post, with the beam 4 mm above the MAX line. An M3 set screw in each post locks the barrel.
- Stick a "LASER - do not stare into the beam - Class 2" label on both posts.
- **Never look into the beam.** Aim it so it always ends on the receiver.
- [Guessing] If you want the laser off between runs as well, switch its 5 V with a small low-side MOSFET from a spare GPIO, on only during the start-up check. The software does not do this yet. With the laser off the gate reads "blocked", so it stays fail-safe.

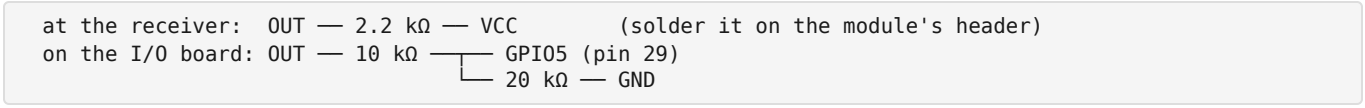
Laser wiring (W46):

- 5 V → the module's + (red) lead; GND → − (black). Confirm with a meter that it lights.
- 24 AWG 2-core, about 1.2 m.

Receiver wiring (W47):

- VCC → 5 V, GND → GND, OUT → I/O board.
- 24 AWG 3-core, about 1.2 m.

Level shift and fail-safe



- Most of these receivers are LM393 boards with an open-collector output and a 10 kΩ pull-up. [Likely] A bare 10k/20k divider on a 10 kΩ pull-up gives only about 2.5 V, which is marginal for a 3.3 V input.
- The extra 2.2 kΩ pull-up raises "high" to about 2.7–3.1 V, and a push-pull output gives 3.3 V. Both are inside the Pi's safe range. [Certain, arithmetic]
- The 2.2 kΩ **must sit at the receiver end**. Then a cut OUT wire leaves only the 20 kΩ, which pulls GPIO5 LOW = "blocked", and the run won't start.

Condition	GPIO5	Meaning
Beam received	about 3 V, HIGH	clear

Condition	GPIO5	Meaning
Beam blocked (stack too tall, or the head is down in the beam plane)	LOW	blocked
OUT wire cut, receiver unpowered, laser dead	LOW	blocked → safe
Receiver GND wire cut	HIGH	**false "clear"**: the selftest must block the beam and see LOW

Check before wiring: measure the receiver's OUT with the beam on and with it blocked. It must be **high when the beam is received**. If your module is the other way round, swap it. Don't invert in software, because a cut wire would then read "clear".

Room light: non-modulated receivers can be held HIGH by bright room light. Adjust the module's trimmer (under the light roof, if you fitted the optional roof) until a blocked beam reads LOW with the room lights on.

9. LED rings

Parts (BOM): two 12 V COB "angel eye" white rings: **40 mm** for the head (a 60 mm ring would cut into the drop arm; the seat in `camera_bracket` and `phase0_top` is a 42.8 mm bore with three centring ribs and a 33 mm lip: it takes a ring of 39.5–42.5 mm OD, up to 5 mm thick, and the ribs are shaved for a ring over 40 mm) and 60 or 63 mm for the station (a 64 mm bore on a shelf in `station_mount` and `phase0_base`). Measure OD, ID and thickness before printing the ring seats (BUILD_GUIDE §3.3). If no 40 mm COB ring can be found, a 40 mm 12-SMD ring that fits the seat is the fallback. **Leads:** the head ring's pads and leads leave through the notch in its seat's lip; the station ring's drop through the shelf slot into the camera ribbon's slot and leave with it. [Certain, CAD]

Switching

- Low side, from GPIO23 (head ring, Q4) and GPIO24 (station ring, Q5).
- Ring + to the 12 V bus through the module's "+" terminal, ring − to the module's "−" terminal.
- If the ring came with a small inline driver box, keep it. Wire the 12 V and module side to its input. [Likely: many angel-eye rings must use their driver]

No dimming by PWM

- Pi 5 GPIO23 and 24 only have software PWM. The rolling-shutter cameras would show dark bands. [Likely]
- Switch the rings fully on and set brightness with camera exposure and gain.
- If a ring is still too bright at minimum exposure, measure its current at 12 V. Then add a series resistor that halves it, rated for at least 4 × the power it burns.

Turn-on time: the ring drivers may soft-start, and camera auto exposure needs time to adapt. `hw.py` already waits 0.3 s after switching a ring on (`PiLights.settle_s`) before the first frame. [Certain, code] If the first frame of each slab still looks dark, raise it.

10. Cameras

Station camera: Camera Module 3 on CAM/DISP 0

- Use the 500 mm Pi 5 camera cable (22-pin 0.5 mm to 15-pin 1 mm, BOM: Raspberry Pi SC1130). **Don't go longer.**
- Mount the Pi on the electronics panel **behind the bed, centred near X ≈ 690** (the station is at X = 686). The ribbon then runs from the station mount's rear slot along the bed and down to the Pi in about 350 mm.
- Fit the ribbon with the power off. Lift the connector latch, insert, and close it. Follow the Raspberry Pi camera guide for which way the contacts face.
- Keep the ribbon away from the motor cables. Cross them at right angles if you have to.

Head camera: ELP 8 MP USB, IMX179

- Its own USB lead joins the BOM's 3 m USB 2.0 extension, which runs through the chain and the Z loop to a **black USB 2.0 port** on the Pi.
- Total length is about 4 m. USB 2.0 allows 5 m [Certain], so add nothing more. If the camera drops out, the first fix is a true drag-chain-rated USB cable (BOM note).

11. Cable routing: drag chain and Z service loop

X drag chain

Part (BOM): an openable chain, 15 × 30 mm inside, 1 m long. Openable links let the USB plug go in. Buy the R38 bend radius or larger if you have the choice. Don't go below R28. [Guessing on the sizes sold]

Fixed end: on the top-back of the X beam, at the **middle of travel, X ≈ 640**.

Moving end: on the X bracket's top tab, with a printed adapter.

Length check [Certain, chain geometry]

- With the fixed end at mid-travel, the chain needs travel/2 + π·R + margin = 1231/2 + π·38 + 100 ≈ 0.84 m, so 1 m is enough.
- With the fixed end at a beam end, it needs travel + π·R ≈ 1.35 m. Buy 1.5 m in that case.

What goes inside

#	Item	About	Carries
1	4-core 22 AWG shielded high-flex	Ø 5.5 mm	Z motor (W51)
2	8-core 24 AWG shielded high-flex	Ø 6 mm	see below
3	USB 2.0 extension	Ø 5 mm	head camera
4	PU tube, 4 mm OD	Ø 4 mm	cup vacuum line

That fills about 30 % of the chain. Lay the items side by side, untwisted, with nothing crossing. Don't tie anything inside the chain.

8-core assignment. Record your cable's real colours here.

Core	Use	Rodent end	Head end
1	+5 V probe and carry sensor	Y-MAX V-Lim	optical board VCC; carry board VCC (spliced at the spine top)
2	probe and carry sensor GND	Y-MAX GND	optical board GND; carry board GND (spliced at the spine top)
3	probe signal	Y-MAX signal	optical board OUT
4	Z switch signal	Z-MAX signal	Z switch NC
5	Z switch GND	Z-MAX GND	Z switch C
6	+12 V head ring	Q4 "+"	ring +
7	head ring return	Q4 "−"	ring −
8	carry sensor OUT	I/O board: 10 kΩ / 20 kΩ divider → Pi GPIO 6 (§6a)	carry board OUT, spliced at the spine top; insulated past the splice
shield	drain	Rodent GND, board end only	cut back, heat-shrink

Off the fixed end:

- The bundle drops behind the beam to the electronics panel.
- Tie it every 100 mm to the rear brace or to printed clips.

- Keep it more than 100 mm behind the head's path.
- The static X motor cable (W50) and X switch cable (W52) run in the beam's back slot from their ends to the middle, then join this bundle. The X switch lead comes up into the lower back slot just right of the left upright (§7).

Z service loop (head moves 263 mm relative to the carriage)

What's in it:

- the 8-core carries on past the carriage to the head;
- at the top of the Z spine, open the jacket over about 40 mm and take cores 4 and 5 out to the Z switch;
- in the same opening, splice the carry-height sensor's lead to cores 1, 2 and 8 (§6a); core 8 ends there;
- heat-shrink over the opening.

The 8-core, the USB extension and the vacuum tube then form one loop wrapped in the BOM's 10 mm spiral wrap.

Loop path: from a tie point at the top of the spine down to the top of the drop arm. Then tie the bundle along the drop arm every 50 mm down to the head.

Loop size:

- Make it about 350 mm long, with no bend tighter than 40 mm radius.
- Check by hand at c = 0 and c = 255. The loop must never hang lower than the top of the drop arm. At c = 0, that point is 350 mm above the pads (`model.ARM_Z`: the Z carriage 295 mm above the cup face, plus 55), well above the 218 mm rod tops. [Certain, layout numbers]

Tube slack: leave 25 mm at the cup's `M-5ALU-4` barb elbow, which points to the back (+Y). The buffer rod rises up to 20 mm on contact (SMC `ZPT25UUK20-B5-A10`, 20 mm stroke). Keep the tube off the head lens and its LED ring: at full stroke it runs about 1 mm under the ring [Guessing; the tube is not in the CAD].

12. Grounding and shielding

Protective earth

- Inlet PE → LRS FG (W4).
- LRS FG → frame (W5): an M5 ring terminal with a serrated washer into a T-nut on the base 2040. Scrape the anodising under it.
- Check that each extrusion joint conducts. Anodising insulates, so a joint may need a serrated washer. [Likely]

0 V bond: one short wire from LRS −V to FG (W6). Nowhere else.

- The Pi's ground joins the 0 V bus through W42 and the USB cable, so the whole low-voltage side has one ground.
- The Pi's own USB-C supply is floating (2-pin), so this creates no mains ground loop. [Likely]

Heavy current returns: pump, valve and motor currents return to the 0 V bus through their own black wires, never through the Pi.

Shields: connect them at the Rodent end only (its VIN GND terminal or any GND pin). Cut back and insulate the far end.

Separation:

- Keep motor cables at least 50 mm from the CSI ribbon, the ADS1115 and the sensor leads.
- Keep the I2C leads under 20 cm.

13. Strain relief and connectors

- **Screw terminals:** use a ferrule on every stranded wire (PSU, Rodent VIN, MOSFET outputs, fuse holders, E-stop). **Never put solder-tinned wire in a screw terminal.** It creeps and goes loose.
- **Wago 221 lever connectors** take stranded wire directly. Use them for the 24 V, 12 V and 0 V buses.
- **Tie every cable to the panel within 50 mm of its connector.** No connector should carry cable weight.
- **Drag chain:** tie each cable to the tie-bar at both chain ends.
- **Head:** use P-clips or tie mounts on the drop arm.
- **Crimps:** crimp JST-XH and Dupont housings with the proper tool and pull-test each one. Or buy pre-crimped leads.
- **Solder joints:** heat-shrink every one (dividers, cable breakouts, LED ring leads).
- **Moving parts:** put the pump on rubber feet and don't hard-mount it to the panel. Fit a short silicone hose section at the pump so vibration doesn't fatigue the fittings.

14. Pre-power checklist (multimeter)

Do this with everything wired, **mains unplugged**, and the E-stop released.

Mains (instructor)

1. Inlet PE pin to the X beam, Z spine and base: **< 0.5 Ω**.
2. Inlet L to PE and N to PE, switch on: **open (> 1 MΩ)**.
3. Inlet L to N, switch on: **not a short (> 10 Ω)**. The supply's input circuit shows some reading.
4. Fuse fitted, T3.15A. **LRS 115/230 switch on your mains voltage.** Terminal cover on. 4a. **Mains box closed, gland tight, insulated receptacles on every inlet tab: no exposed mains metal anywhere with the cord plugged in** (§1.1). Not on the plywood panel.

24 V side

5. 24 V switched bus to 0 V bus: **> 1 kΩ**. It rises as the meter charges the capacitors. Under 10 Ω is a short: find it.
6. E-stop pressed: LRS +V to 24 V switched bus **open**. Released: **< 0.5 Ω**. Check both NC blocks: each one alone must open the circuit when pressed (meter across each block).
7. F1 = 5 A, F2 = 3 A, F3 = 1 A fitted.
8. Rodent VIN polarity: red to "VCC", black to "GND". [Certain, terminal labels]
9. **24 V bus to Pi 5 V pin: open. 0 V bus to Pi GND: < 1 Ω.**

Motors, switches, probe

10. Motor coils, at the motor plug:
    - each coil pair **1–4 Ω** (the 17HS19-2004S1 is about 1.4 Ω [Likely]);
    - coil to coil **open**.
    - X plugged into socket **X** and Z into **Z**. **Nothing in Y or E.**
11. Limit switches, at the unplugged board-end housing: signal to GND **≈ 0 Ω at rest, open when pressed**.

Loads

12. Each MOSFET output: "+" to "−" with the load connected:
    - pump: a few tens of ohms [Guessing];
    - valve coil: hundreds of ohms [Guessing];
    - **never close to 0 Ω**.

13. Divider resistors: measure each 10 kΩ, 20 kΩ and 2.2 kΩ (10k and 100k are easy to mix up), including the carry sensor's pair (§6a): its GPIO 6 node to GND, with the Pi lead and core 8 unplugged, about 20 kΩ. **MOSFET SIG pull-downs:** SIG to GND on each of Q1–Q5, Pi lead unplugged: **about 10 kΩ**. Open means the pull-down is missing: add it (§4) before any power-up.

Board set-up

14. Rodent jumpers as §2. `r_sense_ohms` matches the resistor marking (FIRMWARE.md Step 0).
15. Sensor: pin 1 not connected; pins 2 and 4 to 5 V; pins 3 and 6 to GND.

15. First power-up

1. **Pi only (machine cord unplugged).** Boot the Pi, then check:
   - `i2cdetect -y 1` shows **48**;
   - the sensor gives about 3.0 V at A0 (about 0 kPa);
   - the gate divider gives about 3 V with the beam received and under 0.4 V when blocked. The laser and receiver run from the Pi's 5 V, so this works without 24 V.
2. **Machine cord in, E-stop pressed, rocker on.** Measure the LRS output with no load and set **24.5 V** with V.ADJ.
3. **Release the E-stop.**
   - Rodent VIN ≈ 24.4 V and the Rodent LEDs light;
   - buck output 12.0 V;
   - nothing gets warm in the first minute: Rodent driver area, buck, modules.
4. **Test each load from the Pi** with Raspberry Pi OS's `pinctrl`: `pinctrl set 17 op dh` turns the pump on and `pinctrl set 17 op dl` turns it off. [Likely, Bookworm tool] Do the same for 27, 22, 23 and 24, one at a time. Each output must drive only its own load. 4a. **Carry-height sensor** (it is powered from the Rodent, so the E-stop must be released): `pinctrl get 6` must read low with the slot clear and high with a card in it, and GPIO 6 must measure at least 2.5 V high (§6a). `python -m slabscan selftest` shows the same as its `carry sensor` line.
5. Now follow FIRMWARE.md from Step 1.
6. **After 10 minutes of motors holding:** the motor bodies may be warm but **below about 50 °C** (comfortable to hold a finger on). Hotter than that means the sense-resistor setting is wrong: power off (FIRMWARE.md step 0).

16. Open risks

1. **[Likely] Laser receiver output varies by maker.** The 2.2 kΩ pull-up and divider make both common output types safe, but measure before wiring.
2. **[Likely] Sensor output direction.** It is assumed to be 0.5 V at −100 kPa. Check about 4.5 V at atmosphere on first power-up.
3. **[Certain] E-stop DC rating is below target.** The BOM's blocks are rated DC-13 1.5 A at 24 V, against the ≥ 5 A target and the ≈3.4 A bus peak (§1.2, §1.3). Two NC blocks in series and the monthly E-stop check stay required; a mains-side E-stop or a force-guided safety relay is the fix to decide before unattended use.
4. **[Guessing] Head USB at about 4 m.** It works on paper. A cheap extension in a moving chain is the weakest cable in the machine. Watch for camera dropouts in the soak test.
5. **[Guessing] Whether Adafruit #5648 has a SIG pull-down.** Measured in §14 step 13 before the first power-up; add 10 kΩ if it reads open (§4). BUILD_GUIDE §13.4 ("unplug the Pi, the pump must stop") checks it again.

6. **[Guessing] Probe flag chatter** with `probe_hard_limit: true`: see §6 "Hysteresis".
7. **[Likely] The carry sensor's high level through the divider** may be only about 2.5 V with a 10 kΩ on-board pull-up: measure it (§6a), add 2.2 kΩ at the board if low.
8. **[Certain] A dark carry sensor reads "OK to move".** A dead emitter, dirt or a cut GND is caught only at the next grip (§6a); until then it cannot see a stall. The run checks at every grip, so that window is at most one slab's moves.

Resolved: the BOM now lists Adafruit #5648 modules, bare microswitches, the 1/3/5 A fuses, the mains box, a Class 2 laser and a two-block E-stop used on this page, and the 8-core's spare core now carries the carry-height sensor (§6a). Both limit switches have printed mounts with their lead paths in the CAD (§7). The CAD has printed hard stops that keep both carriages on their rails (CONTRACT), so the old Z-rail margin issue is closed.

SlabRail bill of materials (October 2026)

The line-by-line list with links is in `BOM.csv`. The same parts, grouped into the three orders you actually place, are in SHOPPING_LIST.md. This page has the totals, the order plan, where to save and where not to, the tools, and the design gaps found while pricing.

Confidence tags: **[Certain]** = primary source (for a price: the product page itself, opened on 22 Oct 2026), **[Likely]** = several secondary sources or a dated listing, **[Guessing]** = estimate.

Bottom line

- **The full list costs about $2,182 with a 15% allowance.** [Likely ±10%] You have said $2,500–3,000 is fine if the machine works properly, so there is about $300–800 of headroom.
- **The total did not change this round.** The two homing switches now have printed mounts. They need 4 more M5×10 button heads, 4 more M5 T-nuts and 4 × M2×12 with nuts, all from packs already in the list, and about 33 cm³ more filament. The switch line now says to buy the short hinge lever. In round 4 two lines moved between orders and the confidence tags were aligned; in round 3 the list went up $209 ($182 of parts plus the allowance), almost all of it real product-page prices replacing estimates (below).
- **$1,383 of the $1,897 subtotal is a price seen on the product page** [Certain, 22 Oct 2026], $175 is [Likely] and $339 is [Guessing] (mostly the Misumi extrusions, which block scripts: get that quote first, and Amazon lines whose price was not visible). The allowance covers shipping already seen but not in the line prices (LiMo DDP $39, Digi-Key Marketplace $28, MakerStock $13.49, the endstop boards $5.70) and the ones not shown (ELP express and duty, Proax shipping and brokerage, AFT's small-order fee, Misumi).
- **The worst case fits too.** [Guessing] Add the tools you will probably have to buy ($45–50), the conditional lines Phase 0 is designed to trigger (HT1 cups and the light roof, about $77 with the allowance), and the [Guessing] lines coming in 20% high (about +$78): about **$2,380**, still under $2,500.
- **About $1,862 if the school supplies its usual shop stock** (filament, wire, glue, fuses, common screws, the flag sheet). [Likely] It is the cheapest headroom: the same worst case then comes to about $2,050.
- **About $1,787 lean:** school stock, plus generic direct-acting valves instead of Clippard. [Likely] (The bed is already the Home Depot birch panel: the Baltic birch blank is no longer listed online.)
- **Keep a running tally** (a spreadsheet: line, quoted, paid). Order 1 (Phase 0) is quoted at ≈ $626 and order 2 at ≈ $1,556. If order 1 comes in more than about 10% over its quote, re-quote order 2 before buying it.
- **Spend the headroom only when a test asks for it.** The conditional lines are there for that: HT1 cups if Phase 0B leaves a mark, the light roof if Phase 0A shows glare, and one upgrade per acceptance test that can fail (below). Adding parts before the tests doesn't make the machine work better. If every conditional line triggered at once, the total would reach about $3,330; that many failed tests would mean the design needs rethinking, not more parts. [Guessing, arithmetic]
- **Not in any total: junk graded slabs for the acceptance tests** (15–20, some deliberately dropped). Take your lowest-value slabs from stock and write off 2–3 of them, or budget $150–400 to buy them (BUILD_GUIDE §13).
- **Things that add the most by themselves:**
  - buying every tool new (about +$460, total about $2,640: over $2,500, inside $3,000);
  - the genuine HIWIN rail tier (about +$420, conditional line);
  - a branded pump with a flow curve (about +$200 [Guessing], conditional line).
- **Six lines in the total still have only a search hint, no product link:** the 40 mm COB head ring (a 12-SMD ring that fits the seat is the fallback), the branch fuse holders, the drag chain, the M5×10 button heads and the two shielded moving cables (the igus cable in their notes is unshielded, 4-core and sold by the foot). Their estimates stay in the total, and a few linked prices were not visible (Amazon blocks scripts): SHOPPING_LIST.md marks each one. Look them up before the order they belong to.

Order plan

The design says to pass two Phase 0 gates before buying any motion parts. Each order, with a link per line, is in SHOPPING_LIST.md.

Order	What	Cost incl. 15%
1a — Phase 0A, camera test	Pi 5 with PSU, microSD and cooler, both cameras, the 3 m USB extension, both LED rings, a 12 V wall adapter, M2 screws and standoffs, two 2020 jig posts and the 66 mm arm stub, the optical endstop boards, the flag sheet, hookup wire, heat shrink, filament. Rows marked "PHASE 0A" in the CSV. **Plus, from school stock:** a handful of M5/M3 screws (6 × M5×10, 6 M5 washers, 6 M5 T-nuts or plain M5 nuts, 1 × M5×30 + nylock, 2 × M3×16 + 4 washers + 2 nuts) and a strip of microfiber with spray adhesive or double-sided tape for the tray ledges: their packs are order-2 lines.	**≈ $470**; ≈ $281 with a borrowed school Pi; ≈ $202 if the school also supplies filament, wire, heat shrink and the flag sheet
1b — Phase 0B, cup bench test	Two SMC cups and buffers (one is the spare), spare pads, the M5 barb elbows, PU tube (with its fittings kit), PTFE tape, IPA, a hand vacuum pump with gauge. No mains wiring, no electronics. Rows marked "PHASE 0B". Buy after 0A passes, or order the two cups and the two elbows with 0A: Proax shows the cups back-ordered with no lead time, and cup, buffer and elbow are the jig's real occluder.	**≈ $155**
2 — Machine	Extrusions, rails and a spare block, motors, belt, brackets, fasteners, bed, dowels, Rodent, valves, pump, reservoir and its push fittings, vacuum sensor and ADC, 24 V PSU, mains inlet and mains box, MOSFET modules, fuses, I/O board, limit switches (short hinge lever), laser gate, drag chain, flex cables, E-stop, and the screws for the carry-height sensor and the two switch mounts (the sensor's board is already in the 0A endstop packs; the switches' M2 screws in the 0A M2 kit). Add the light-roof lines only if Phase 0 showed glare, the HT1 cups only if the urethane cup marked, and an upgrade line only when its test failed.	**≈ $1,556** (+ ≈ $190 for the Pi set if you borrowed one for Phase 0)

If Phase 0A fails, you have spent at most ≈ $470 (≈ $281 with a borrowed Pi). If Phase 0B fails, ≈ $626. The first fix to try would be the HT1 cup (+ ≈ $41), but it needs an M10×1-to-G1/8 adapter nobody has sourced yet: until one is found, a cup that marks means stop and ask. Nothing from order 2 has been bought at that point. [Certain, by construction of the tags]

Totals by category

Category	Total	Usually at school	Biggest line
Frame (extrusion)	$158.00	—	2 × 2040 × 1416 base rails, $48
Motion	$191.69	$10.99	VXB Z motor 3-pack, $99.99
Fasteners & brackets	$268.99	$49.13	CNC Kitchen heat-set insert kit, $26.99
Bed, rods & light roof	$158.38	$35.74	PureBond birch panel, $60.17
Controller & compute	$225.80	—	Raspberry Pi 5 4 GB, $110
Cameras & lighting	$137.30	—	ELP 8 MP head camera, $55
Vacuum & gripper	$294.61	$12.00	2 × Clippard EV-2-24 valves, $77.90
Sensors	$92.42	$11.99	5 × XGZP6847A, $36
Power & switching	$197.28	$47.12	AutomationDirect E-stop set, $36.75
Wiring & cable management	$146.09	$84.13	22 AWG hookup wire set, $19.95
Printed parts (filament)	$26.78	$26.78	2 kg PETG
Acceptance tests	$0.00	—	junk slabs, conditional (qty 0)
Subtotal	**$1,897.34**	**$277.88**	
Spares, shipping, duties (15%)	$284.60		
Grand total	**$2,181.94**		
Minus school stock (+ its 15%)	−$319.56		**≈ $1,862**
Lean swap (generic valves), with school stock			**≈ $1,787**

What changed in this round ($0.00):

- **Homing switches:** the CAD now has printed mounts (`x_switch_mount`, `z_switch_mount`), so the 2020 corner brackets no longer carry the switches. The line now asks for the short hinge lever (Omron SS-5GL size, a lever ending at the body's end): the mounts take a lever up to 23 mm from the hinge end, and a longer one is cut back to 20 mm (BUILD_GUIDE §6.8).
- **M5×10 button heads:** 6 needed, not 2 (2 per switch mount). Same pack, same price [Guessing, a small pack of 10 or more].
- **M5 T-nuts:** about 130 needed, not 126 (2 per switch mount). The 150-pack still covers them and the 6 for the Phase 0 jig.
- **M2 kit:** 4 × M2×12 and 4 nuts more, for the switches. No price change.
- **M2 standoffs:** the jig's bridge is now 10.6 mm thick, so a standoff is the board-to-lens length minus 10.0 mm, with 2.0 mm of plastic around each hole.
- **Head LED ring:** the seat is now a 42.8 mm bore with three centring ribs: it takes a ring of 39.5–42.5 mm OD.
- **Filament:** 801 cm³ in 27 prints (was 768 in 25). The 2 kg still cover it.

What changed in round 4 ($0.00):

- **Push fittings kit** ($11.99) from order 1b to order 2: its 1/8 NPT fittings go only into the reservoir cap. Phase 0B uses the tube and the barb elbow.
- **PTFE tape** ($2.36) split out of the reservoir set ($19.93 → $17.57) into its own order-1b line: Phase 0B seals the elbow's M5 thread with it.
- **Confidence tags** now follow each line's price evidence, in the CSV and the shopping list alike. In the CSV, six lines whose price was never seen went from [Likely] to [Guessing] (the two M2 kits, the M5×12 button heads, the M2.5/M3 standoff kit, the resistor kit, the awl pin) and three whose price was seen went to [Certain] (the M-5ALU-4 elbows, the vacuum sensors, the MOSFET modules). In the shopping list, the mains inlet and the mains wire went from [Likely] to [Guessing], as in the CSV.
- **Specs and notes:** the awl pin is one spec now (steel or stainless, hardened not needed); the station ring seat takes a 60 or a 63 mm ring; the head ring line has a fallback; the moving cables link to a search, not to the per-foot igus page; "measure before printing" names the Phase 0 part, which is printed first.

What changed in round 3 (+$182.08, +$209.39 with the allowance):

- **Links and prices from a sourcing run** (22 Oct 2026), spot-checked line by line. Every line in the total now carries a product link except the six above. Where the link is [Likely] or [Guessing], the old search hint stays after it. Price changes of $10 or more: up, the Z motor +$69.99, vacuum sensors +$20.00, E-stop +$19.75, mains and earth wire +$15.00 (still an estimate), microSD +$13.95, pad cloth +$13.48, pad foam +$10.99, heat-set inserts +$10.99, check valves +$10.60; down, the LiMo rails and block −$23.90, the hand pump −$14.01, the two SMC cups −$13.76 (Proax), the head camera −$11.00 (the ELP store), the dowels −$10.00. Every other line moved by less than $10.
- **Spec corrections from the sourcing run, applied because they are plainly right:**
  - **Bed:** the Woodcraft 12 × 60 in Baltic birch blank is no longer listed; the line is now a Home Depot 3/4 in PureBond birch 2 × 8 ft panel (17.9 mm), one rip and one crosscut.
  - **Dowels:** 5/16 in (7.94 mm) × 12 in, so the brad-point bit and collar are 5/16 in.
  - **Class 2 laser:** the only module found (Quarton VLM-650-03) has a 7 mm body. The left gate post now prints with a 7.2 mm barrel hole for it (BUILD_GUIDE §5.9).
  - **Pump "upgrade":** the VXB 10 L/min pump is probably the main pump itself (VXB's SKU is the Amazon ASIN) [Likely]. The upgrade line is now an unsourced branded pump, about $200 [Guessing].
  - **Optical endstop boards:** the 3-pack is out of stock; two 2-packs give 4 boards (probe, carry-height sensor, 2 spares). The listing does not name the TCST2103: measure it.
  - **Flag sheet:** at least 50 × 60 mm, for the probe flag and the new carry flag.

- **Not applied:** the sourcing run's igus chainflex cable for both moving cables. It is unshielded and 4-core, so the head cable would become two runs with renumbered cores. The shielded 8-core line stays [Guessing, price not seen]; the igus cable is listed as a verified alternative in the CSV.
- **New lines for the carry-height sensor** (BUILD_GUIDE §7.4a, WIRING.md §6a): the board (the second one in the endstop packs, $0), the carry flag (cut from the flag sheet, $0), 2 × M5×10 button heads ($6, new), the M3 screws, washers, nuts and T-nuts (from the existing packs, $0), its 0.2 m lead (the board's own 3-pin cable, $0) and the two printed parts ($0). A 10 kΩ + 20 kΩ divider for GPIO 6 comes from the resistor kit.
- **Phase 0 jig:** six M5 washers, not twelve (one under each M5×10: the jig walls are now 4 mm).

What changed in the earlier rounds (+$266 in all): the second SMC cup and buffer, the M5 barb elbows, a spare rail block and the jig's arm stub (+$60), and +$112 of parts the first review found the design needs (a closed mains box, a Class 2 laser, a second E-stop contact block, a hand vacuum pump, a 12 V adapter, separate head and station LED rings, more screw sizes, M2 standoffs, set screws), plus the allowance going from 10% to 15% (+$94).

Conditional lines are in the CSV with qty 0 and are not in the total. Buy them only if a test fails:

- Schmalz HT1 cups, 2 × $18 [Guessing: the price needs a Schmalz login]. Buy if the urethane cup marks slabs. The 25 mm HT1 cups are G1/8-threaded and need an M10×1-to-G1/8 adapter nobody has sourced yet: find it before Phase 0B. Until then, a cup that marks means stop and ask.
- Stepper extension cable, $8. Buy (or make) it if the X motor's lead is too short.
- Light roof (4 foam boards $5, gaffer tape about $12, two 2020 × ~500 mm posts $14). The roof is optional (CONTRACT): buy it only if Phase 0 or the first runs show room-light glare.
- 1N5819 flyback diodes, $6. Only if you swap the Adafruit #5648 modules for ones without an on-board diode (for example D4184 modules).
- Junk graded slabs for the acceptance tests (above).

Upgrades, also qty 0: buy one only when its test fails (about $920 if all of them). [Likely for the failure each one removes; prices as tagged]

Upgrade	About	Buy only if
Genuine HIWIN MGNR12 rail + block (replace both)	$420 [Certain in EUR, Guessing on the exchange rate]	The generic X rail fails the §6.1 roll test (notches or binding after cleaning and greasing), or binds in §13.1
Z motor with a published torque curve	$55 [Guessing]	The §7.8 lift test still fails after the free fixes: Z at 3000 mm/min and Z current raised to about 60% of the motor's rating, motor under 50 °C. No off-the-shelf motor with an 8 mm lead and a 311–356 mm screw was found: ask for a custom screw length.
Constant-force spring for Z	$20.36 [Likely]	The §7.8 hold test creeps. Its mount is not in the CAD.
High-flex USB cable (L-com CAUAAHFX-3M + coupler)	$50 [Likely]	Head-camera dropouts in §13.1, §13.2 or later runs that the spare cable does not cure. It is rated 10,000 flex cycles, not continuous drag-chain duty.
Branded 24 V vacuum pump with a flow curve	$200 [Guessing]	The generic pump fails the §8.3 seal or reservoir test although the line holds
Schneider Harmony XB5AS8444 E-stop (2 NC) + box	$175 [Likely]	The monthly E-stop check ever finds the 24 V bus live. [Likely] Even these blocks are rated below the capacitor inrush, so keep two in series and the check.

Not bought, on purpose: a closed-loop Z motor (the Rodent has no step/dir header for one, and FluidNC uses StallGuard only for homing [Likely]); a spare Rodent (a dead board is a safe stop, and its Y and E driver sockets are already spares [Certain, CONTRACT]); a better head camera (it gives 0.055 mm/px against the 0.14 mm/px the back QR needs [Certain, synthetic labels]; buy one only if Phase 0A fails on sharpness, not on glare). **The Z stall detector is now in the design** at about $6: the spare TCST2103 board on the Z spine with a flag on `z_plate`

(BUILD_GUIDE §7.4a). It sees a loaded head that ended more than about 2.5 mm low before any sideways move; it does not make a weak Z motor strong, so the §7.8 lift test still has to pass.

The allowance is 15%. [Likely] The US removed the de minimis duty exemption in 2025, so parcels from abroad (ELP from China, LiMo, Proax from Canada) carry duty, and several shops add shipping the line prices leave out (above).

The 5 biggest cost items

1. **Raspberry Pi 5 4 GB, $110** [Certain, CanaKit, board only]. DRAM costs pushed the list price from $60 to $70 (Dec 2025), then $85 (Feb 2026), then about $110 (Apr 2026). [Likely]
2. **VXB NEMA17 + TR8×8 400 mm, 3-pack, $99.99** [Certain, VXB]. 1.7 A, 1.7 Ω, 3.1 mH, 0.44 N·m: every number the Z motor rule asks for is on the page. You need one; the other two are spares. A single motor with the same published specs would save about $70.
3. **Two Clippard EV-2-24 direct-acting NC valves, $77.90** [Certain, Clippard; AHE listed $25.90 each on sale].
4. **Two SMC `ZPT25UUK20-B5-A10` cups and buffers, $70.24** [Certain, Proax: CAD 49.34 each, back-ordered]. Canadian shipping and brokerage are extra.
5. **The bed panel, $60.17** [Certain, Home Depot], then the Rodent ($56, in stock at BIQU) and the ELP head camera ($55, ELP's own store).

The 15% allowance ($285) is bigger than any single part.

Where to save

- **Valves.** Generic 24 V direct-acting 2/2 NC micro valves cost about $6 each instead of $39, saving about $66.
  - They work if they are truly direct-acting. Test them when order 2 arrives (BUILD_GUIDE §8.3, reservoir test): hold −60 kPa for 60 s with the pump off and watch the sensor.
  - Their orifice is small, so venting is slower. Lengthen `vent_s` if release lags.
- **The Z motor.** A single motor whose listing gives 1.5–2.0 A, ≤ 4 mH and an 8 mm lead saves about $70 over the VXB 3-pack. Measure 1–2 Ω per phase on arrival; reject anything around 10 Ω.
- **Rails.** The generic MGN12 rail is enough here. Pockets have ±1.85 mm of play and the slabs sit in rods, so you don't need HIWIN precision.
  - Clean the rail.
  - Count the balls in the carriage before installing it, because clones lose balls when the carriage slides off the rail.
  - Re-grease it.
- **Extrusions.** Amazon 20-series V-slot in stock lengths, cut at school, costs about the same ($130–150). Only do it if the school saw cuts square. Misumi's free cut to ±0.5 mm saves an afternoon of squaring.
- **Fasteners.** One large M5 T-nut and screw kit plus one M3 kit can replace the separate packs. Count first: about 130 M5 T-nuts (plus 6 for the Phase 0 jig) and 69 M3×10 are needed, and the kit must contain M5×10/12/16/20/25/30, M5×12 and M5×10 button heads and M3×8/10/12/16/20 (BOM.csv notes say where each goes). AFT charges a small-order fee under $75.
- **Small shop items.** The flag sheet ($11.99 + $13.49 delivery for a 12 × 12 in sheet) and the wood screws (Parts Express shipping) are cheaper as school offcuts or a local hardware-store pack.
- **Pi 5 2 GB.** It would probably run the software. [Guessing] It saves maybe $30–40, but the contract specifies 4 GB, so test memory use in `sim` before switching.

Where NOT to save

- **The cup and buffer.** Buy a genuine urethane pad and spare pads. The cup is the only part that touches your slabs. A cheap silicone cup can leave silicone residue rings on glossy plastic. [Likely] Run the 24 h marking test on a junk slab before any real inventory goes in (Phase 0B, then again on the machine, BUILD_GUIDE §13.2).
  - Buy exactly `ZPT25UUK20-B5-A10` (or the rotating `J20`). The flag, its guide and the sensor post are built on that buffer's catalog heights. [Certain, CAD] A generic AliExpress buffer or a Schmalz or Piab spring plunger has other heights, and the Piab bodies are not M10. [Likely]
- **The Z motor's specs.** 1.5–2.0 A, ≤ 4 mH, an 8 mm lead. A 0.4–0.6 A "integrated" motor stalls at the speed the firmware runs it. The carry-height sensor stops the run after a stall; it does not prevent one.
- **Valve type.** Never use pilot-operated, "servo-assisted" or diaphragm water valves. They need a pressure difference of 0.2–0.5 bar to work, so they won't seal or open reliably on vacuum. [Likely] The listing must say "direct acting" and give a vacuum rating, or a "0 bar" minimum.
- **The MOSFET modules.** Use modules specified for 3.3 V logic. The BOM's Adafruit #5648 is listed for 3–20 V logic and has its flyback diode on board (WIRING.md §4). IRF520 modules barely switch from a 3.3 V Pi pin. [Likely] If you substitute modules without an on-board diode (D4184 type), add a flyback diode across the pump and each valve coil (the conditional 1N5819 line).
- **The 24 V PSU and mains side.** Buy a Mean Well, not an unbranded "24 V 10 A" brick, and the closed mains box for the IEC inlet: a snap-in inlet on a 6 mm plywood panel leaves live tabs open. Have an instructor check the inlet, fuse, box and earth wiring before first power-up.
- **The laser.** Buy a ≤ 1 mW (Class 2) module. The KY-008 that comes with the receiver kit is usually sold as 5 mW (Class 3R), and the beam is on whenever the Pi is on, at face height when you bend over the stacks. [Likely]
- **The E-stop's second contact block.** A single block switching the 24 V capacitor inrush can weld closed without you noticing (WIRING.md §1.3). The AutomationDirect set has three.
- **The 4040 profile type.** It must be 20-series, slot 6, with two slots per face. Amazon's "4040 European standard" is often 40-series slot 8 with one slot per face, and the printed mounts won't fit it; so are Amazon's "4040 brackets". [Likely]
- **Dowel straightness.** Roll every dowel on a flat table and reject bowed ones. A bowed rod narrows the pocket at the top, where the cup lifts a full stack.

Required tools (not in the machine total)

Most of these are standard in a tech-school shop. The "buy" column is the price if the school doesn't have one.

Tool	Used for	Buy price if missing	At school?
Hand drill (cordless)	The 82 rod holes, drilled in place on the assembled machine (§5B), and the bed bolts	$60–100	Likely
Drill press	The rod-hole guide block, the drop arm's and the jig stub's cross holes	—	Likely
5/16 in brad-point bit, new and sharp, at least 4 1/2 in (115 mm) long	Rod holes, for the BOM's 5/16 in dowels (8 mm if you buy 8 mm dowels). The depth collar sits 57 mm above the cutting lips, and the chuck needs about 30 mm of shank above the collar [Likely]	$10	**Buy**
Depth-stop collar, 5/16 in	All rod holes the same 15 mm deep	$5	**Buy**
Drill guide (Milescraft DrillBlock type) or a drill press with long-table support	Holes square to the bed. The printed `drill_template` sets the position, not the angle.	$15–40	Likely
Metric hex keys, 1.5–10 mm, ball end	All socket screws. The probe board's front M3 screw needs the **2.5 mm ball end**, tilted about 20° (BUILD_GUIDE §7.6).	$15	Likely

Tool	Used for	Buy price if missing	At school?
Digital calipers, 150 mm	Measuring 5 of your slabs (`dims.py`), dowels, ring OD/ID, camera board holes, the switch levers; setting each switch 0.3 mm past its click (§6.8)	$20	Likely
Ferrule crimper (self-adjusting)	Screw-terminal wire ends	$25	Maybe
JST/Dupont crimper (IWISS SN-28B or SN-2549)	Sensor and GPIO connectors	$25	Maybe
Wire stripper, flush cutters	Wiring; cutting a long switch lever back to 20 mm (BUILD_GUIDE §6.8)	$20	Likely
Multimeter	Wiring checks, sensor voltages, beam receiver level, setting the carry-sensor trip	$30	Likely
Temperature-controlled soldering iron	Diodes, divider resistors, LED ring leads	$30–100	Likely
Heat-set insert tips (ruthex set fits 900M/T18 irons; TS100-style tips for a Pinecil)	M5 insert in `awl_adapter`, spares	$10–16	**Buy** unless the shop has them
Fine saw + miter box, or a bandsaw	Cutting 82 dowels to 236 mm	$15	Likely
Junior hacksaw or piercing saw, needle files (one about 1.5 mm thick), a vise	The probe flag and the carry flag (BUILD_GUIDE §7.4a, §7.6); clearing the 1.9 mm flag slot in `head_foot` (a standard 12.7 mm hacksaw blade does not fit it)	$15	Likely
7.2 mm (or 9/32 in) drill	Only if your left gate post was printed from an older STL with a 6.2 mm laser hole	$3	Likely
Jigsaw	41 × 41 mm notches in the bed for the uprights (CONTRACT)	$50	Likely
M5/M3 taps + tap handle; **1/8 NPT tap**	Extrusion end taps; reservoir caps	$8 (NPT)	Taps likely; NPT maybe
Dial indicator + magnetic base	Beam-twist check (DESIGN.md)	$25	Likely (machine shop)
Heat gun, clamps, square, level, 120/220 sandpaper	General; doming the rod tops	$25	Likely
Drills and small items			
Metric drill bits 1.8, 3.4, 5.2 and 5.5 mm	1.8: new board holes in the jig's bridge (§1.5); 3.4: M3 clearance holes and the carry flag (§3.4, §7.4a); 5.2: the probe flag (§7.6); 5.5: M5 clearance holes and the cross holes (§3.4, §1.5, §7.5)	$15	Likely
11/32 in drill	Tap drill for the 1/8 NPT reservoir ports (§8.2)	$6	Maybe
Sharp knife and steel rule	Cutting the pads (§5.5), the PU tube square (§8.2)	$10	Likely
Small spring scale, 0–500 g (or 0–5 N)	Belt tension (§6.6)	$8	Maybe
CA glue (superglue)	The awl pin (§3.4)	$5	Likely
Shims (card, thin wood) or a small lab jack	Raising the slab under the cup in Phase 0B (§1.10)	$0–25	Likely
microSD card reader	Flashing Raspberry Pi OS (§11.1)	$10	Likely
M2 flat washers, about 0.3 mm, a few	The camera-aim shim (§7.7): the M2 kit has screws and nuts only	$3	**Buy**

Tools you will probably have to buy: about $45–50. That is the brad-point bit, depth collar, heat-set tips, NPT tap, M2 washers and the spring scale. **If you had to buy every tool new, the total would be about $2,640** ($2,182 + about $460), so use the school's where you can.

How the quantities were derived

- **Extrusions.** These are `model.cut_list()` values (`cad/out/cut_list.csv`), plus an optional 2020 × 600 mm rear brace and, only if you build the optional light roof, two 2020 × ~500 mm roof posts:
  - 2040 × 1416 (2)
  - 4040 × 401 (2)
  - 4040 × 1416 (1)
  - 2040 × 340, both ends tapped M5 (1)
  - 2020 × 285 (1)
  - Phase 0 jig: 2020 × 312 (2) and the 2020 × 66 arm stub (1)
  - plus the Z lead screw, cut to 340 mm below the motor face, and two 1 mm aluminium flags: the probe flag blank, 24.4 × 55 mm, and the carry flag, 19.9 × 36 mm
- **Rods.** `len(layout.rods())` = 82. Each is 218 mm above the pad + 3 mm pad + 15 mm embedded = 236 mm, so 12 in (305 mm) stock leaves room to trim. Buy 100 so you can reject bowed ones.
- **Pads.** There are 13 pockets, and each pad is 83 × 138 mm.
- **M3 rail screws.** MGN12 rails have holes at 25 mm pitch: 1300 mm needs 52 and 330 mm needs 13. The carriages take 4 × M3 each on a 20 × 20 mm pattern. [Certain] The carry sensor adds 4 × M3×10 and 2 × M3×8.
- **M5 T-nuts, about 130.** That covers:
  - brackets and joining plates (about 80–90 nuts);
  - 20 for the printed mounts on the extrusions: x_motor_mount 4, x_idler_mount 2, x_bracket 4 (the spine), z_plate 2 (the drop arm), x_bumper_L 1, z_motor_mount tab 1, carry_sensor_mount 2, x_switch_mount 2, z_switch_mount 2. The z_motor_mount and z_end_cap screw into the spine's tapped end bores instead;
  - 16 bed bolts;
  - 6 feet (both base rails at each end and at midspan; the OpenBuilds feet bring their own).
- **Wood screws.** station_mount 4, gate posts 4 and the electronics panel.
- **Filament.** `print_list.json` totals 801 cm³ of solid parts in 27 prints from 25 STL files, including the Phase 0 jig and its occluder mount, the drill template, two fill gauges, the two carry-sensor parts and the two switch mounts. That is about 1020 g of PETG if printed solid, or about 720 g at the listed walls and infill. Add a Pi case, electronics trays, cable clips and reprints, and it comes to about 1.0–1.3 kg, so the list has 2 kg. [Likely]

Design gaps found while pricing (for the CAD and wiring owners)

These are not BOM problems, but they decide what gets bought.

Resolved since pricing (the CAD decisions in CONTRACT.md):

- The MGN12 rails sit on slots: the X rail centre is at Z = 410 on the 4040's upper front slot, and the Z rail is in the 2040 spine's right slot, on the cup axis. [Certain, `cad/out/checks.txt`]
- The bed has 41 × 41 mm notches at the two upright positions. [Certain, `model.py`]
- The light roof is optional and left out of the CAD (conditional lines in the CSV).
- The 2020 drop arm is 285 mm with two 5.5 mm cross holes, 19 and 84 mm from its lower end.
- The Z machinery sits above the highest point a held slab can reach, and the lead screw is cut to 340 mm. Printed end bumpers and a `z_end_cap` are hard stops on both axes. [Certain]
- The mains side has a closed box, the laser is Class 2, the E-stop has two contact blocks, and the MOSFET modules and limit switches match WIRING.md.
- The buffer code is `ZPT25UUK20-B5-A10` [Certain, SMC ZP catalog Table 2; Proax lists it], the probe flag, its guide and the barb elbow are in the CAD, and the jig posts and arm stub are in `cut_list.csv`.

- **The head camera now sees the face-down card window** [Certain, CAD frustum check]: the lens sits 5 mm toward the label from the cup axis and the foot was trimmed. Nothing to buy; the `camera_bracket` and `phase0_top` STLs changed, so print the current ones.
- **Both LED rings are in the CAD** [Certain, `cad/out/checks.txt`; Likely for the ring sizes]. The head ring (modelled 40 × 30 × 5 mm) sits in a 42.8 mm bore with three centring ribs, against a 33 mm lip, its LED face 1 mm under the lens front, and the barb elbow clears it by 3.75 mm at the full 20 mm stroke. The seat takes rings of 39.5–42.5 mm OD (shave the ribs for one over 40). The station seat is a 64 mm bore on a shelf that takes a 60 or a 63 mm ring. `station_mount`, `camera_bracket`, `phase0_base` and `phase0_top` changed: print the current ones.
- **The fastener margins are closed in the CAD**: the probe board takes M3×16 with a washer under the head and under the nut, and every Phase 0 jig wall an M5×10 passes through is 4 mm (one washer per screw).
- **The Z stall guard is designed** (carry-height sensor, above).
- **The homing switches have printed mounts** (`x_switch_mount`, `z_switch_mount`) [Certain, `cad/out/checks.txt`]: a finger on `x_bracket` and a chamfer on `z_plate` slide across each lever and hold it 0.6 mm past its click point to the hard stop. Nothing new to buy but the screws above (was open gap 7).

Still open:

1. **[Certain, CAD] There are only 18 mm between each rail end and the upright's inner face.** Use back-face brackets and flat joining plates on the outer ends (BUILD_GUIDE §4.3).
2. **[Likely] The laser receiver module is a 5 V board.** Divide its output before GPIO 5. It is also non-modulated, so room light can hold it HIGH. Test it with the room lights on.
3. **[Likely] Angel-eye LED rings are 12 V parts, sold by OD only.** Measure the ring OD, ID and thickness before printing the ring seats. Head: OD 39.5–42.5 mm (42.8 mm bore with ribs), ID about 24 mm up to OD − 3 (33 mm lip), 5 mm thick or less (each mm thicker takes 1 mm off the 3.75 mm barb clearance). Station: OD 59–63.5 mm (64 mm bore; some "60 mm" rings are 63 mm). No 40 mm COB listing is live: a 40 mm 12-SMD ring that fits is the fallback (check it for glare in Phase 0A).
4. **[Guessing] Some mounting patterns in the CAD are not verified.**
   - The head camera board's hole pattern is assumed to be 32 mm square (the ELP board is 38 × 38 mm), and its board-to-lens-front length 20 mm (`HEAD_LENS_PROJ`).
   - The optical endstop boards: the TCST2103 itself is from Vishay's drawing (19 mm flange holes, symmetric about the slot) [Certain], but the listing does not name the sensor, and the board's header position and component heights are unknown. This now matters twice: the probe and the carry-height sensor.
   - Measure these parts before printing `camera_bracket`, `phase0_top`, `head_foot` and `carry_sensor_mount`.
5. **[Certain, sourcing; resolved in CAD] The only Class 2 laser found is 7 mm**, so the left gate post's barrel hole is now 7.2 mm (the right post keeps 6.2 mm for the receiver).
6. **[Certain, forum report] BTT has shipped Rodents with an old FluidNC build.** Reflash the current FluidNC release before loading `firmware/fluidnc/config.yaml`. Also check the Rodent's limit and motor connector types (XH or other) before buying crimp housings.
7. **[Guessing] Which lever the homing switch 10-pack has.** The switch mounts are now in the CAD (resolved above), drawn for the Omron SS-5GL size with a short hinge lever. A lever that ends more than 23 mm from the hinge end must be cut back to 20 mm (BUILD_GUIDE §6.8); a roller lever does not fit. The listing was not opened: check the photos, or measure on arrival.

Main sources

- **Sourcing run, 22 Oct 2026:** every product link in BOM.csv; the per-line notes say which pages were opened ([Certain]) and which prices come from search text ([Likely]) or the old estimate.
- **Raspberry Pi 2026 prices:** Liliputing/Heise/TechSpot reports, The Register (Feb 2026)

- **Rodent:** BIQU product page
- **LRS-150-24:** TRC Electronics
- **Clippard EV-2-24:** Clippard
- **SMC ZPT/ZP codes and prices:** ZPT25UUK20-B5-A10 at Proax, ZP25UU at automationdistribution, SMC USA ZPT (J,K) family page with configurator and CAD
- **SMC ZP catalog (stroke table, buffer dimensions):** RS copy of the ZP catalog, pages 13-11-9 (Table 2) and 13-11-12
- **SMC M-5ALU-4 barb elbow:** automationdistribution
- **TCST2103 drawing:** Vishay doc 81147, drawing 6.550-5040.01-4
- **VXB Z motor 3-pack:** VXB
- **XGZP6847A100KPGN:** Digi-Key
- **E-stop set:** AutomationDirect GCX1131
- **Schmalz HT1:** Schmalz PFG 25 HT1
- **HIWIN MGN12 price tier:** DamenCNC MGNR12
- **Laser receiver behaviour:** Acxico KY-008 listing
- **D4184 module notes:** ProtoSupplies
- **FluidNC on Rodent:** V1E forum thread